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Kwon et al. (43) **Pub. Date: Sep. 4, 2025**(54) **ORGANOMETALLIC COMPOUND,
ORGANIC LIGHT-EMITTING DEVICE
INCLUDING THE SAME, AND ELECTRONIC
APPARATUS INCLUDING THE ORGANIC
LIGHT-EMITTING DEVICE**(71) Applicant: **Samsung Electronics Co., Ltd.,**
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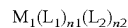
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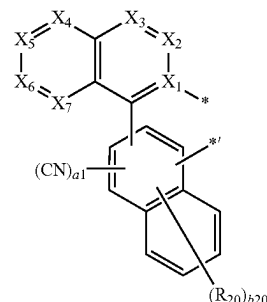
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(57) **ABSTRACT**

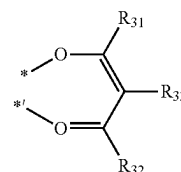
An organometallic compound represented by Formula 1:



Formula 1

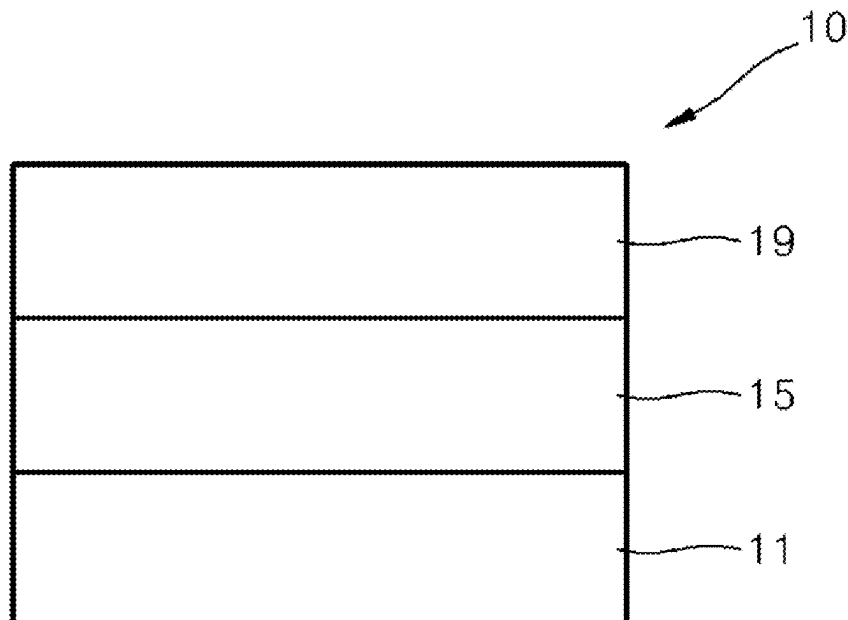
wherein M_1 is a transition metal, L_1 is a ligand represented
by Formula 1A, L_2 is a ligand represented by Formula
1B, and $n1$ and $n2$ are each independently 1 or 2:

Formula 1A

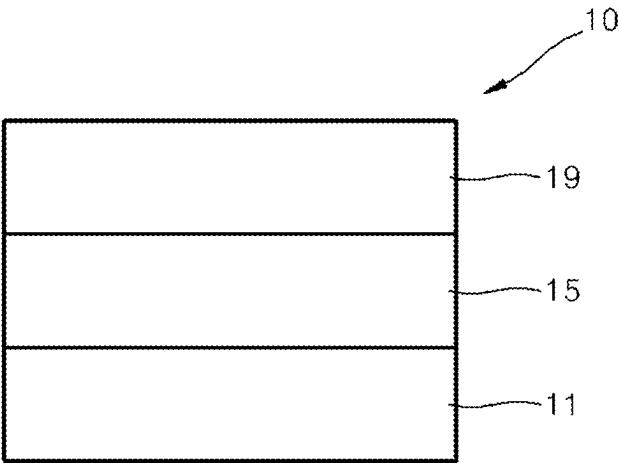


Formula 1B

wherein X_2 is C(R_2) or N, X_3 is C(R_3) or N, X_4 is C(R_4)
or N, X_5 is C(R_5) or N, X_6 is C(R_6) or N, and X_7 is
C(R_7) or N, wherein at least one of X_2 to X_7 is not N;
at least one of R_2 to R_7 is —F, —Si(Q_1)(Q_2)(Q_3), or
—Ge(Q_1)(Q_2)(Q_3); or a C_1 - C_{60} alkyl, a C_2 - C_{60} alkenyl,
a C_2 - C_{60} alkynyl, a C_3 - C_{10} cycloalkyl, a C_1 - C_{10} het-
erocycloalkyl, a C_6 - C_{60} aryl, or a C_1 - C_{60} heteroaryl,
each substituted with at least one of —F, —Si(Q_{31})
(Q_{32})(Q_{33}), —Ge(Q_{31})(Q_{32})(Q_{33}), or a combination
thereof, and the remaining substituents are as described
herein.



FIGURE



**ORGANOMETALLIC COMPOUND,
ORGANIC LIGHT-EMITTING DEVICE
INCLUDING THE SAME, AND ELECTRONIC
APPARATUS INCLUDING THE ORGANIC
LIGHT-EMITTING DEVICE**

CROSS-REFERENCE TO RELATED
APPLICATION

[0001] This application claims priority to and the benefit of Korean Patent Application No. 10-2024-0030111, filed on Feb. 29, 2024, in the Korean Intellectual Property Office, and all the benefits accruing therefrom under 35 U.S.C. § 119, the entire content of which is incorporated by reference herein.

BACKGROUND

1. Field

[0002] The disclosure relates to an organometallic compound, an organic light-emitting device including the same, and an electronic apparatus including the organic light-emitting device.

2. Description of the Related Art

[0003] Organic light-emitting devices (OLEDs) are self-emissive devices, have excellent characteristics in terms of viewing angles, response times, luminance, driving voltage, and response speed. OLEDs can also produce full-color images.

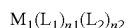
[0004] For example, an organic light-emitting device may include an anode, a cathode, and an organic layer arranged between the anode and the cathode and including an emission layer. A hole transport region may be arranged between the anode and the emission layer, and an electron transport region may be arranged between the emission layer and the cathode. Holes provided from the anode move toward the emission layer through the hole transport region, and electrons provided from the cathode move toward the emission layer through the electron transport region. The holes and the electrons recombine in the emission layer to produce excitons. The excitons transition from an excited state to a ground state, thereby generating light.

SUMMARY

[0005] Provided are an organometallic compound, an organic light-emitting device including the same, and an electronic apparatus including the organic light-emitting device.

[0006] Additional aspects will be set forth in part in the detailed description which follows and, in part, will be apparent from the detailed description, or may be learned by practice of the presented exemplary embodiments herein.

[0007] According to an aspect, provided is an organometallic compound represented by Formula 1.



Formula 1

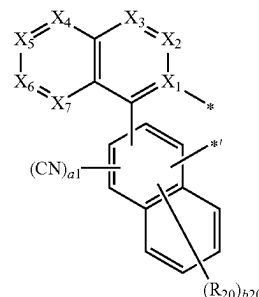
wherein, in Formula 1,

[0008] M_1 is a transition metal,

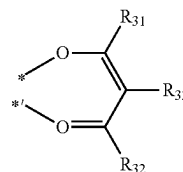
[0009] L_1 is a ligand represented by Formula 1A,

[0010] L_2 is a ligand represented by Formula 1B, and

[0011] $n1$ and $n2$ are each independently 1 or 2,



Formula 1A



Formula 1B

wherein, in Formulae 1A and 1B,

[0012] X_1 is C or N,

[0013] X_2 is C(R_2) or N, X_3 is C(R_3) or N, X_4 is C(R_4) or N, X_5 is C(R_5) or N, X_6 is C(R_6) or N, and X_7 is C(R_7) or N, wherein at least one of X_2 to X_7 is not N,

[0014] $a1$ is 1 or 2,

[0015] R_2 to R_7 , R_{20} , and R_{31} to R_{33} are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, —SF₅, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C₁-C₆₀ alkyl group, a substituted or unsubstituted C₂-C₆₀ alkenyl group, a substituted or unsubstituted C₂-C₆₀ alkynyl group, a substituted or unsubstituted C₁-C₆₀ alkoxy group, a substituted or unsubstituted C₁-C₆₀ alkylthio group, a substituted or unsubstituted C₃-C₁₀ cycloalkyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkyl group, a substituted or unsubstituted C₃-C₁₀ cycloalkenyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkenyl group, a substituted or unsubstituted C₆-C₆₀ aryl group, a substituted or unsubstituted C₇-C₆₀ alkyl aryl group, a substituted or unsubstituted C₇-C₆₀ aryl alkyl group, a substituted or unsubstituted C₆-C₆₀ aryloxy group, a substituted or unsubstituted C₆-C₆₀ arylthio group, a substituted or unsubstituted C₁-C₆₀ heteroaryl group, a substituted or unsubstituted C₂-C₆₀ alkyl heteroaryl group, a substituted or unsubstituted C₂-C₆₀ heteroaryl alkyl group, a substituted or unsubstituted C₁-C₆₀ heteroaryloxy group, a substituted or unsubstituted C₁-C₆₀ heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q_1)(Q_2)(Q_3), —Ge(Q_1)(Q_2)(Q_3), —N(Q_4)(Q_5), —B(Q_6)(Q_7), —P(Q_8)(Q_9), or —P(=O)(Q_8)(Q_9),

[0016] at least one of R₂ to R₇ is:

[0017] —F, —Si(Q₁)(Q₂)(Q₃), or —Ge(Q₁)(Q₂)(Q₃),
or

[0018] a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₆-C₆₀ aryl group, or a C₁-C₆₀ heteroaryl group, each substituted with at least one of —F, —Si(Q₃₁)(Q₃₂)(Q₃₃), —Ge(Q₃₁)(Q₃₂)(Q₃₃), or a combination thereof,

[0019] b₂₀ is an integer from 1 to 5,

[0020] two or more of R₂ to R₇ are optionally linked to each other to form a C₅-C₃₀ carbocyclic group that is unsubstituted or substituted with at least one R_{10a}, or two or more of R₂ to R₇ are optionally linked to each other to form a C₁-C₃₀ heterocyclic group that is unsubstituted or substituted with at least one R_{10a},

[0021] two or more of a plurality of R₂₀ groups are optionally linked to each other to form a C₅-C₃₀ carbocyclic group that is unsubstituted or substituted with at least one R_{10a}, or two or more of the plurality of R₂₀ groups are optionally linked to each other to form a C₁-C₃₀ heterocyclic group that is unsubstituted or substituted with at least one R_{10a},

[0022] R_{10a} is as defined for R₁,

[0023] * and *' each indicate a binding site to M₁,

[0024] a substituent of the substituted C₅-C₃₀ carbocyclic group, the substituted C₁-C₃₀ heterocyclic group, the substituted C₁-C₆₀ alkyl group, the substituted C₂-C₆₀ alkenyl group, the substituted C₂-C₆₀ alkynyl group, the substituted C₁-C₆₀ alkoxy group, the substituted C₁-C₆₀ alkylthio group, the substituted C₁-C₆₀ alkylthio group, the substituted C₃-C₁₀ cycloalkyl group, the substituted C₁-C₁₀ heterocycloalkyl group, the substituted C₃-C₁₀ cycloalkenyl group, the substituted C₁-C₁₀ heterocycloalkenyl group, the substituted C₆-C₆₀ aryl group, the substituted C₇-C₆₀ alkyl aryl group, the substituted C₇-C₆₀ aryl alkyl group, the substituted C₆-C₆₀ aryloxy group, the substituted C₆-C₆₀ arylthio group, the substituted C₁-C₆₀ heteroaryl group, the substituted C₂-C₆₀ alkyl heteroaryl group, the substituted C₂-C₆₀ heteroaryl alkyl group, the substituted C₁-C₆₀ heteroaryloxy group, the substituted C₁-C₆₀ heteroarylthio group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed heteropolycyclic group is each independently:

[0025] deuterium, —F, —Cl, —Br, —I, —SF₅, —CD₃, —CD₂H, —CDH₂, —CF₃, —CF₂H, —CFH₂, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, or a C₁-C₆₀ alkylthio group,

[0026] a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, or a C₁-C₆₀ alkylthio group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF₅, —CD₃, —CD₂H, —CDH₂, —CF₃, —CF₂H, —CFH₂, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid

group or a salt thereof, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₇-C₆₀ alkyl aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a C₂-C₆₀ alkyl heteroaryl group, a C₁-C₆₀ heteroaryloxy group, a C₁-C₆₀ heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q₁₁)(Q₁₂)(Q₁₃), —Ge(Q₁₁)(Q₁₂)(Q₁₃), —N(Q₁₄)(Q₁₅),

[0027] —B(Q₁₆)(Q₁₇), —P(Q₁₈)(Q₁₉), —P(=O)(Q₁₈)(Q₁₉), or a combination thereof,

[0028] a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₇-C₆₀ alkyl aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a C₂-C₆₀ alkyl heteroaryl group, a C₁-C₆₀ heteroaryloxy group, a C₁-C₆₀ heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, or a monovalent non-aromatic condensed heteropolycyclic group,

[0029] a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₇-C₆₀ alkyl aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a C₂-C₆₀ alkyl heteroaryl group, a C₁-C₆₀ heteroaryloxy group, a C₁-C₆₀ heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, or a monovalent non-aromatic condensed heteropolycyclic group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF₅, —CD₃, —CD₂H, —CDH₂, —CF₃, —CF₂H, —CFH₂, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, a C₁-C₆₀ alkylthio group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₇-C₆₀ alkyl aryl group, a C₇-C₆₀ aryl alkyl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a C₂-C₆₀ alkyl heteroaryl group, a C₁-C₆₀ heteroaryloxy group, a C₁-C₆₀ heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q₂₁)(Q₂₂)(Q₂₃), —Ge(Q₂₁)(Q₂₂)(Q₂₃), —N(Q₂₄)(Q₂₅), —B(Q₂₆)(Q₂₇),

[0030] —P(Q₂₈)(Q₂₉), —P(=O)(Q₂₈)(Q₂₉), or a combination thereof, or

[0031] —Si(Q₃₁)(Q₃₂)(Q₃₃), —Ge(Q₃₁)(Q₃₂)(Q₃₃), —N(Q₃₄)(Q₃₅), —B(Q₃₆)(Q₃₇), —P(Q₃₈)(Q₃₉), or

[0032] —P(=O)(Q₃₈)(Q₃₉), and

[0033] Q₁ to Q₉, Q₁₁ to Q₁₉, Q₂₁ to Q₂₉, and Q₃₁ to Q₃₉ are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, —SF₅, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid

group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C_1 - C_{60} alkyl group, a substituted or unsubstituted C_2 - C_{60} alkenyl group, a substituted or unsubstituted C_2 - C_{60} alkynyl group, a substituted or unsubstituted C_1 - C_{60} alkoxy group, a substituted or unsubstituted C_1 - C_{60} alkylthio group, a substituted or unsubstituted C_3 - C_{10} cycloalkyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3 - C_{10} cycloalkenyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkenyl group, a substituted or unsubstituted C_6 - C_{60} aryl group, a substituted or unsubstituted C_7 - C_{60} alkyl aryl group, a substituted or unsubstituted C_7 - C_{60} aryl alkyl group, a substituted or unsubstituted C_6 - C_{60} aryloxy group, a substituted or unsubstituted C_6 - C_{60} arylthio group, a substituted or unsubstituted C_1 - C_{10} heteroaryl group, a substituted or unsubstituted C_2 - C_{60} alkyl heteroaryl group, a substituted or unsubstituted C_2 - C_{60} heteroaryl alkyl group, a substituted or unsubstituted C_1 - C_{60} heteroaryloxy group, a substituted or unsubstituted C_1 - C_{60} heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, or a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group.

[0034] According to another aspect, an organic light-emitting device includes a first electrode, a second electrode, and an organic layer arranged between the first electrode and the second electrode, wherein the organic layer includes an emission layer, and wherein the organic layer further includes at least one of the organometallic compounds represented by Formula 1.

[0035] The organometallic compound represented by Formula 1 may be included in the emission layer in the organic layer, and the organometallic compound included in the emission layer may act as a dopant.

BRIEF DESCRIPTION OF THE DRAWING

[0036] The above and other aspects, features, and advantages of certain exemplary embodiments of the disclosure will be more apparent from the following detailed description taken in conjunction with the accompanying drawing, in which the FIGURE is a schematic cross-sectional view of an organic light-emitting device according to one or more embodiments.

DETAILED DESCRIPTION

[0037] Reference will now be made in detail to exemplary embodiments, examples of which are illustrated in the accompanying drawing, wherein like reference numerals refer to like elements throughout the specification. In this regard, the present embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described below, by referring to the figures, to explain aspects. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

[0038] The terminology used herein is for the purpose of describing one or more exemplary embodiments only and is not intended to be limiting. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. The term “or” means “and/or.” It will be further understood that the terms “comprises” and/or “comprising,” or “includes” and/or “including” when used in this specification, specify the presence of stated features, regions, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, regions, integers, steps, operations, elements, components, and/or groups thereof.

[0039] It will be understood that, although the terms first, second, third etc. may be used herein to describe various elements, components, regions, layers, and/or sections, these elements, components, regions, layers, and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer, or section from another element, component, region, layer, or section. Thus, a first element, component, region, layer, or section discussed below could be termed a second element, component, region, layer, or section without departing from the teachings of the present embodiments.

[0040] Exemplary embodiments are described herein with reference to cross section illustrations that are schematic illustrations of idealized embodiments. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments described herein should not be construed as limited to the particular shapes of regions as illustrated herein but are to include deviations in shapes that result, for example, from manufacturing. For example, a region illustrated or described as flat may, typically, have rough and/or nonlinear features. Moreover, sharp angles that are illustrated may be rounded. Thus, the regions illustrated in the figures are schematic in nature and their shapes are not intended to illustrate the precise shape of a region and are not intended to limit the scope of the present claims.

[0041] It will be understood that when an element is referred to as being “on” another element, it can be directly in contact with the other element or intervening elements may be present therebetween. In contrast, when an element is referred to as being “directly on” another element, there are no intervening elements present.

[0042] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this general inventive concept belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and the present disclosure, and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0043] “About” or “approximately” as used herein is inclusive of the stated value and means within an acceptable range of deviation for the particular value as determined by one of ordinary skill in the art, considering the measurement in question and the error associated with measurement of the particular quantity (i.e., the limitations of the measurement system). For example, “about” can mean within one or more standard deviations, or within $\pm 30\%$, 20% , 10% , 5% of the stated value.

[0044] An organometallic compound according to an aspect is represented by Formula 1:



wherein, in Formula 1, M_1 is a transition metal.

[0045] For example, M_1 may be a first-row transition

metal of the Periodic Table of Elements, a second-row

transition metal of the Periodic Table of Elements, or a

third-row transition metal of the Periodic Table of Elements.

[0046] In one or more embodiments, M_1 may be iridium (Ir), platinum (Pt), osmium (Os), palladium (Pd), gold (Au), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), thulium (Tm), or rhodium (Rh).

[0047] In one or more embodiments, M_1 may be iridium (Ir), platinum (Pt), osmium (Os), or rhodium (Rh).

[0048] In one or more embodiments, M_1 may be iridium (Ir).

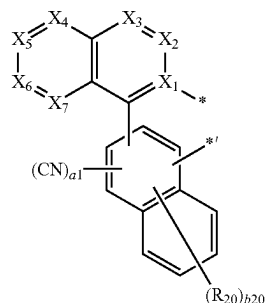
[0049] In Formula 1, n_1 is 1 or 2, and n_2 is 1 or 2.

[0050] In one or more embodiments, a sum of n_1 and n_2 may be 2 or 3.

[0051] In one or more embodiments, M_1 may be iridium (Ir), and the sum of n_1 and n_2 may be 3.

[0052] In one or more embodiments, M_1 may be platinum (Pt), and the sum of n_1 and n_2 may be 2.

[0053] In Formula 1, L_1 is a ligand represented by Formula 1A.



Formula 1A

[0054] In Formula 1A, X_1 is C or N. For example, X_1 may be N.

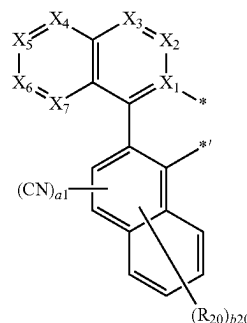
[0055] In Formula 1A, X_2 is C(R_2) or N, X_3 is C(R_3) or N, X_4 is C(R_4) or N, X_5 is C(R_5) or N, X_6 is C(R_6) or N, and X_7 is C(R_7) or N.

[0056] At least one of X_2 to X_7 is not N.

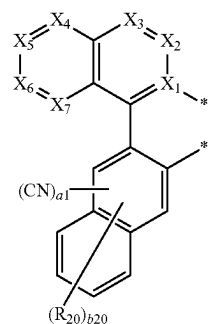
[0057] In one or more embodiments, in Formula 1A, X_2 may be C(R_2), X_3 may be C(R_3), X_4 may be C(R_4), X_5 may be C(R_5), X_6 may be C(R_6), and X_7 may be C(R_7).

[0058] In Formula 1A, a_1 is 1 or 2. For example, a_1 may be 1.

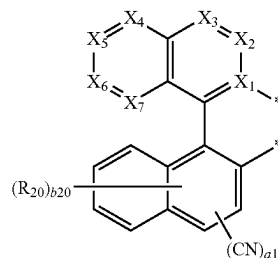
[0059] In one or more embodiments, L_1 may be a ligand represented by any one of Formulae 1A-1 to 1A-3.



1A-1



1A-2



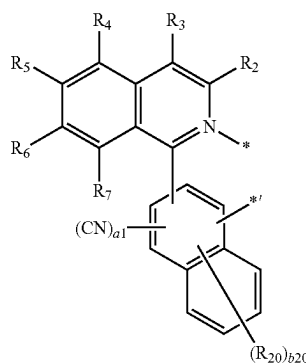
1A-3

[0060] In Formulae 1A-1 to 1A-3,

[0061] X_1 to X_7 , a_1 , R_{20} , and b_{20} are each as described herein, and

[0062] * and *' each indicate a binding site to M_1 .

[0063] In one or more embodiments, L_1 may be a ligand represented by Formula 1A(1).



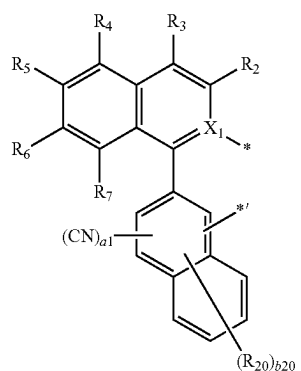
Formula 1A(1)

[0064] In Formula 1A(1),

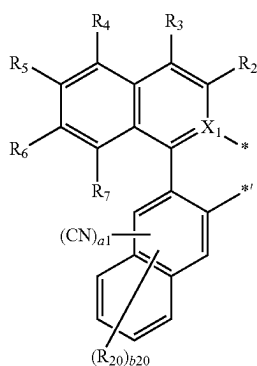
[0065] R_2 to R_7 , $a1$, R_{20} , and $b20$ are each as described herein, and

[0066] * and *' each indicate a binding site to M_1 .

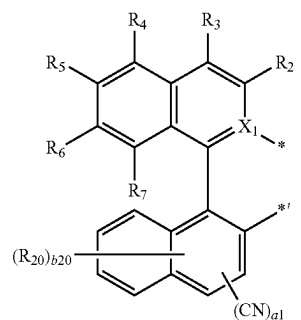
[0067] In one or more embodiments, L_1 may be a ligand represented by one of Formulae 1A(1)-1 to 1A(1)-3.



1A(1)-1



1A(1)-2



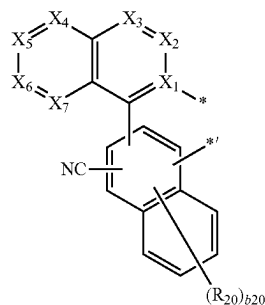
1A(1)-3

[0068] In Formulae 1A(1)-1 to 1A(1)-3,

[0069] X_1 , R_2 to R_7 , $a1$, R_{20} , and $b20$ are each as described herein, and

[0070] * and *' each indicate a binding site to M_1 .

[0071] In one or more embodiments, L_1 may be a ligand represented by Formula 1A(2).



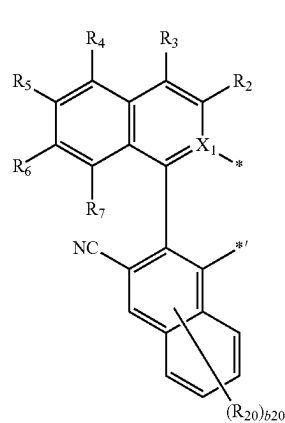
Formula 1A(2)

[0072] In Formula 1A(2),

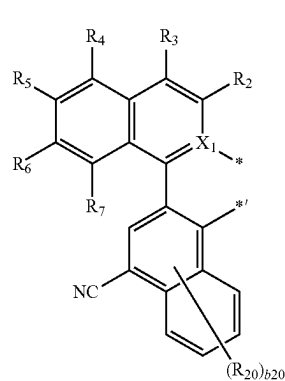
[0073] X_1 to X_7 , R_{20} , and $b20$ are each as described herein, and

[0074] * and *' each indicate a binding site to M_1 .

[0075] In one or more embodiments, L_1 may be a ligand represented by one of Formulae 1A(2)-1 to 1A(2)-6.

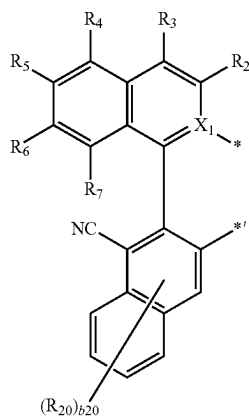


1A(2)-1



1A(2)-2

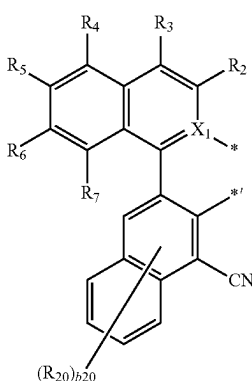
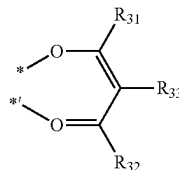
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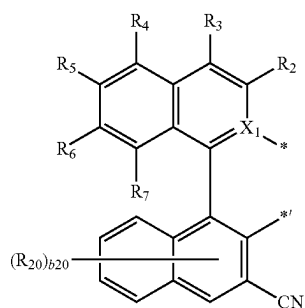
1A(2)-3

[0079] L_2 in Formula 1 is a ligand represented by Formula 1B.

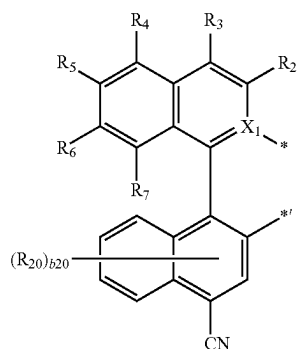
Formula 1B



1A(2)-4



1A(2)-5



1A(2)-6

[0076] In Formulae 1A(2)-1 to 1A(2)-6,

[0077] X_1 , R_2 to R_7 , a_1 , R_{20} , and b_{20} are each as described in claim 1, and

[0078] * and *' each indicate a binding site to M_1 .

[0080] In Formulae 1A and 1B, R_2 to R_7 , R_{20} , and R_{31} to R_{33} are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, —SF₅, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, —SF₅, a substituted or unsubstituted C_1 - C_{60} alkyl group, a substituted or unsubstituted C_2 - C_{60} alkenyl group, a substituted or unsubstituted C_2 - C_{60} alkynyl group, a substituted or unsubstituted C_1 - C_{60} alkoxy group, a substituted or unsubstituted C_1 - C_{60} alkylthio group, a substituted or unsubstituted C_3 - C_{10} cycloalkyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3 - C_{10} cycloalkenyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkenyl group, a substituted or unsubstituted C_6 - C_{60} aryl group, a substituted or unsubstituted C_7 - C_{60} alkyl aryl group, a substituted or unsubstituted C_7 - C_{60} aryl alkyl group, a substituted or unsubstituted C_6 - C_{60} aryloxy group, a substituted or unsubstituted C_6 - C_{60} arylthio group, a substituted or unsubstituted C_1 - C_{60} heteroaryl group, a C_2 - C_{60} alkyl heteroaryl group, a C_2 - C_{60} heteroaryl alkyl group, a substituted or unsubstituted C_1 - C_{60} heteroaryloxy group, a substituted or unsubstituted C_1 - C_{60} heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q_1)(Q_2)(Q_3), —Ge(Q_1)(Q_2)(Q_3), —N(Q_4)(Q_5), —B(Q_6)(Q_7), —P(Q_8)(Q_9), or —P(=O)(Q_8)(Q_9).

[0081] In Formula 1A, two or more of R_2 to R_7 are optionally linked to each other to form a C_5 - C_{30} carbocyclic group that is unsubstituted or substituted with at least one R_{10a} , or two or more of R_2 to R_7 are optionally linked to each other to form a C_1 - C_{30} heterocyclic group that is unsubstituted or substituted with at least one R_{10a} , wherein R_{10a} is as defined for R_1 .

[0082] In Formula 1A, two or more of a plurality of R_{20} are optionally linked to each other to form a C_5 - C_{30} carbocyclic group that is unsubstituted or substituted with at least one R_{10a} , or two or more of the plurality of R_{20} are optionally linked to each other to form a C_1 - C_{30} heterocyclic group that is unsubstituted or substituted with at least one R_{10a} , wherein R_{10a} is as defined for R_1 .

[0083] In one or more embodiments, in Formulae 1A and 1B, R_2 to R_7 , R_{20} , and R_{31} to R_{33} may each independently be:

[0084] hydrogen, deuterium, —F, —Cl, —Br, —I, —SF₅, —CD₃, —CD₂H, —CDH₂, —CF₃, —CF₂H, —CFH₂, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a

salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, $-\text{SF}_5$, a $\text{C}_1\text{-C}_{20}$ alkyl group, a $\text{C}_1\text{-C}_{20}$ alkoxy group, or a $\text{C}_1\text{-C}_{20}$ alkylthio group;

[0085] a $\text{C}_1\text{-C}_{20}$ alkyl group, a $\text{C}_1\text{-C}_{20}$ alkoxy group, or a $\text{C}_1\text{-C}_{20}$ alkylthio group, each substituted with at least one of deuterium, $-\text{F}$, $-\text{Cl}$, $-\text{Br}$, $-\text{I}$, $-\text{SF}_5$, $-\text{CD}_3$, $-\text{CD}_2\text{H}$, $-\text{CDH}_2$, $-\text{CF}_3$, $-\text{CF}_2\text{H}$, $-\text{CFH}_2$, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a $\text{C}_1\text{-C}_{10}$ alkyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a pyridinyl group, a pyrimidinyl group, or a combination thereof;

[0086] a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzoimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, or an imidazopyrimidinyl group;

[0087] a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzoimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a tri-

azolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, or an imidazopyrimidinyl group, each substituted with at least one of deuterium, $-\text{F}$, $-\text{Cl}$, $-\text{Br}$, $-\text{I}$, $-\text{SF}_5$, $-\text{CD}_3$, $-\text{CD}_2\text{H}$, $-\text{CDH}_2$, $-\text{CF}_3$, $-\text{CF}_2\text{H}$, $-\text{CFH}_2$, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a $\text{C}_1\text{-C}_{20}$ alkyl group, a $\text{C}_1\text{-C}_{20}$ alkoxy group, a $\text{C}_1\text{-C}_{20}$ alkylthio group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzoimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, or a combination thereof; or

[0088] $-\text{Si}(\text{Q}_1)(\text{Q}_2)(\text{Q}_3)$, $-\text{Ge}(\text{Q}_1)(\text{Q}_2)(\text{Q}_3)$, $-\text{N}(\text{Q}_4)(\text{Q}_5)$, $-\text{B}(\text{Q}_6)(\text{Q}_7)$, $-\text{P}(\text{Q}_8)(\text{Q}_9)$, or

[0089] $-\text{P}(=\text{O})(\text{Q}_8)(\text{Q}_9)$, and

[0090] Q_1 to Q_9 may each independently be:

[0091] $-\text{CH}_3$, $-\text{CD}_3$, $-\text{CD}_2\text{H}$, $-\text{CDH}_2$, $-\text{CH}_2\text{CH}_3$, $-\text{CH}_2\text{CD}_3$, $-\text{CH}_2\text{CD}_2\text{H}$, $-\text{CH}_2\text{CDH}_2$, $-\text{CHDCD}_3$, $-\text{CHDCD}_2\text{H}$, $-\text{CHDCDH}_2$, $-\text{CHDCD}_3$, $-\text{CD}_2\text{CD}_3$, $-\text{CD}_2\text{CD}_2\text{H}$, or $-\text{CD}_2\text{CDH}_2$;

[0092] an n-propyl group, an iso-propyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, an n-pentyl group, an isopentyl group, a sec-pentyl group, a tert-pentyl group, a phenyl group, or a naphthyl group; or

[0093] an n-propyl group, an iso-propyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, an n-pentyl group, an isopentyl group, a sec-pentyl group, a tert-pentyl group, a phenyl group, or a naphthyl group, each substituted with at least one of deuterium, a $\text{C}_1\text{-C}_{10}$ alkyl group, a phenyl group, or a combination thereof.

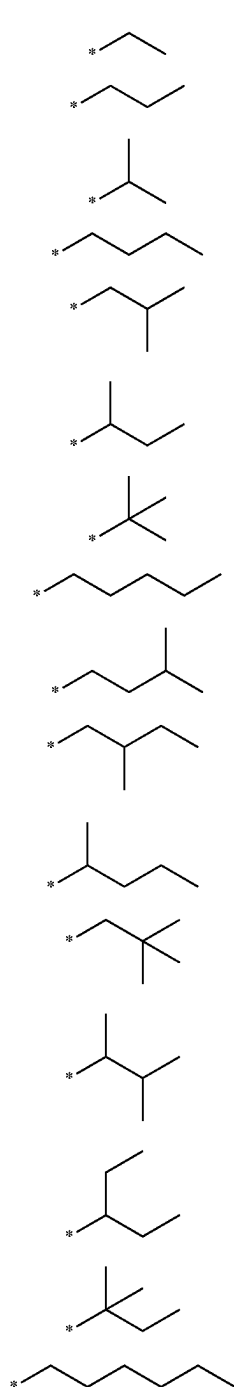
[0094] In one or more embodiments, in Formulae 1A and 1B, R_2 to R_7 , R_{20} , and R_{31} to R_{33} may each independently be:

[0095] hydrogen, deuterium, $-\text{F}$, $-\text{Cl}$, $-\text{Br}$, $-\text{I}$, $-\text{SF}_5$, $-\text{CD}_3$, $-\text{CD}_2\text{H}$, $-\text{CDH}_2$, $-\text{CF}_3$, $-\text{CF}_2\text{H}$,

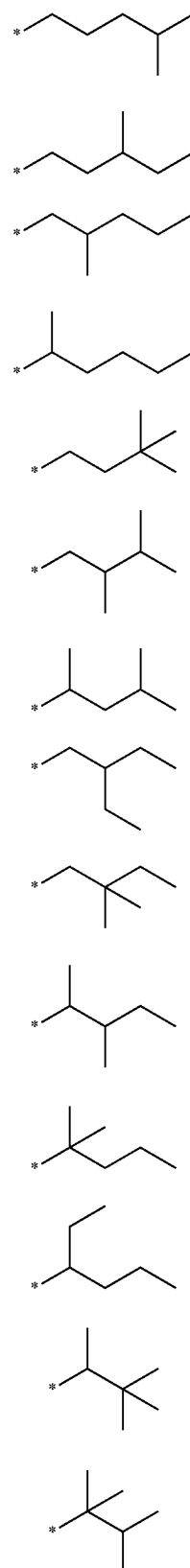
—CFH₂, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, or a C₁-C₆₀ alkylthio group;

[0096] a group represented by one of Formulae 9-1 to 9-39, 9-44 to 9-61, 9-201 to 9-237, 10-1 to 10-129, or 10-201 to 10-353; or

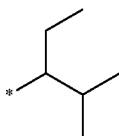
[0097] —Si(Q₁)(Q₂)(Q₃), —Ge(Q₁)(Q₂)(Q₃), or —N(Q₄)(Q₅);



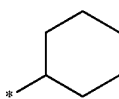
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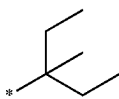
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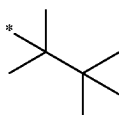
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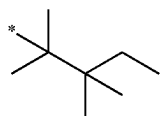
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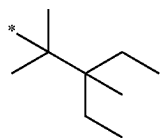
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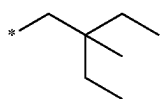
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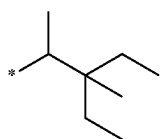
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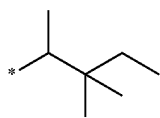
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9-37



9-38



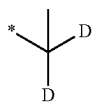
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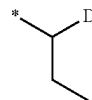


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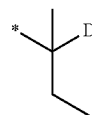


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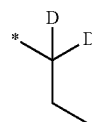
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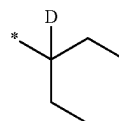
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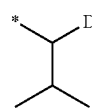
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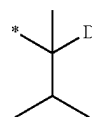
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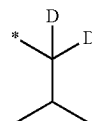
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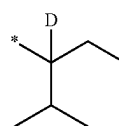
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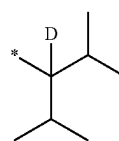
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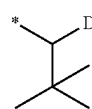
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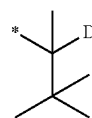
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9-55

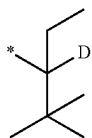


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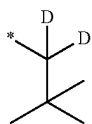


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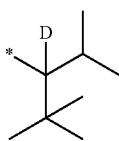
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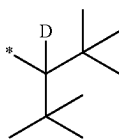
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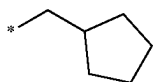
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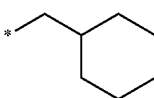
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9-61



9-201



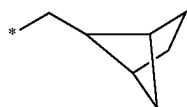
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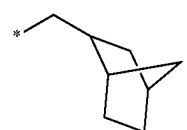
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9-204



9-205

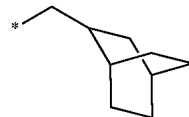


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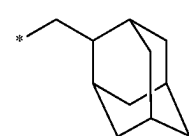


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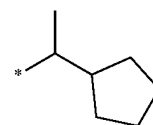
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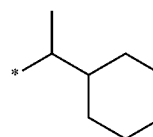
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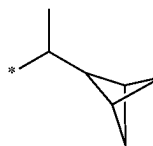
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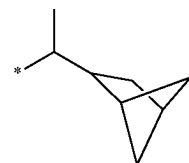
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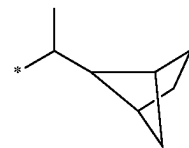
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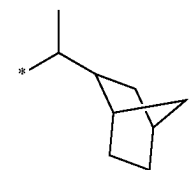
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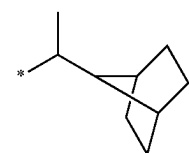
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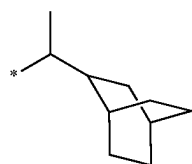


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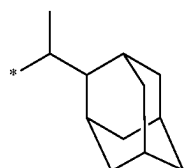


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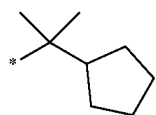
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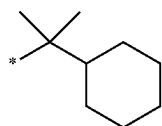
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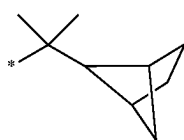
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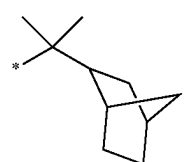
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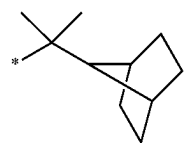
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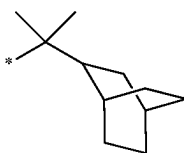
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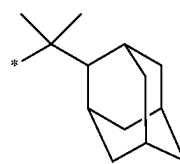


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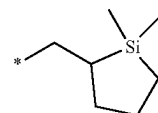


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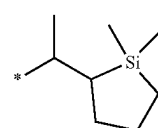
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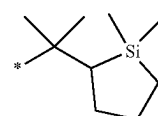
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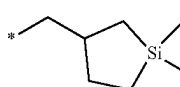
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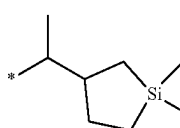
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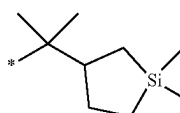
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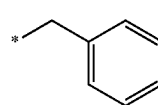
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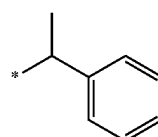
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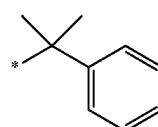
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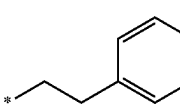
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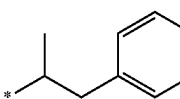
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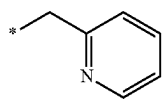


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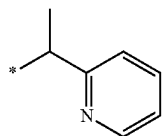


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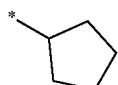
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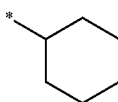
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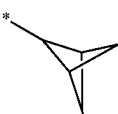
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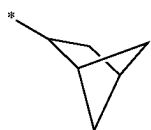
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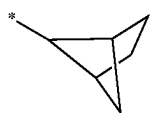
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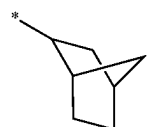
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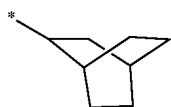
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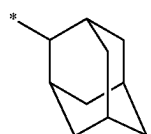
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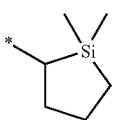
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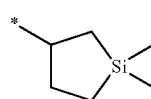


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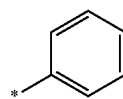


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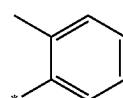
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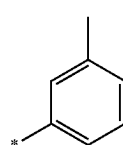
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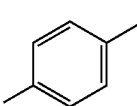
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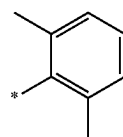
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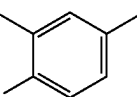
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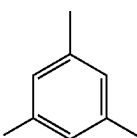
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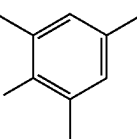
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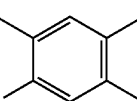
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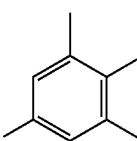
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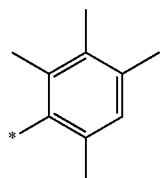


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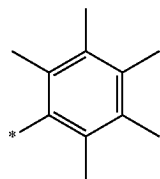


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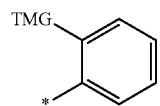
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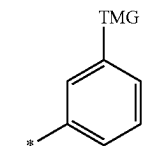
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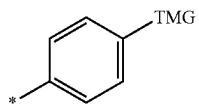
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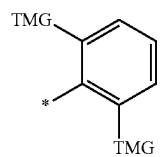
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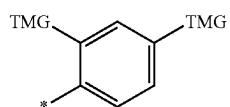
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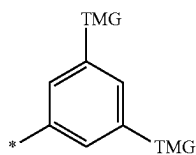
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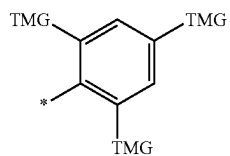
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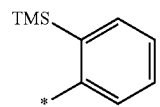
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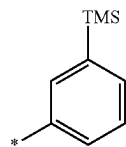


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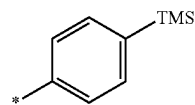


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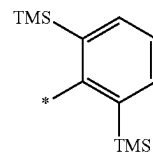
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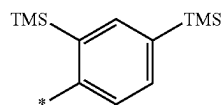
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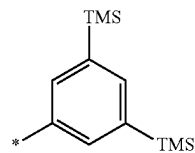
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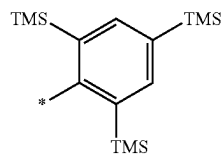
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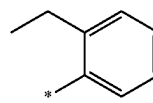
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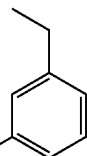
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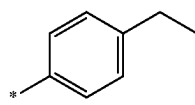
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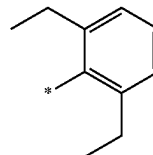
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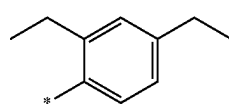
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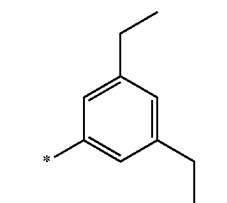


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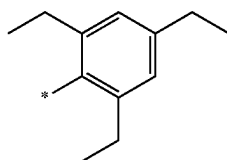


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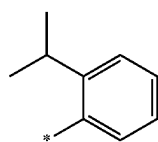
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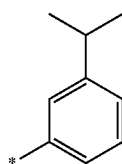
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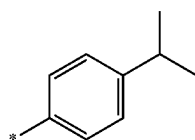
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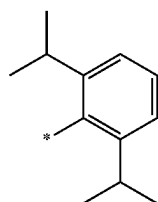
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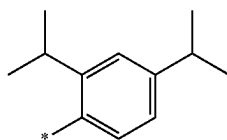
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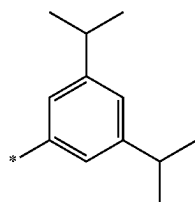
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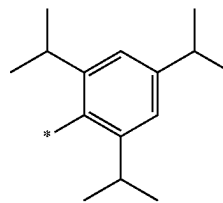


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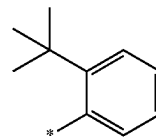


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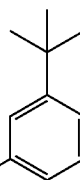
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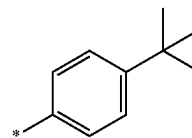
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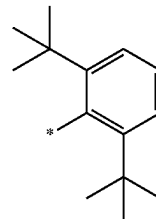
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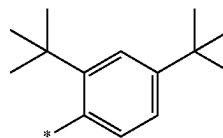
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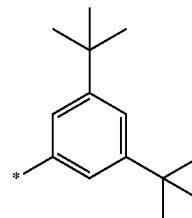
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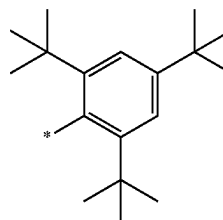
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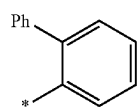


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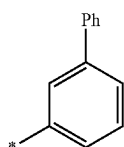


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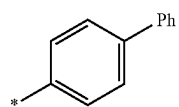
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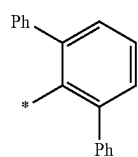
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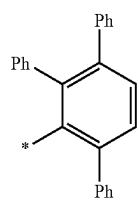
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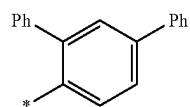
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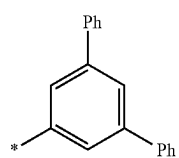
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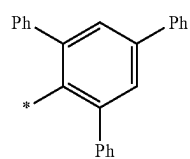
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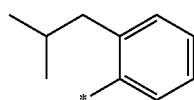
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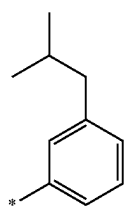
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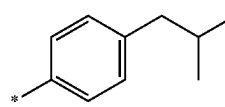


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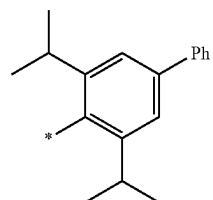


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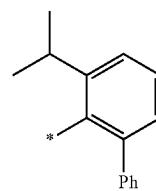
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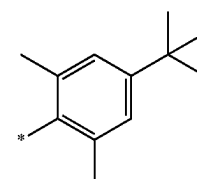
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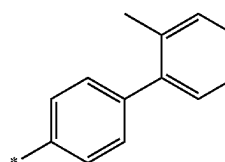
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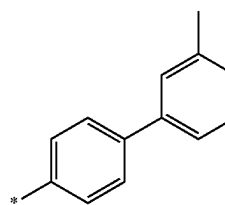
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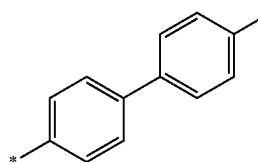
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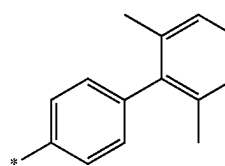
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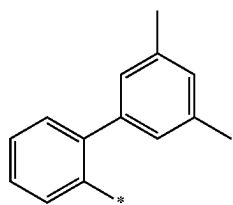


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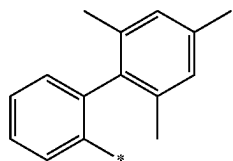


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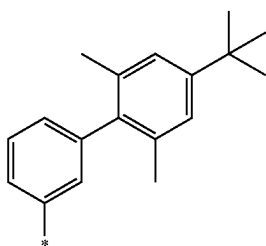
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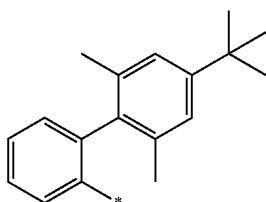
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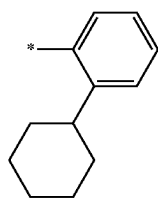
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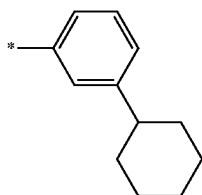
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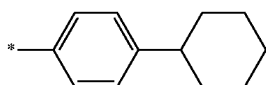
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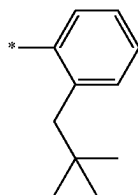
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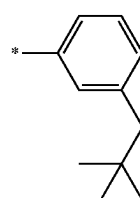


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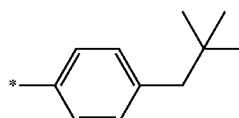


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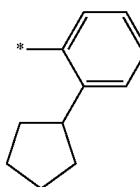
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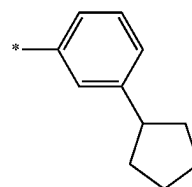
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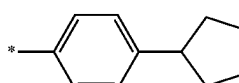
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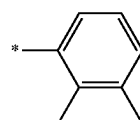
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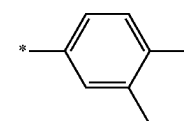
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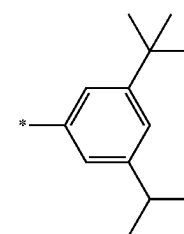
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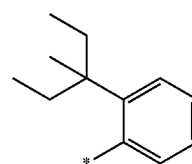
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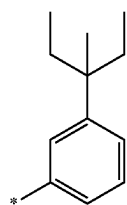


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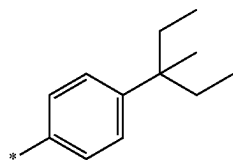


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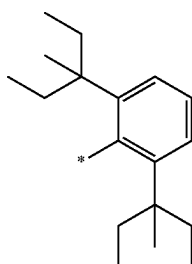
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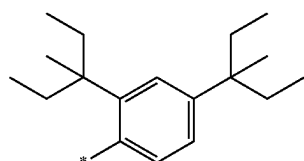
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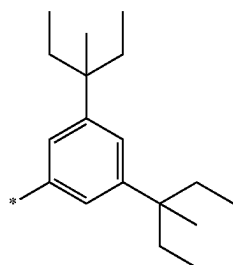
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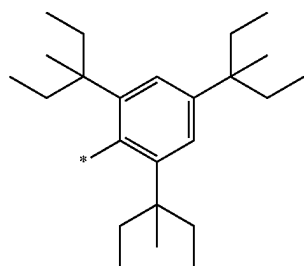
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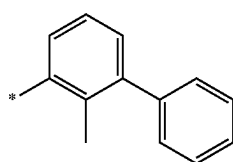
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10-113

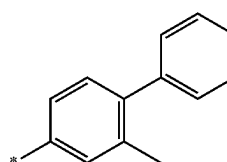


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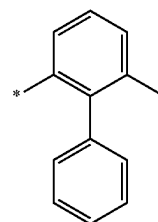


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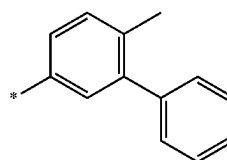
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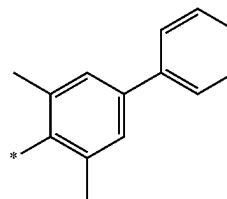
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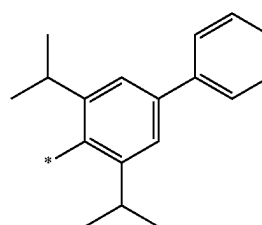
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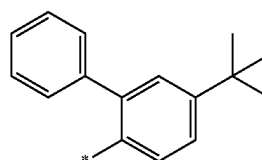
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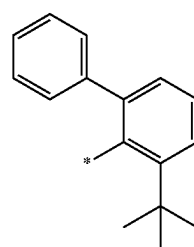
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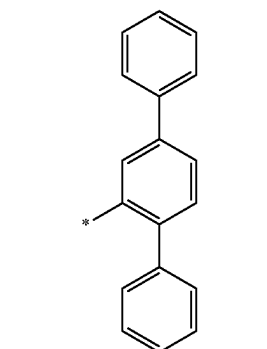


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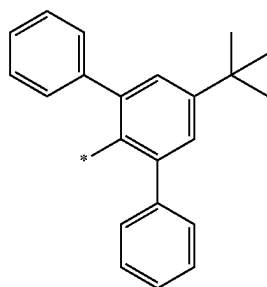


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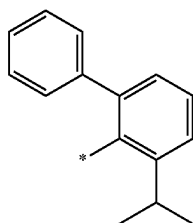
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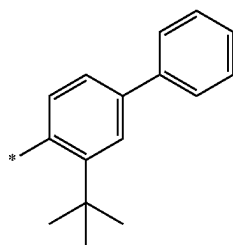
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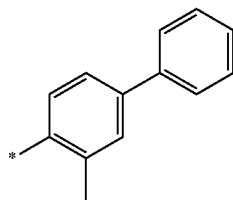
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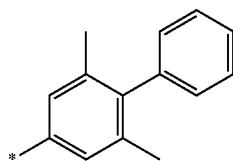
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10-126

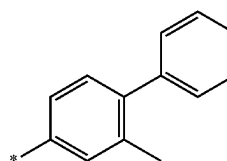


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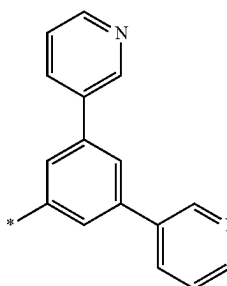


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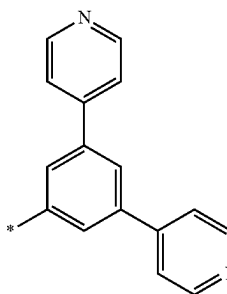
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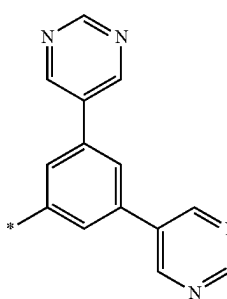
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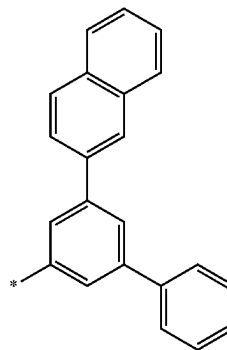
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10-202

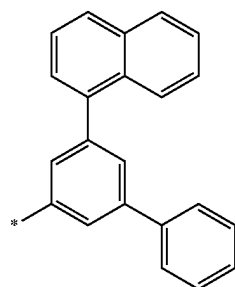


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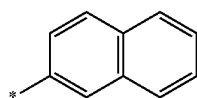


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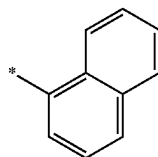
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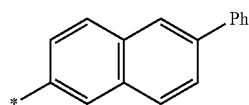
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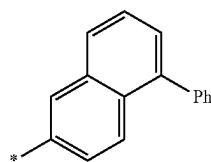
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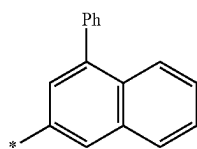
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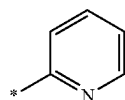
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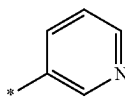
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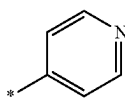
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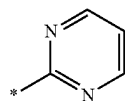
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10-212

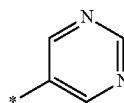


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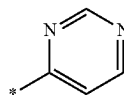


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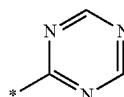
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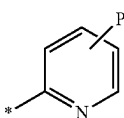
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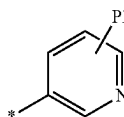
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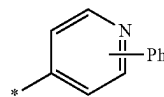
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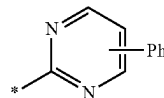
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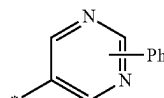
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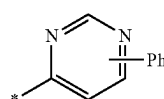
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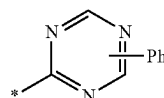
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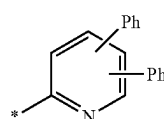
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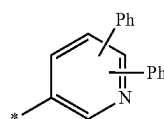
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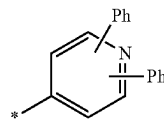
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10-225

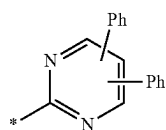


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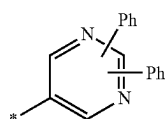


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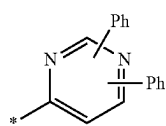
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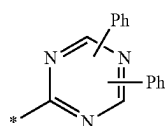
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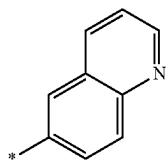
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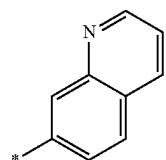
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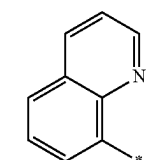
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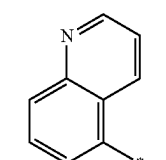
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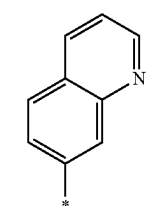
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10-234

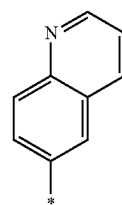


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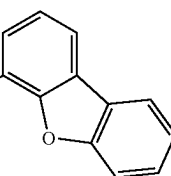


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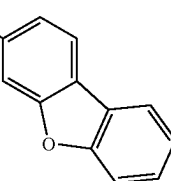
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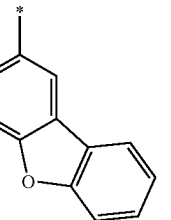
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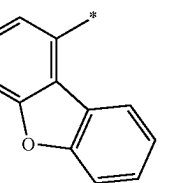
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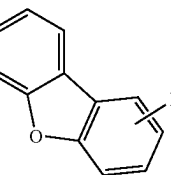
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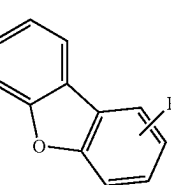
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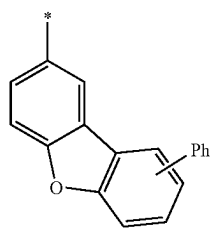


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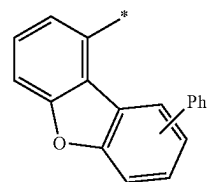


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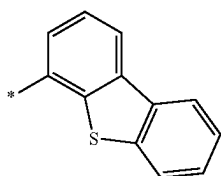
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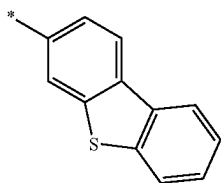
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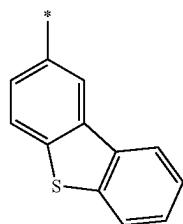
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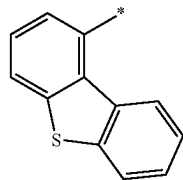
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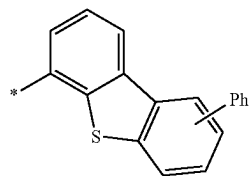
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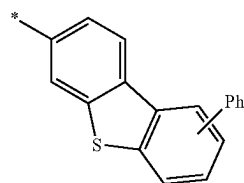


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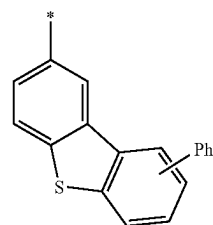


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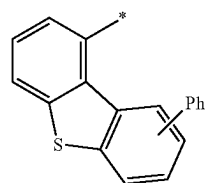
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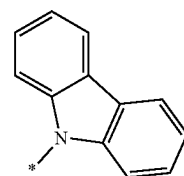
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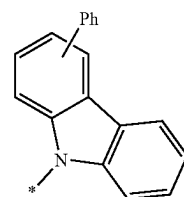
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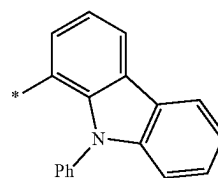
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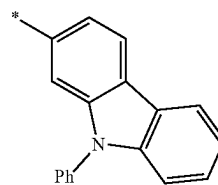
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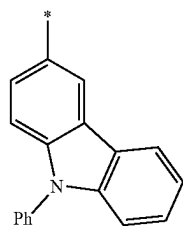


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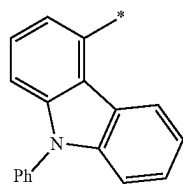


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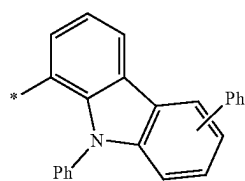
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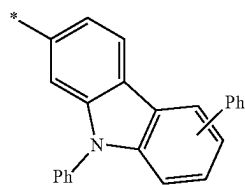
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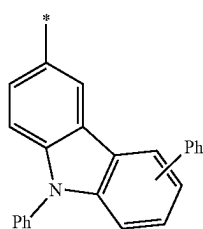
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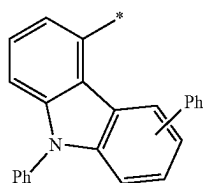
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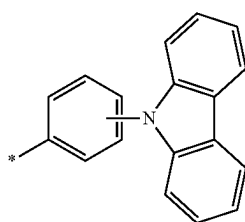
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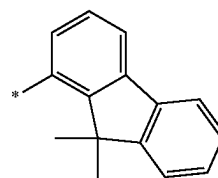


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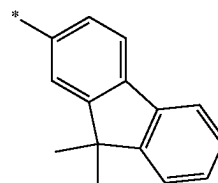


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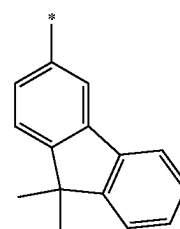
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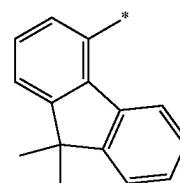
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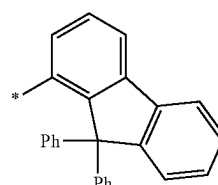
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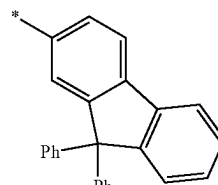
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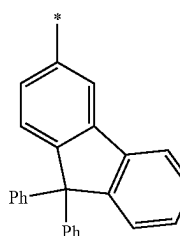
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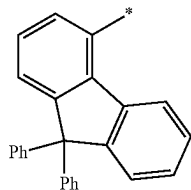


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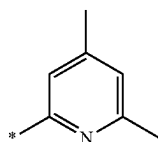


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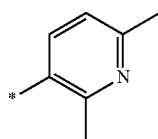
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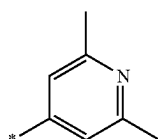
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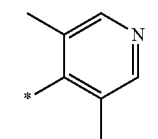
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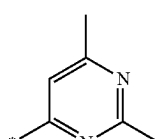
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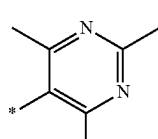
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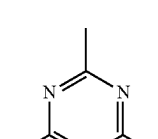
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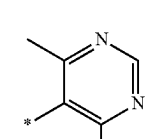
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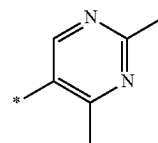


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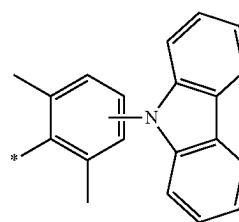


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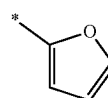
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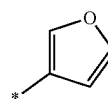
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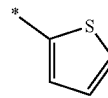
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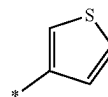
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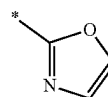
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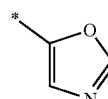
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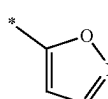
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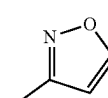
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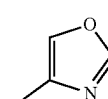
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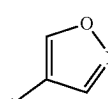
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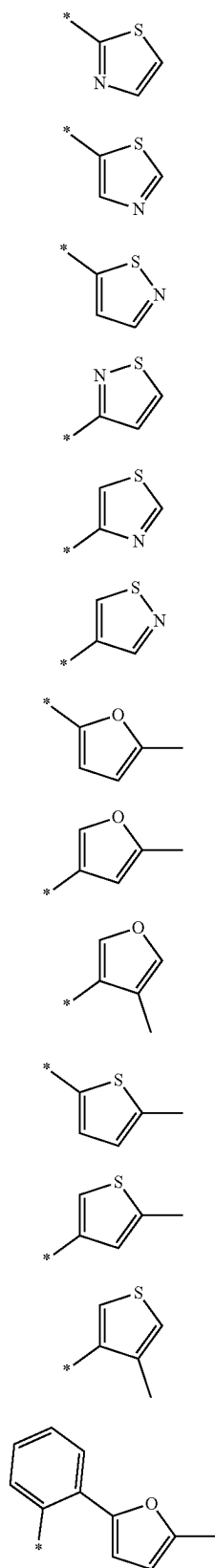


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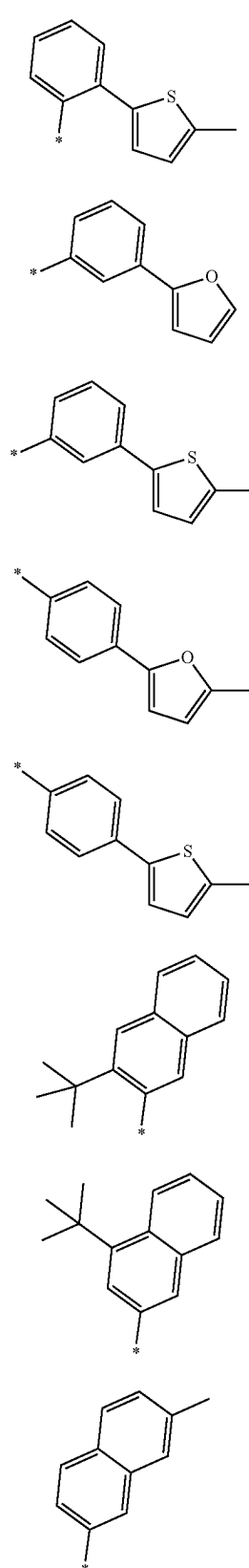
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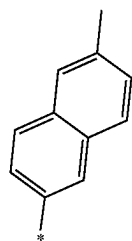
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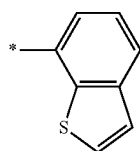
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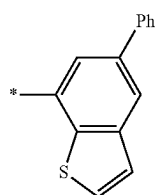
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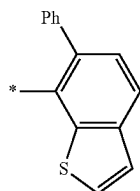
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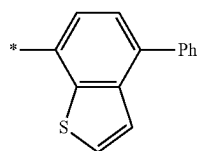
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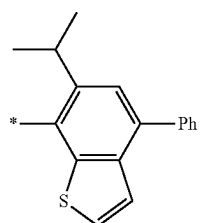
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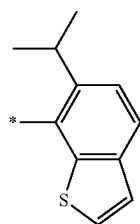
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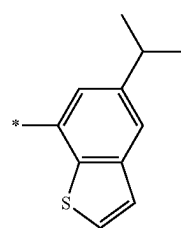


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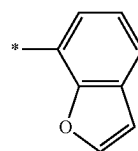


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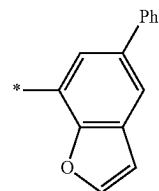
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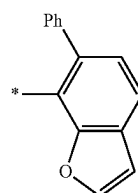
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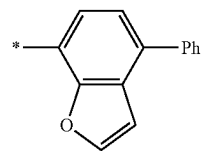
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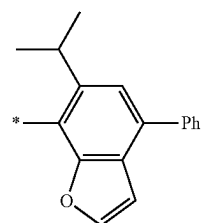
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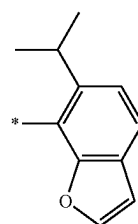
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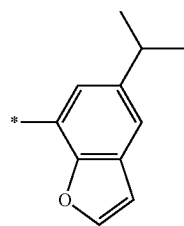


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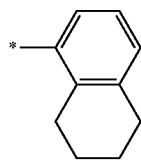


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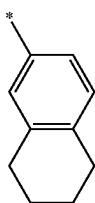
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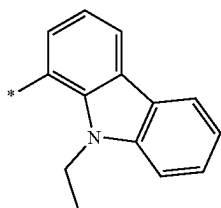
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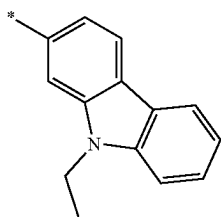
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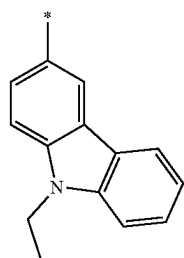
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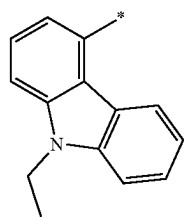
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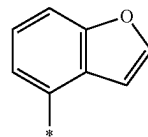


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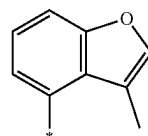


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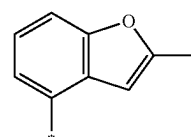
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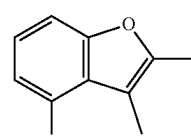
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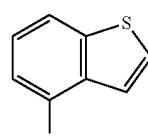
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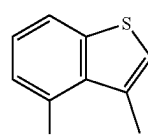
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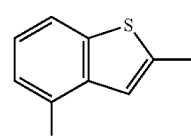
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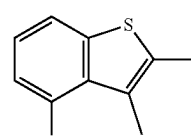
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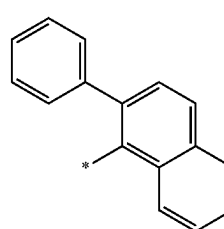
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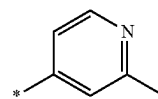
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10-342

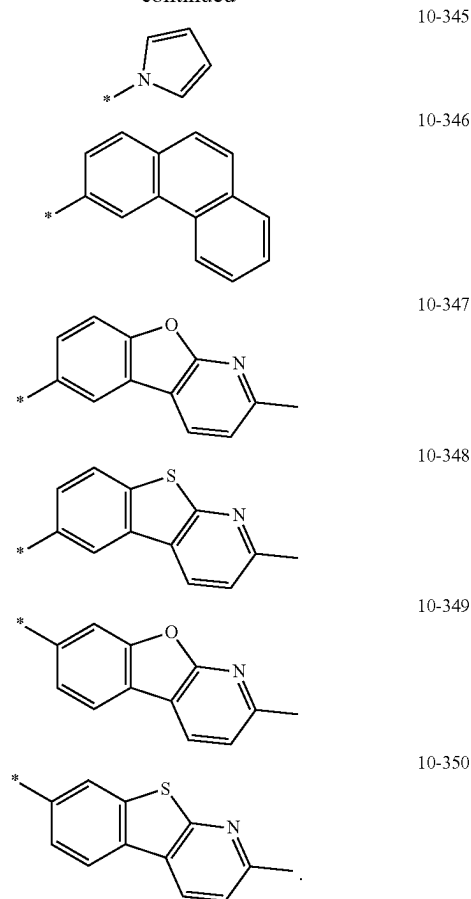


10-343



10-344

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[0098] In Formulae 9-1 to 9-39, 9-44 to 9-61, 9-201 to 9-240, 10-1 to 10-129, and 10-201 to 10-350, * indicates a binding site to a neighboring atom, “Ph” indicates a phenyl group, “TMS” indicates a trimethylsilyl group, and “TMG” indicates a trimethylgermyl group.

[0099] In Formula 1A, at least one of R_2 to R_7 is:

[0100] $-F$, $-Si(Q_1)(Q_2)(Q_3)$, or $-Ge(Q_1)(Q_2)(Q_3)$; or

[0101] a C_1 - C_{60} alkyl group, a C_2 - C_{60} alkenyl group, a C_2 - C_{60} alkynyl group, a C_3 - C_{10} cycloalkyl group, a C_1 - C_{10} heterocycloalkyl group, a C_6 - C_6 aryl group, or a C_1 - C_{60} heteroaryl group, each substituted with at least one of $-F$, $-Si(Q_{31})(Q_{32})(Q_{33})$, $-Ge(Q_{31})(Q_{32})(Q_{33})$, or a combination thereof.

[0102] In one or more embodiments, at least one of R_2 to R_7 may be:

[0103] $-F$, $-Si(Q_1)(Q_2)(Q_3)$, or $-Ge(Q_1)(Q_2)(Q_3)$; or

[0104] a C_1 - C_{20} alkyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thi-

azolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolynyl group, an isoquinolynyl group, a benzoquinolynyl group, a quinoxalynyl group, a quinazolinyl group, a cinnolynyl group, a carbazolyl group, a phenanthrolinyl group, a benzoimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, or an imidazopyrimidinyl group, each substituted with at least one of $-F$, $-Si(Q_{31})(Q_{32})(Q_{33})$, $-Ge(Q_{31})(Q_{32})(Q_{33})$, or a combination thereof, and

[0105] Q_1 to Q_3 and Q_{31} to Q_{33} may each independently be:

[0106] $-CH_3$, $-CD_3$, $-CD_2H$, $-CDH_2$, $-CH_2CH_3$, $-CH_2CD_3$, $-CH_2CD_2H$, $-CH_2CDH_2$, $-CHDCD_3$, $-CHDCD_2H$, $-CHDCDH_2$, $-CHDCD_3$, $-CD_2CD_3$, $-CD_2CD_2H$, or $-CD_2CDH_2$;

[0107] an n-propyl group, an iso-propyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, an n-pentyl group, an isopentyl group, a sec-pentyl group, a tert-pentyl group, a phenyl group, or a naphthyl group; or

[0108] an n-propyl group, an iso-propyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, an n-pentyl group, an isopentyl group, a sec-pentyl group, a tert-pentyl group, a phenyl group, or a naphthyl group, each substituted with at least one of deuterium, a C_1 - C_{10} alkyl group, a phenyl group, or a combination thereof.

[0109] In one or more embodiments, at least one of R_2 to R_7 may be:

[0110] $-F$, $-CF_3$, $-CF_2H$, $-CFH_2$, $-Si(Q_1)(Q_2)(Q_3)$, or $-Ge(Q_1)(Q_2)(Q_3)$;

[0111] a group represented by any one of Formulae 9-1 to 9-39 in which at least one hydrogen atom is substituted with $-F$, $-Si(Q_1)(Q_2)(Q_3)$, or $-Ge(Q_1)(Q_2)(Q_3)$; or

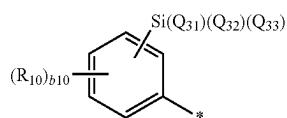
[0112] a group represented by one of Formulae 10-1 to 10-129 or 10-201 to 10-350 in which at least one hydrogen atom is substituted with $-F$, $-Si(Q_1)(Q_2)(Q_3)$, or $-Ge(Q_1)(Q_2)(Q_3)$.

[0113] In one or more embodiments, at least one of R_2 to R_7 may be:

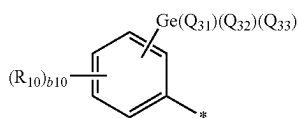
[0114] $-F$, $-CF_3$, $-CF_2H$, $-CFH_2$, $-Si(Q_1)(Q_2)(Q_3)$, or $-Ge(Q_1)(Q_2)(Q_3)$;

[0115] a C_1 - C_{20} alkyl group that is substituted with at least one $-F$; or

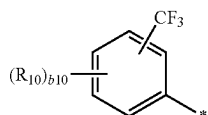
[0116] a group represented by one of Formulae 2A-1 to 2A-3:



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2A-2



2A-3

[0117] In Formulae 2A-1 to 2A-3,

[0118] R_{10} may be hydrogen, deuterium, $-F$, $-Cl$, $-Br$, $-I$, $-SF_5$, a hydroxyl group, a cyano group, a nitro group, an amino group, $-CD_3$, $-CD_2H$, $-CDH_2$, $-CF_3$, $-CF_2H$, $-CFH_2$, a C_1 - C_{10} alkyl group, a C_2 - C_{10} alkenyl group, a C_2 - C_{60} alkynyl group, a C_3 - C_{10} cycloalkyl group, a C_1 - C_{10} heterocycloalkyl group, a C_3 - C_{10} cycloalkenyl group, a C_1 - C_{10} heterocycloalkenyl group, a C_6 - C_{30} aryl group, a C_1 - C_{30} heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group,

[0119] $-Si(Q_1)(Q_2)(Q_3)$, $-Ge(Q_1)(Q_2)(Q_3)$, $-N(Q_4)(Q_5)$, $-B(Q_6)(Q_7)$, $-P(Q_8)(Q_9)$, or $-P(=O)(Q_8)(Q_9)$,

[0120] b_{10} may be an integer from 1 to 4, and

[0121] Q_1 to Q_9 and Q_{31} to Q_{33} may each independently be:

[0122] $-CH_3$, $-CD_3$, $-CD_2H$, $-CDH_2$, $-CH_2CH_3$, $-CH_2CD_3$, $-CH_2CD_2H$, $-CH_2CDH_2$, $-CHDCH_3$, $-CHDCD_2H$, $-CHDCH_2H$, $-CHDCD_3$, $-CD_2CD_3$, $-CD_2CD_2H$, or $-CD_2CDH_2$;

[0123] an n-propyl group, an iso-propyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, an n-pentyl group, an isopentyl group, a sec-pentyl group, a tert-pentyl group, a phenyl group, or a naphthyl group; or

[0124] an n-propyl group, an iso-propyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, an n-pentyl group, an isopentyl group, a sec-pentyl group, a tert-pentyl group, a phenyl group, or a naphthyl group, each substituted with at least one of deuterium, a C_1 - C_1 alkyl group, a phenyl group, or a combination thereof.

[0125] In one or more embodiments, R_{31} to R_{33} may each independently be a C_1 - C_{20} alkyl group, a C_3 - C_{10} cycloalkyl group, or a C_6 - C_{30} aryl group, each substituted with at least one of deuterium, $-F$, $-Cl$, $-Br$, $-I$, $-SF_5$, $-CD_3$, $-CD_2H$, $-CDH_2$, $-CF_3$, $-CF_2H$, $-CFH_2$, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1 - C_1 alkyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a pyridinyl group, a pyrimidinyl group, or a combination thereof.

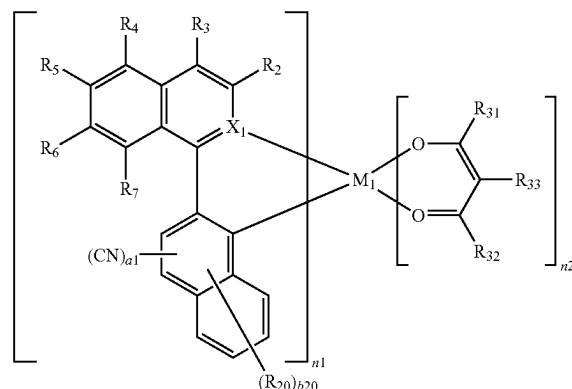
[0126] In one or more embodiments, R_{33} may be hydrogen or deuterium.

[0127] In Formula 1A, b_{20} is an integer from 1 to 5.

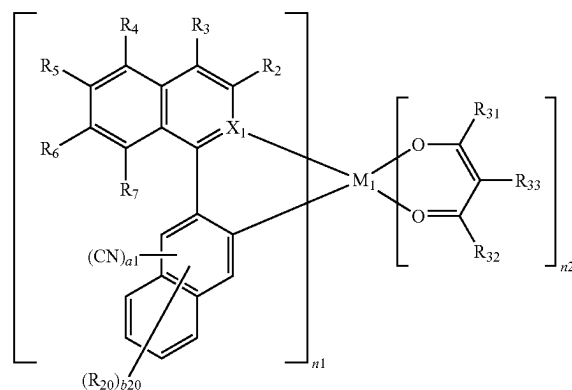
[0128] In Formulae 1A and 1B, * and *[†] each indicate a binding site to M_1 .

[0129] In one or more embodiments, the organometallic compound may be represented by one of Formulae 5-1 to 5-3.

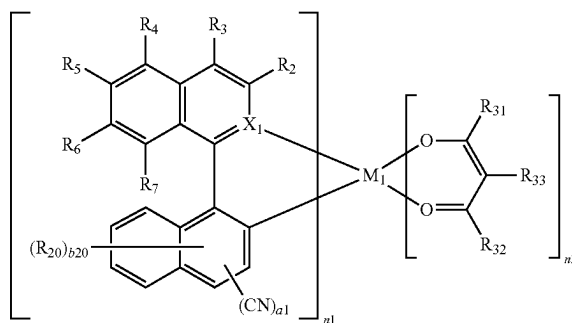
Formula 5-1



Formula 5-2



Formula 5-3



[0130] In Formulae 5-1 to 5-3,

[0131] M_1 , n_1 , n_2 , X_1 , R_2 to R_7 , a_1 , R_{20} , b_{20} , and R_{31} to R_{33} are each as described herein.

[0132] In one or more embodiments, in Formulae 5-1 to 5-3, at least one of R_2 to R_7 may be:

[0133] $-F$, $-CF_3$, $-CF_2H$, $-CFH_2$, $-Si(Q_1)(Q_2)(Q_3)$, or $-Ge(Q_1)(Q_2)(Q_3)$;

[0134] a C_1 - C_{20} alkyl group that is substituted with at least one $-F$; or

[0135] a group represented by one of Formulae 2A-1 to 2A-3.

[0136] In one or more embodiments,

[0137] R_5 may be $-F$, $-CF_3$, $-CF_2H$, $-CFH_2$, $-Si(Q_1)(Q_2)(Q_3)$, or $-Ge(Q_1)(Q_2)(Q_3)$;

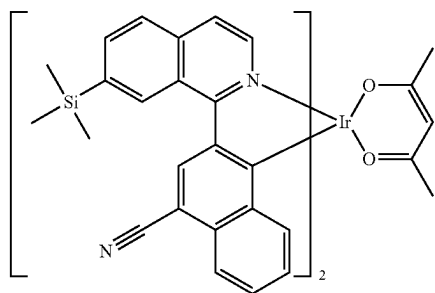
[0138] a_1 may be 1;

[0139] R_{31} and R_{33} may each independently be a C_1 - C_{20} alkyl group, a C_3 - C_{10} cycloalkyl group, or a C_6 - C_{60} aryl group, each substituted with at least one of deuterium, $-F$, $-Cl$, $-Br$, $-I$, $-SF_5$, $-CD_3$, $-CD_2H$, $-CDH_2$, $-CF_3$, $-CF_2H$, $-CFH_2$, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1 - C_{10} alkyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a pyridinyl group, a pyrimidinyl group, or a combination thereof; and

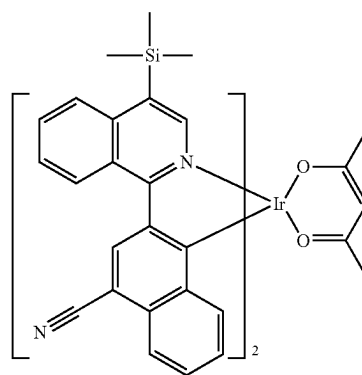
[0140] R_{32} may be hydrogen or deuterium.

[0141] In one or more embodiments, the organometallic compound represented by Formula 1 may be at least one of Compounds 1 to 72:

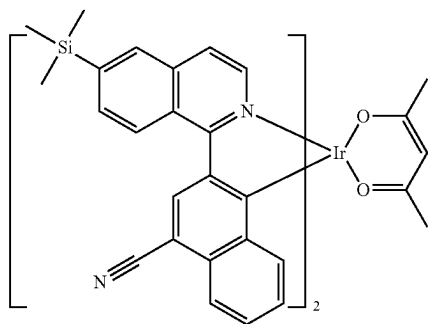
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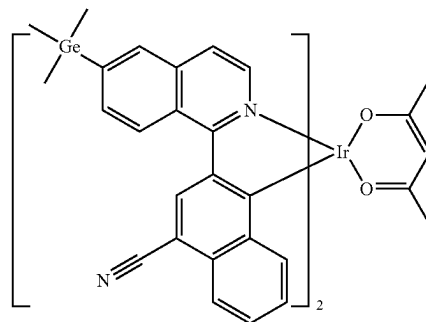
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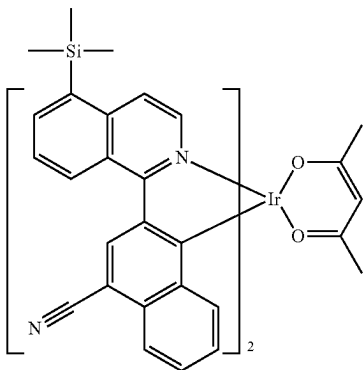
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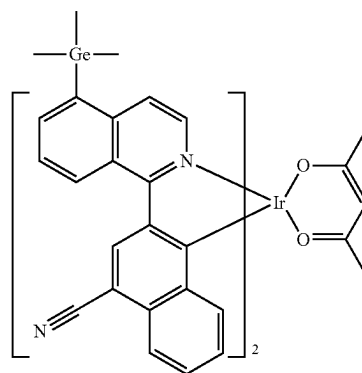
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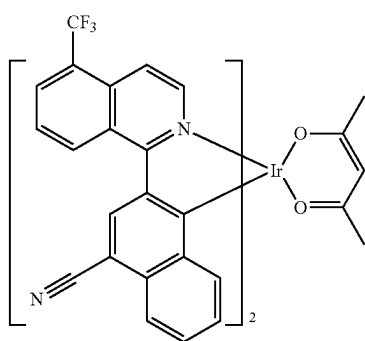
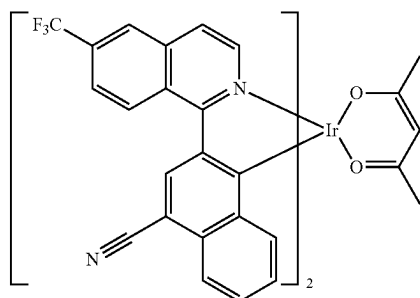
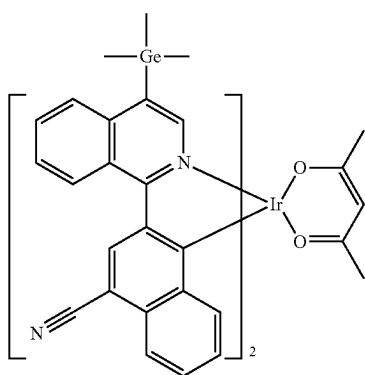
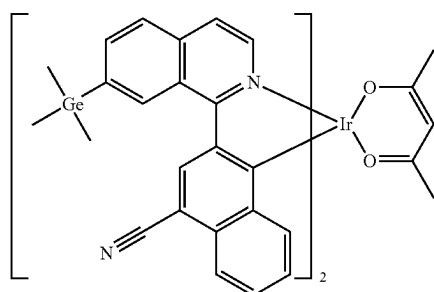


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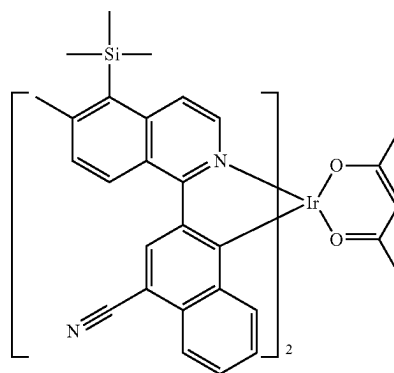
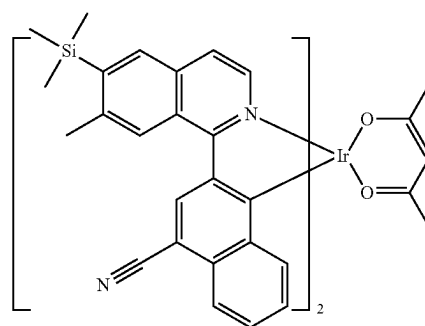
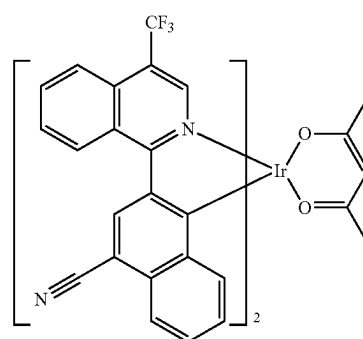
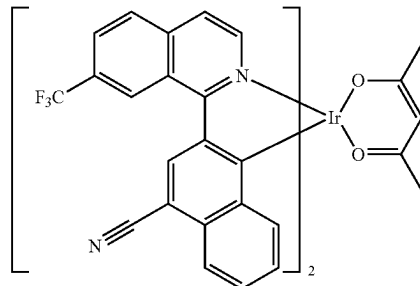


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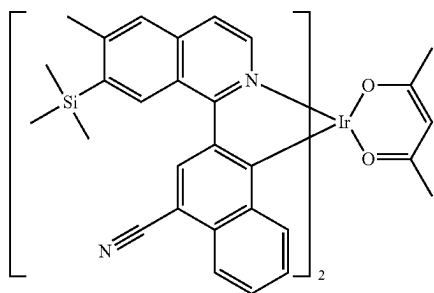
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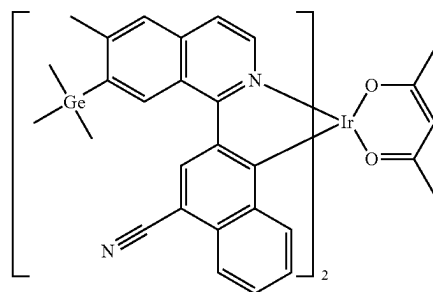


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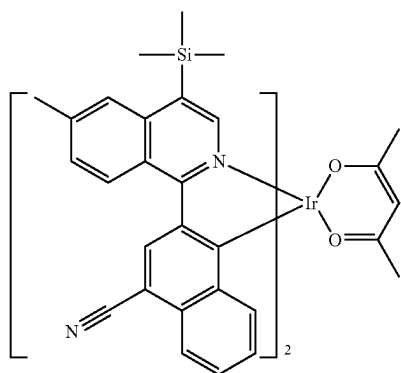
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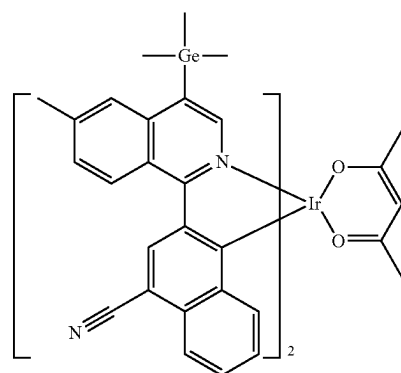


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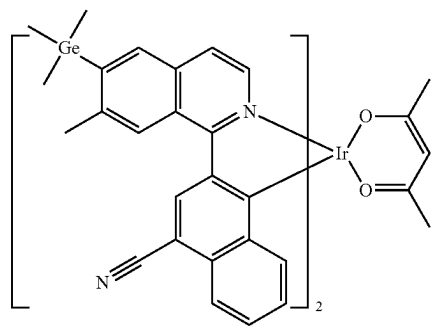
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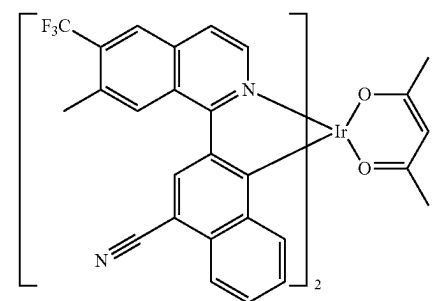
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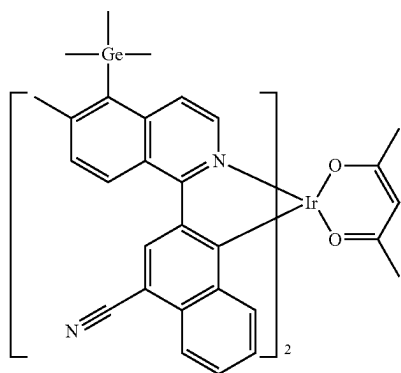
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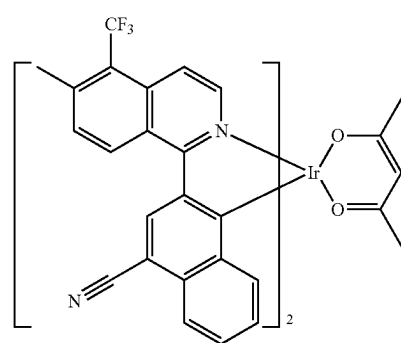
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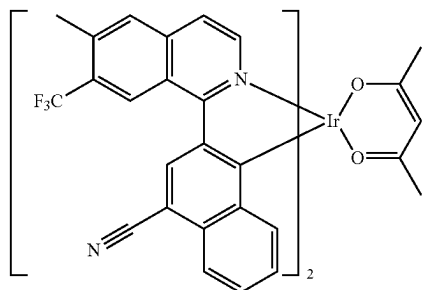
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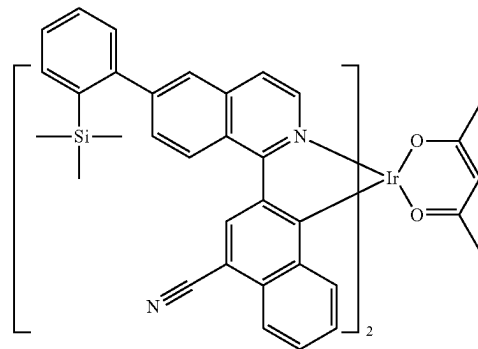


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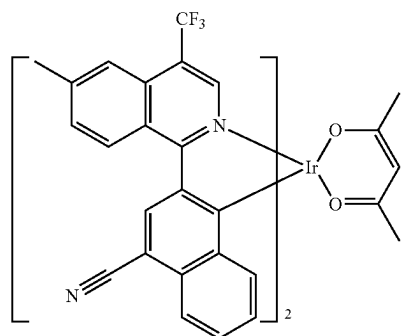
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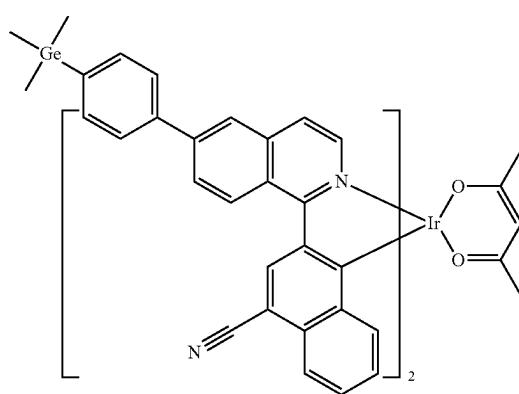


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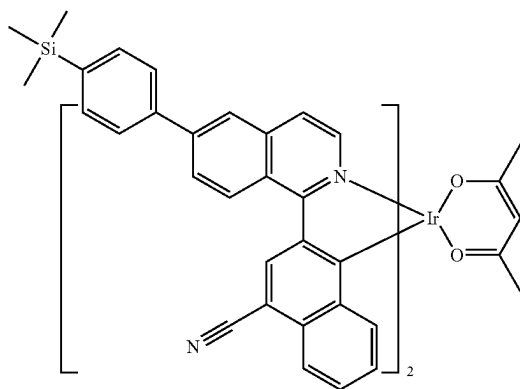
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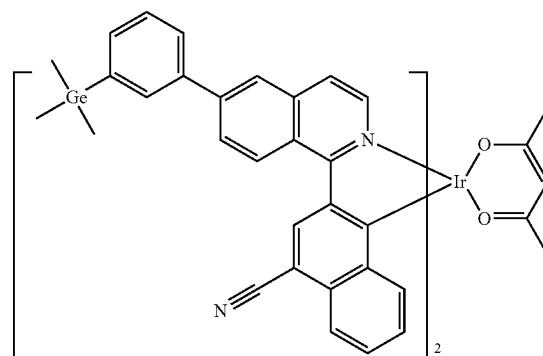
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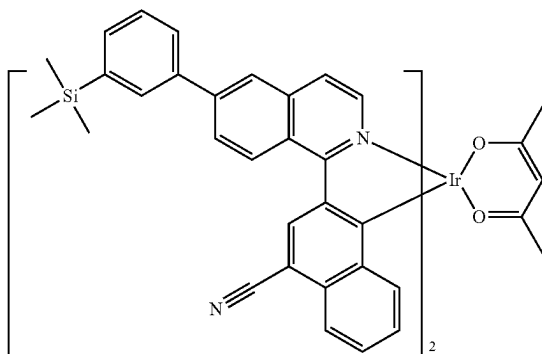
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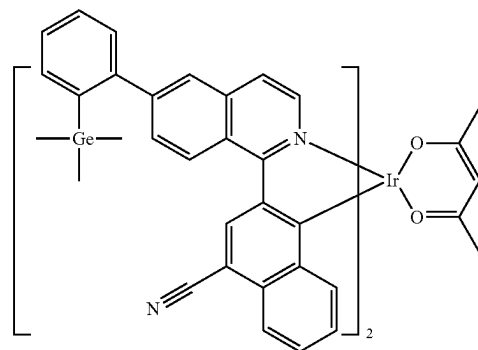
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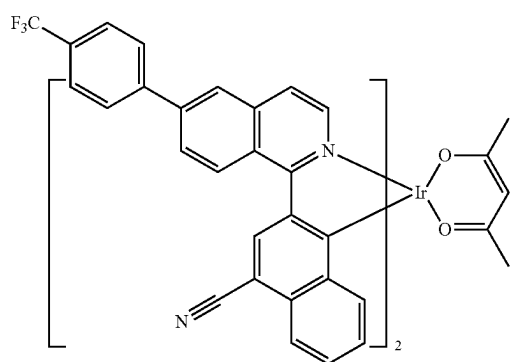
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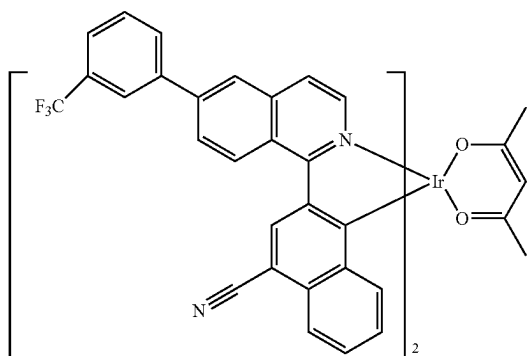
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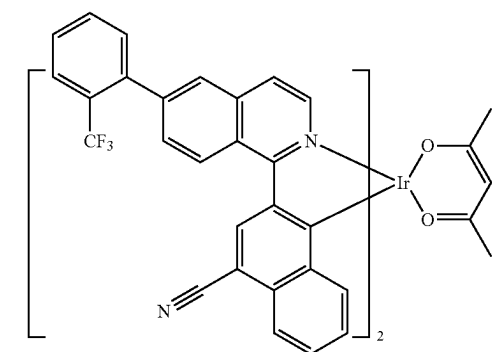
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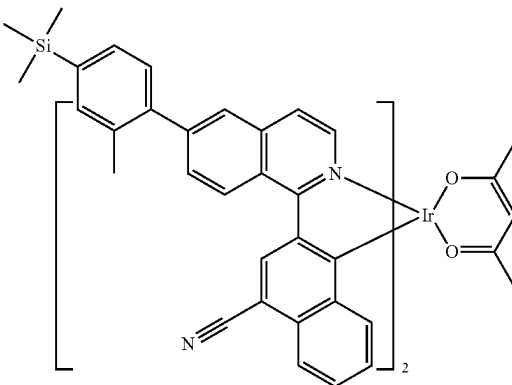
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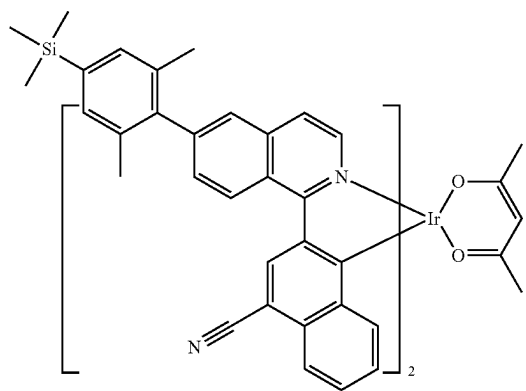
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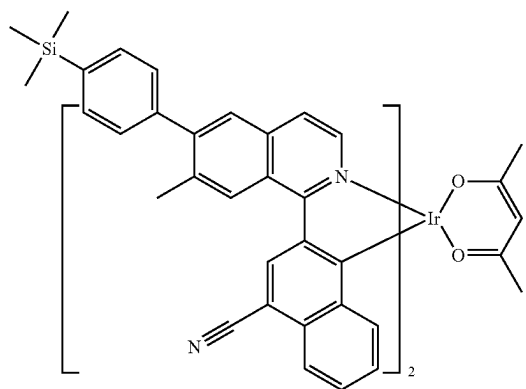
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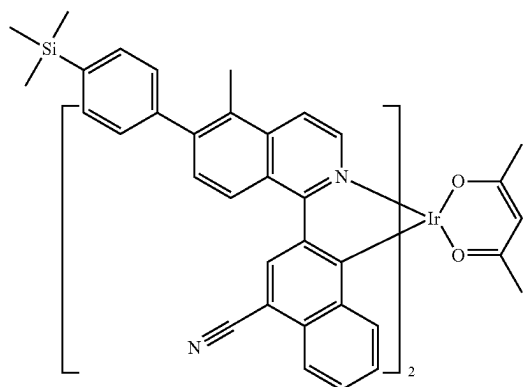
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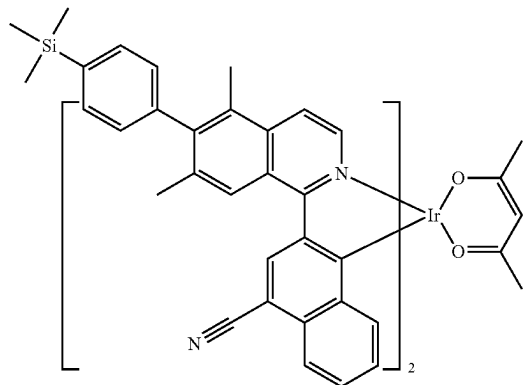
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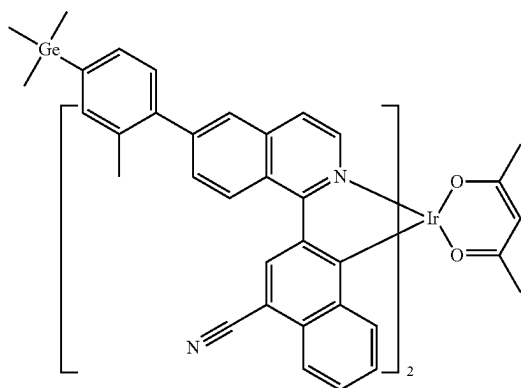
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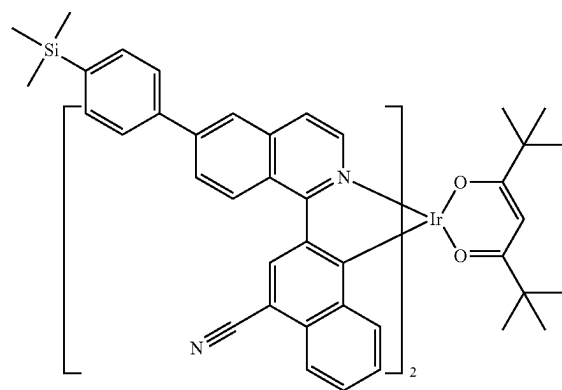
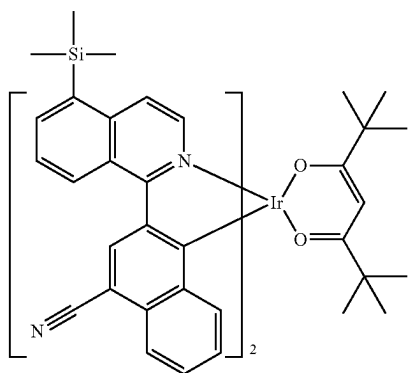
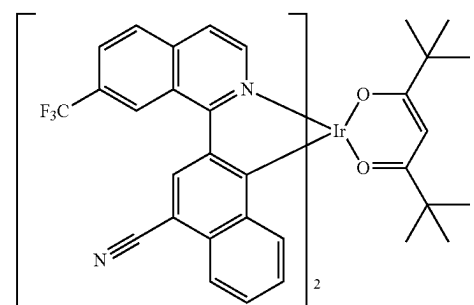
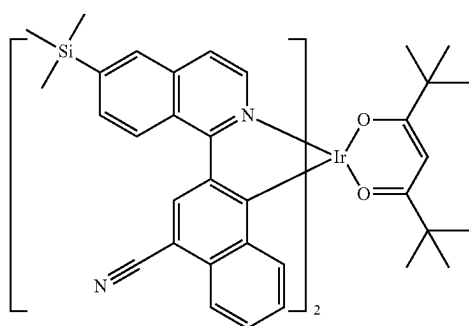
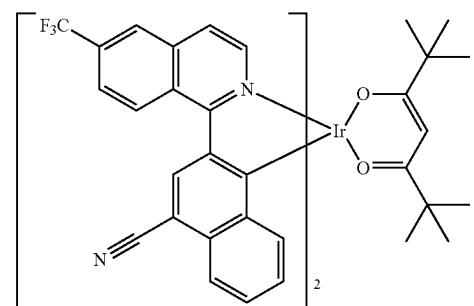
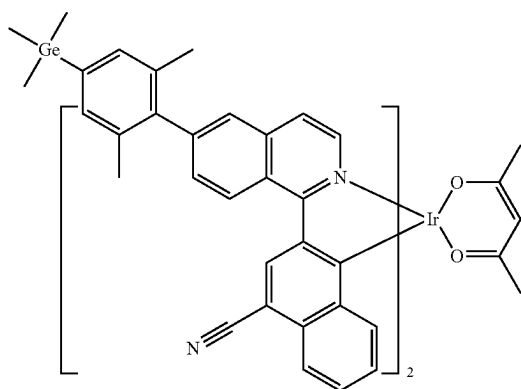
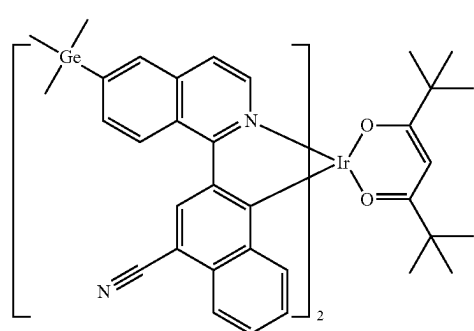
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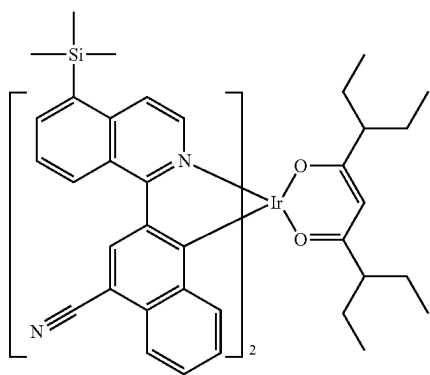
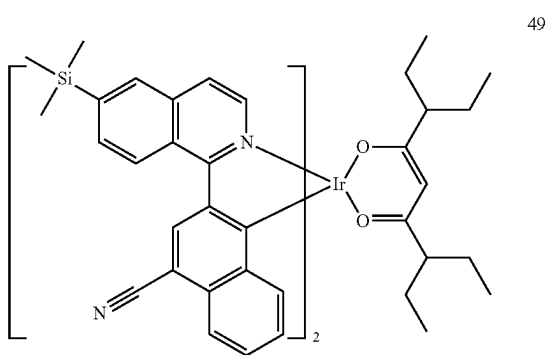
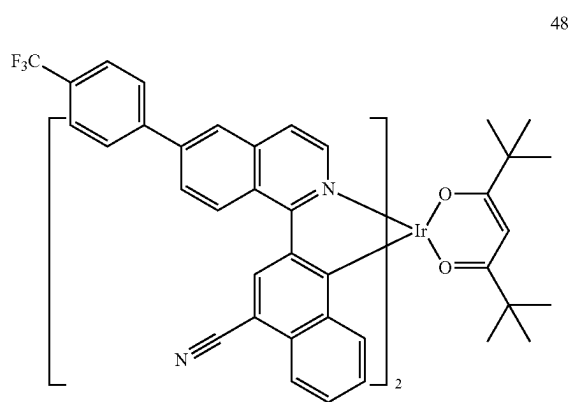
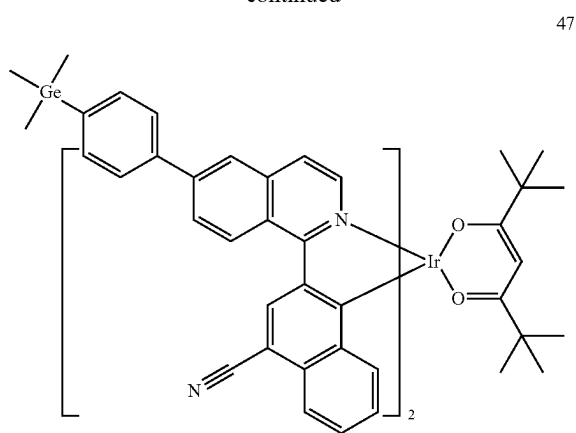
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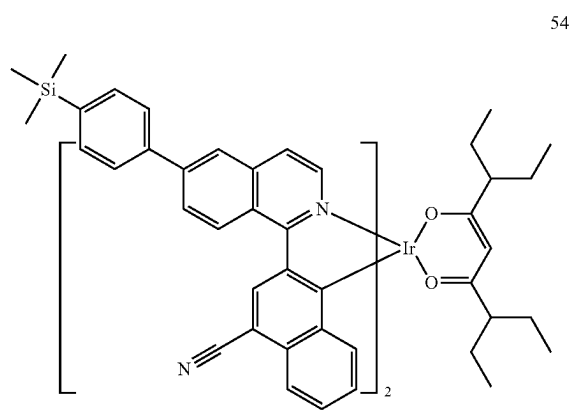
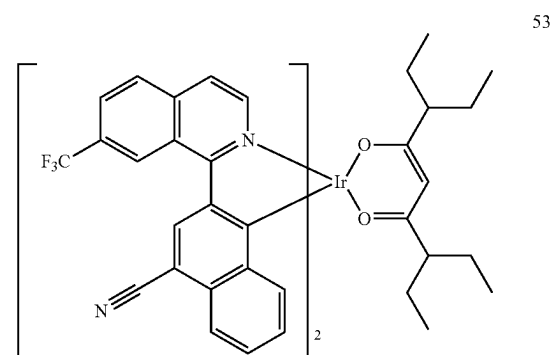
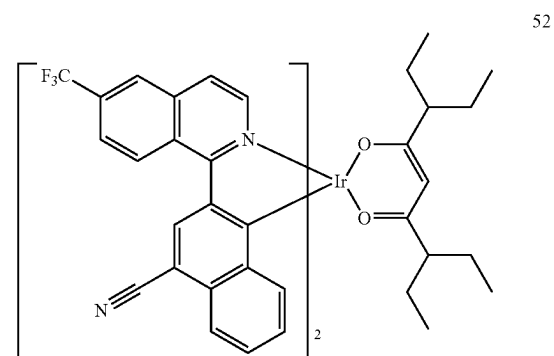
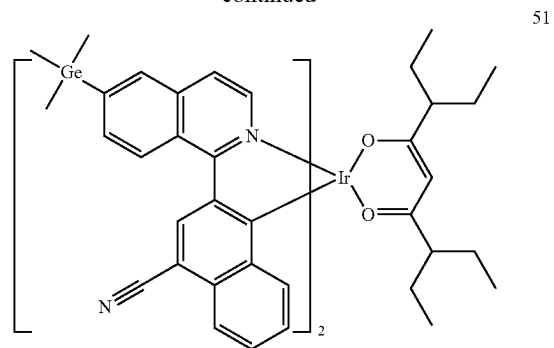
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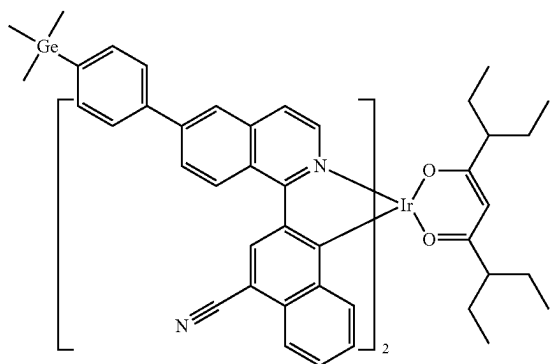


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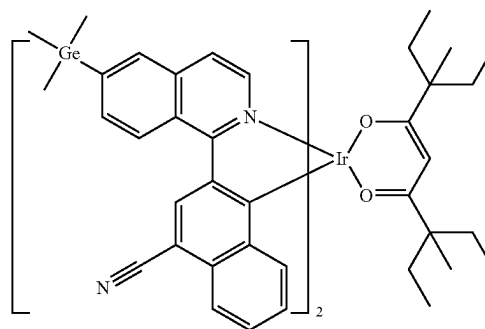
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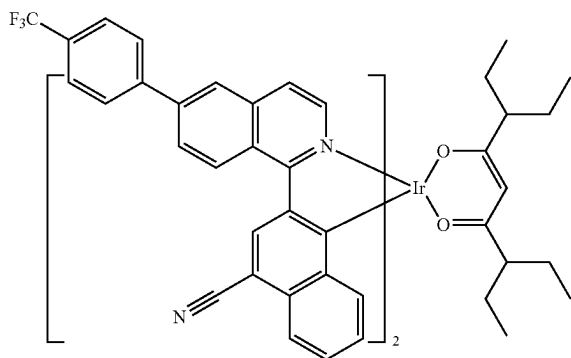


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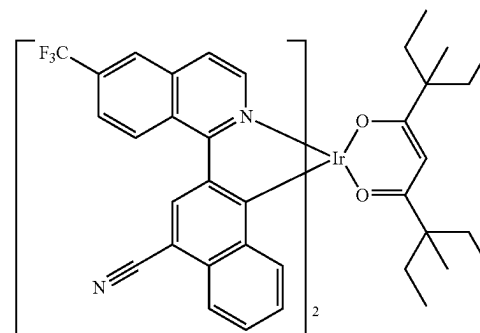
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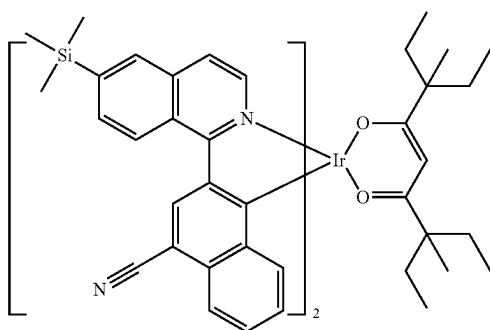
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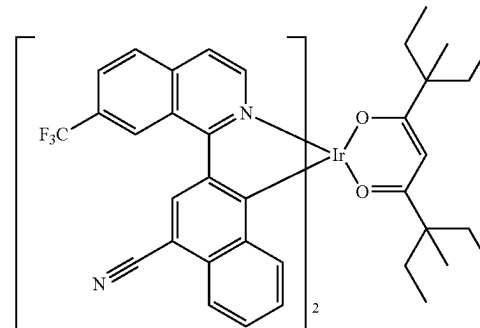
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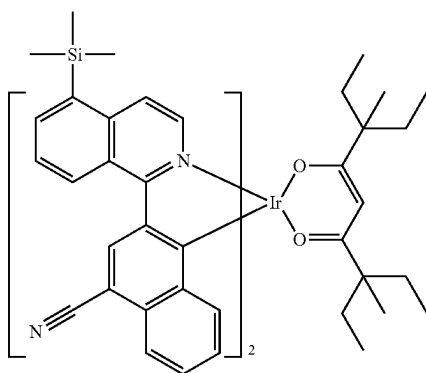
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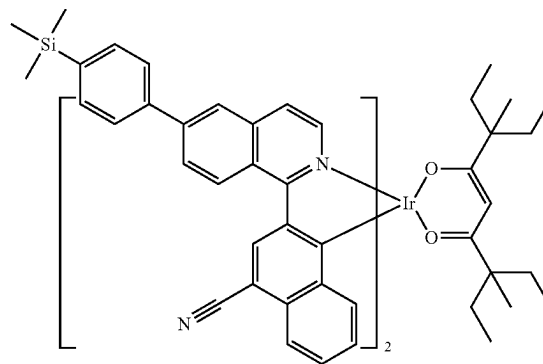
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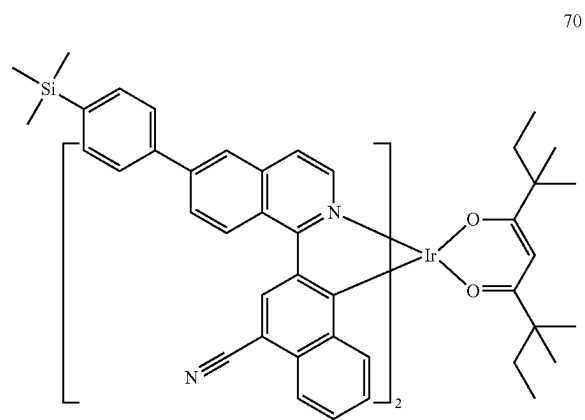
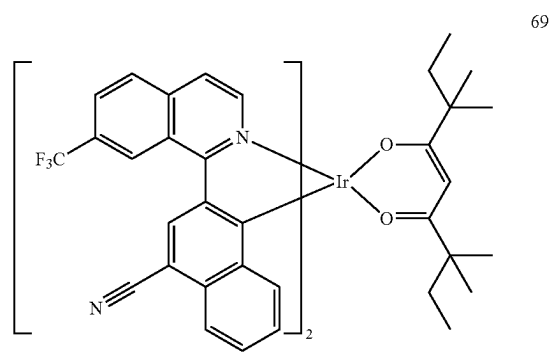
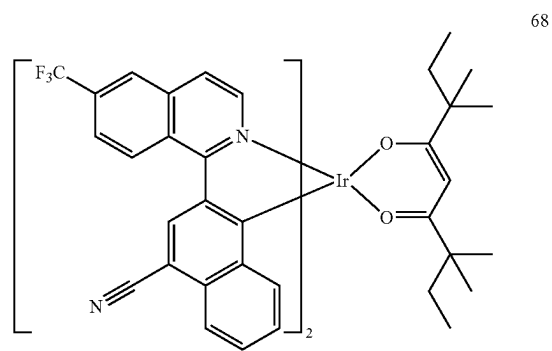
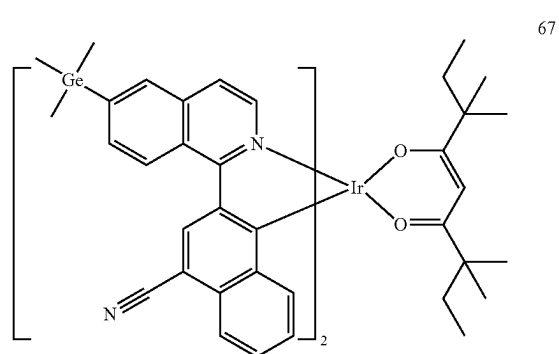
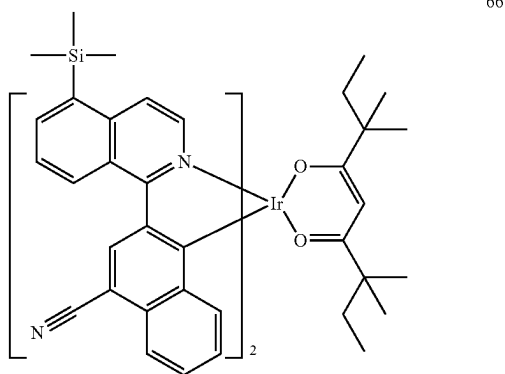
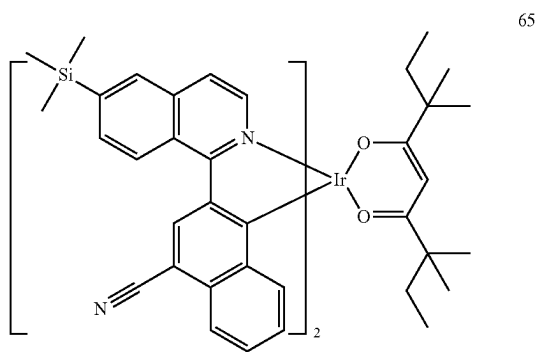
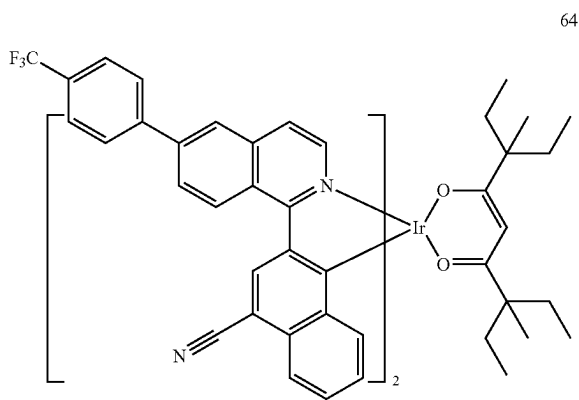
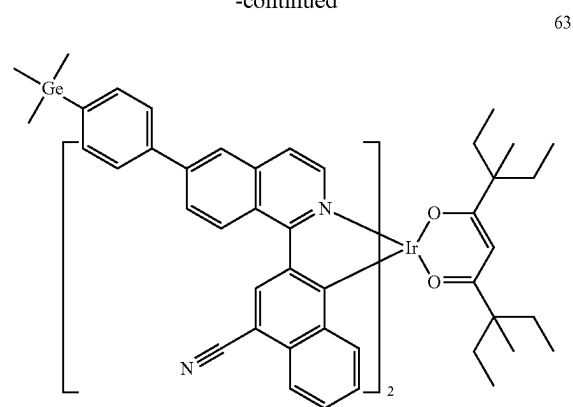
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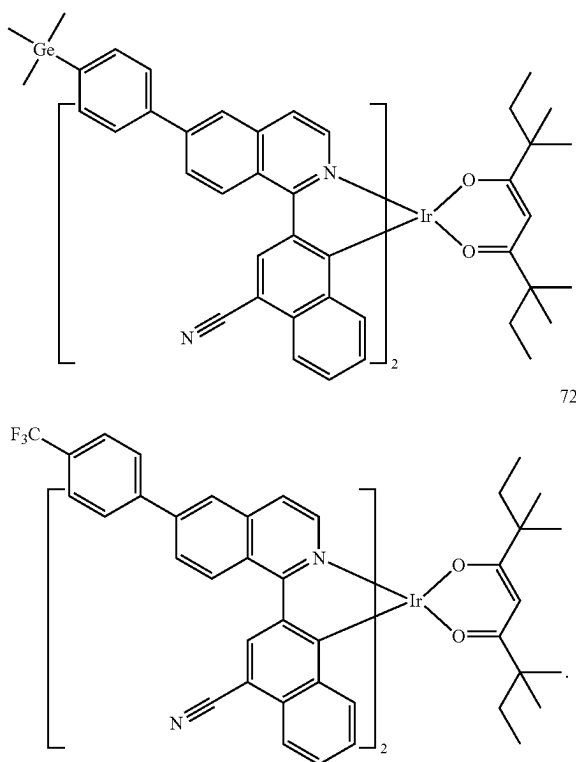
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[0142] In one or more embodiments, the organometallic compound represented by Formula 1 may be electrically neutral.

[0143] The organometallic compound represented by Formula 1 includes ligands having structures represented by Formula 1A and Formula 1B, wherein, in Formula 1A, a ligand including X₁ includes —F, —Si(Q₁)(Q₂)(Q₃), or —Ge(Q₁)(Q₂)(Q₃) as a substituent, and a naphthyl ligand includes at least one cyano group as a substituent. Due to this structure, the organometallic compound represented by Formula 1 has excellent luminescence characteristics, and thus may have characteristics suitable for use as a luminescent material with high color purity by controlling the emission wavelength range and full width at half maximum (FWHM).

[0144] Therefore, an electronic device, e.g., an organic light-emitting device, including at least one of the organometallic compounds represented by Formula 1 may have a low driving voltage, high maximum external quantum efficiency, and/or a low roll-off ratio.

[0145] In one or more embodiments, the maximum emission wavelength (emission peak wavelength, λ_{max}) of the emission peak in an emission spectrum or an electroluminescence (EL) spectrum of the organometallic compound represented by Formula 1 may be about 590 nanometers (nm) to about 650 nm, about 600 nm to about 640 nm, or about 610 nm to about 630 nm.

[0146] Synthesis methods of the organometallic compound represented by Formula 1 may be recognizable by one of ordinary skill in the art and by referring to Synthesis Examples provided below.

[0147] The organometallic compound represented by Formula 1 is suitable for use in an organic layer of an organic

light-emitting device, for example, for use as a dopant in an emission layer of the organic layer. Thus, another aspect provides an organic light-emitting device that includes a first electrode; a second electrode; an organic layer arranged between the first electrode and the second electrode, wherein the organic layer includes an emission layer; and wherein the organic layer further includes at least one of the organometallic compounds represented by Formula 1.

[0148] The organic light-emitting device includes the organic layer including at least one of the organometallic compounds represented by Formula 1, and thus, the organic light-emitting device may have excellent driving voltage, excellent maximum external quantum efficiency, and/or excellent lifespan characteristics.

[0149] The organometallic compound represented by Formula 1 may be used between a pair of electrodes of the organic light-emitting device. For example, at least one of the organometallic compounds represented by Formula 1 may be included in the emission layer.

[0150] At this time, the organometallic compound(s) represented by Formula 1 may act as a dopant, and the emission layer may further include a host. When the emission layer further includes a host, the amount of the host in the emission layer may be greater than the amount of the at least one organometallic compound represented by Formula 1 in the emission layer, on a weight basis.

[0151] In one or more embodiments, the emission layer may emit a red light. For example, the emission layer may emit an orange light or red light having a maximum emission wavelength of about 590 nm to about 650 nm, about 600 nm to about 640 nm, or about 610 nm to about 630 nm.

[0152] The expression “(organic layer) includes at least one of the organometallic compound represented by Formula 1” as used herein may mean that the (organic layer) may include one kind of organometallic compound represented by Formula 1 or two or more different kinds of organometallic compounds, each represented by Formula 1.

[0153] For example, the organic layer may include, as the at least one organometallic compound represented by Formula 1, only Compound 1. In this embodiment, Compound 1 may be included in the emission layer of the organic light-emitting device. In one or more embodiments, the organic layer may include, as the at least one organometallic compound represented by Formula 1, Compound 1 and Compound 2. In this embodiment, Compound 1 and Compound 2 may exist in an identical layer (for example, Compound 1 and Compound 2 all may exist in an emission layer).

[0154] The first electrode may be an anode, which is a hole injection electrode, and the second electrode may be a cathode, which is an electron injection electrode; or the first electrode may be a cathode, which is an electron injection electrode, and the second electrode may be an anode, which is a hole injection electrode.

[0155] In one or more embodiments, in the organic light-emitting device, the first electrode may be an anode, and the second electrode may be a cathode, the organic layer may further include a hole transport region located between the first electrode and the emission layer, and an electron transport region located between the emission layer and the second electrode, the hole transport region may include a hole injection layer, a hole transport layer, an electron blocking layer, a buffer layer, or a combination thereof, and

the electron transport region may include a hole blocking layer, an electron transport layer, an electron injection layer, or a combination thereof.

[0156] The term “organic layer” as used herein refers to a single layer and/or a plurality of layers that are located between the first electrode and the second electrode of the organic light-emitting device. The “organic layer” may include, in addition to an organic compound, an organometallic complex including metal.

[0157] The FIGURE is a schematic cross-sectional view of an organic light-emitting device **10** according to one or more embodiments. Hereinafter, the structure of an organic light-emitting device according to one or more embodiments and a method of manufacturing an organic light-emitting device according to one or more embodiments will be described in connection with FIGURE. The organic light-emitting device **10** includes a first electrode **11**, an organic layer **15**, and a second electrode **19**, which are sequentially stacked in the stated order.

[0158] A substrate may be additionally disposed under the first electrode **11** or on the second electrode **19**. For use as the substrate, any substrate that is used in organic light-emitting devices available in the art may be used, and the substrate may be a glass substrate or a transparent plastic substrate, each having excellent mechanical strength, thermal stability, transparency, surface smoothness, ease of handling, and/or water resistance.

[0159] The first electrode **11** may be formed by, for example, depositing or sputtering, onto the substrate, a material for forming the first electrode **11**. The first electrode **11** may be an anode. The material for forming the first electrode **11** may be selected from materials with a high work function for easy hole injection. The first electrode **11** may be a reflective electrode, a semi-transmissive electrode, or a transmissive electrode. The material for forming the first electrode may be indium tin oxide (ITO), indium zinc oxide (IZO), tin oxide (SnO₂), or zinc oxide (ZnO). In one or more embodiments, the material for forming the first electrode may be metal, such as magnesium (Mg), aluminum (Al), silver (Ag), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), or magnesium-silver (Mg—Ag).

[0160] The first electrode **11** may have a single-layer structure or a multilayer structure including two or more layers. For example, the first electrode **11** may have a three-layered structure of ITO/Ag/ITO, but the structure of the first electrode **11** is not limited thereto.

[0161] The organic layer **15** is located on the first electrode **11**.

[0162] The organic layer **15** may include a hole transport region, an emission layer, and an electron transport region.

[0163] The hole transport region may be arranged between the first electrode **11** and the emission layer.

[0164] The hole transport region may include a hole injection layer, a hole transport layer, an electron blocking layer, a buffer layer, or a combination thereof.

[0165] The hole transport region may include only either a hole injection layer or a hole transport layer. In one or more embodiments, the hole transport region may have a hole injection layer/hole transport layer structure, or a hole injection layer/hole transport layer/electron blocking layer structure, wherein, for each structure, respective layers are sequentially stacked in this stated order from the first electrode **11**.

[0166] When the hole transport region includes a hole injection layer, the hole injection layer (HIL) may be formed on the first electrode **11** by using one or more suitable methods, for example, vacuum deposition, spin coating,

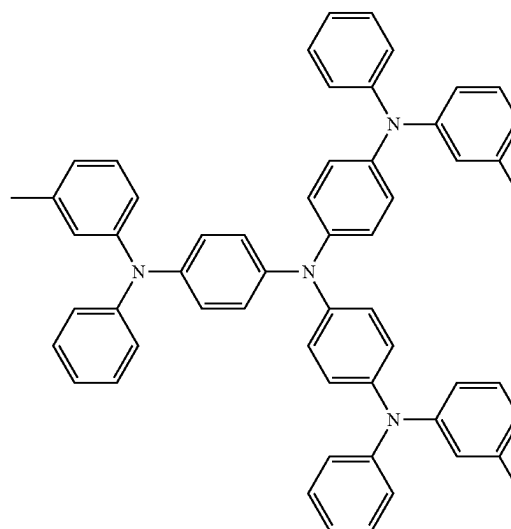
casting, and/or Langmuir-Blodgett (LB) deposition, but embodiments are not limited thereto.

[0167] When a hole injection layer is formed by vacuum deposition, the deposition conditions may vary according to a material that is used to form the hole injection layer, and the structure and thermal characteristics of the hole injection layer. For example, the deposition conditions may include a deposition temperature of about 100° C. to about 500° C., a vacuum pressure of about 10⁻⁸ torr to about 10⁻³ torr, and a deposition rate of about 0.01 angstroms per second (Å/sec) to about 100 Å/sec. However, the deposition conditions are not limited thereto.

[0168] When the hole injection layer is formed using spin coating, coating conditions may vary according to the material used to form the hole injection layer, and the structure and thermal properties of the hole injection layer. For example, a coating speed may be from about 2,000 revolutions per minute (rpm) to about 5,000 rpm, and a temperature at which a heat treatment is performed to remove a solvent after coating may be from about 80° C. to about 200° C. However, the coating conditions are not limited thereto.

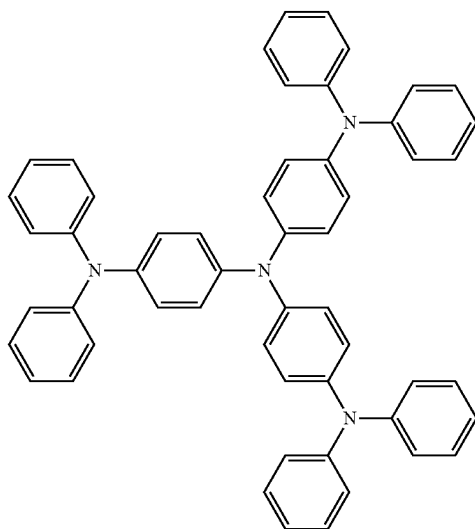
[0169] The conditions for forming a hole transport layer and an electron blocking layer may be understood by referring to conditions for forming the hole injection layer.

[0170] The hole transport region may include, for example, at least one of 4,4',4''-tris(3-methylphenylphenylamino)triphenylamine (m-MTDATA), 4,4',4''-tris(N,N-diphenylamino)triphenylamine (TDATA), 4,4',4''-tris[N-(2-naphthyl)-N-phenylamino]-triphenylamine (2-TNATA), N,N'-di(1-naphthyl)-N,N'-diphenylbenzidine (NPB), β-NPB, N,N'-bis(3-methylphenyl)-N,N'-diphenyl-1,1'-biphenyl-4,4'-diamine (TPD), Spiro-TPD, Spiro-NPB, methylated NPB, 4,4'-cyclohexylidene bis[N,N-bis(4-methylphenyl)benzenamine](TAPC), 4,4'-bis[N,N'-(3-tolyl)amino]-3,3'-dimethylbiphenyl (HMTDP), 4,4',4''-tris(N-carbazolyl)triphenylamine (TCTA), polyaniline/dodecylbenzenesulfonic acid (PANI/DBSA), poly(3,4-ethylenedioxythiophene)/poly(4-styrenesulfonate) (PEDOT/PSS), polyaniline/camphor sulfonic acid (PANI/CSA), polyaniline/poly(4-styrenesulfonate) (PANI/PSS), a compound represented by Formula 201, or a compound represented by Formula 202, but embodiments are not limited thereto:



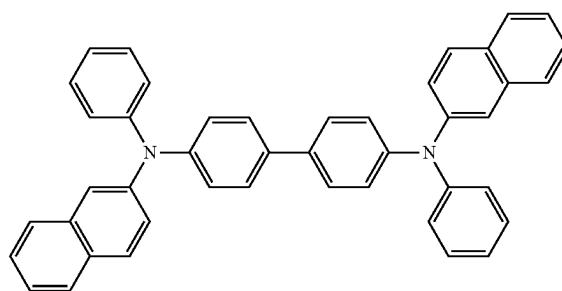
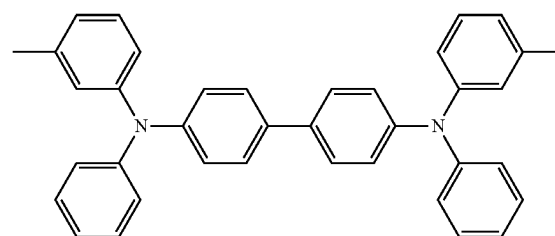
m-MTDATA

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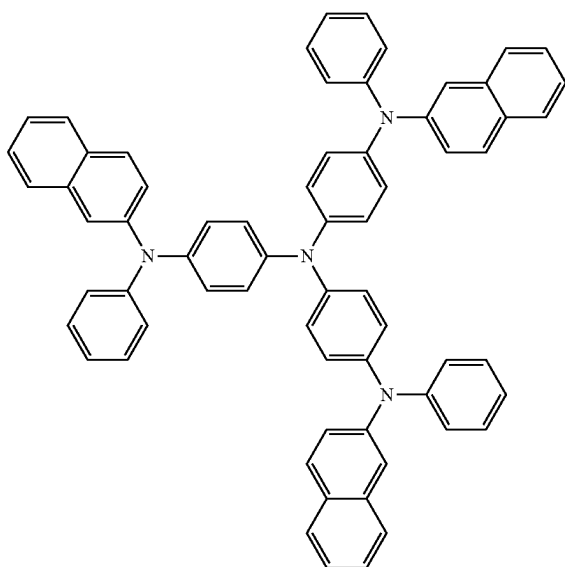


TDATA

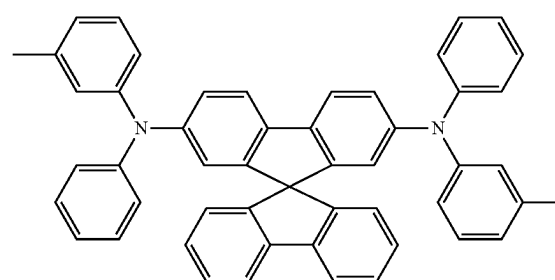
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 β -NPB

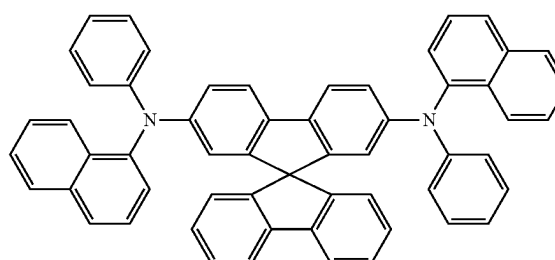
TPD



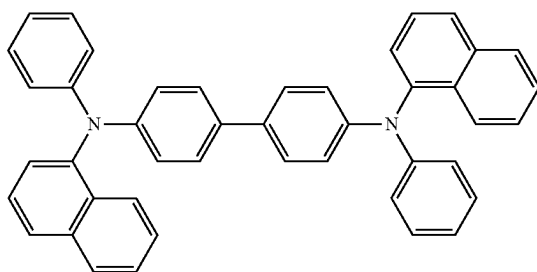
2-TNATA



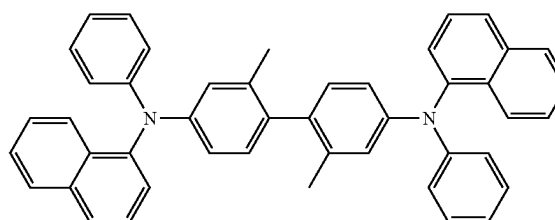
Spiro-TPD



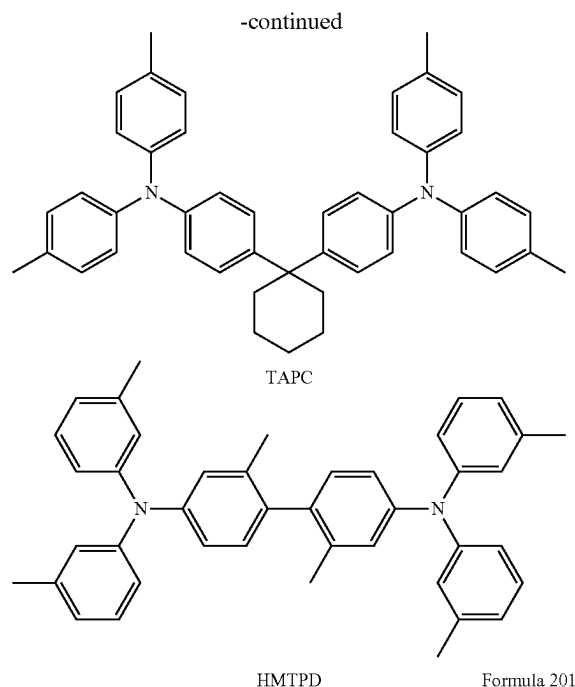
Spiro-NPB



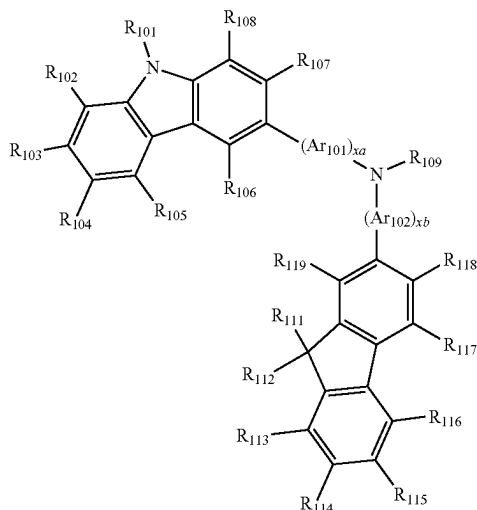
NPB



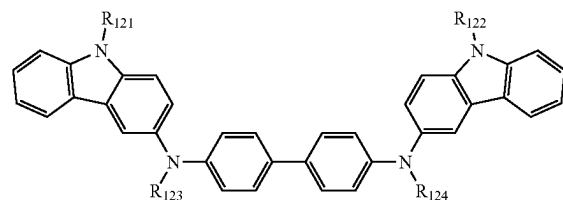
methylated NPB



Formula 201



Formula 202



wherein, in Formula 201, Ar₁₀₁ and Ar₁₀₂ may each independently be:

[0171] a phenylene group, a pentalenylene group, an indenylene group, a naphthylene group, an azulenylenylene group, a heptalenylene group, an acenaphthylenylene group, a fluorenylenylene group, a phenalenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylenylene group, a naph-

thacenylenylene group, a picenylene group, a perylenylene group, or a pentacenylene group; or

[0172] a phenylene group, a pentalenylene group, an indenylene group, a naphthylene group, an azulenylenylene group, a heptalenylene group, an acenaphthylenylene group, a fluorenylenylene group, a phenalenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylenylene group, a naphthacenylenylene group, a picenylene group, a perylenylene group, or a pentacenylene group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF₅, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, a C₁-C₆₀ alkylthio group, a C₃-C₁₀ cycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₇-C₆₀ alkyl aryl group, a C₇-C₆₀ aryl alkyl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a C₂-C₆₀ alkyl heteroaryl group, a C₂-C₆₀ heteroaryl alkyl group, a C₁-C₆₀ heteroaryloxy group, a C₁-C₆₀ heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, or a combination thereof.

[0173] xa and xb in Formula 201 may each independently be an integer from 0 to 5, or 0, 1, or 2. For example, xa may be 1 and xb may be 0, but embodiments are not limited thereto.

[0174] R₁₀₁ to R₁₀₈, R₁₁₁ to R₁₁₉, and R₁₂₁ to R₁₂₄ in Formulae 201 and 202 may each independently be:

[0175] hydrogen, deuterium, —F, —Cl, —Br, —I, —SF₅, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C₁-C₁₀ alkyl group (e.g., a methyl group, an ethyl group, a propyl group, a butyl group, pentyl group, a hexyl group, or the like), a C₁-C₁₀ alkoxy group (e.g., a methoxy group, an ethoxy group, a propoxy group, a butoxy group, a pentoxy group, or the like), or a C₁-C₁₀ alkylthio group;

[0176] a C₁-C₁₀ alkyl group, a C₁-C₁₀ alkoxy group, a C₁-C₁₀ alkylthio group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF₅, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, or a combination thereof;

[0177] a phenyl group, a naphthyl group, an anthracenyl group, a fluorenyl group, or a pyrenyl group; or

[0178] a phenyl group, a naphthyl group, an anthracenyl group, a fluorenyl group, or a pyrenyl group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF₅, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C₁-C₁₀ alkyl group, a C₁-C₁₀ alkoxy group, a C₁-C₁₀

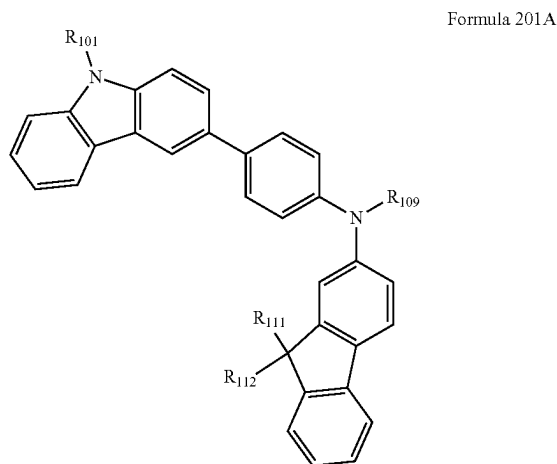
alkylthio group, or a combination thereof, but embodiments are not limited thereto.

[0179] R_{109} in Formula 201 may be:

[0180] a phenyl group, a naphthyl group, an anthracenyl group, or a pyridinyl group; or

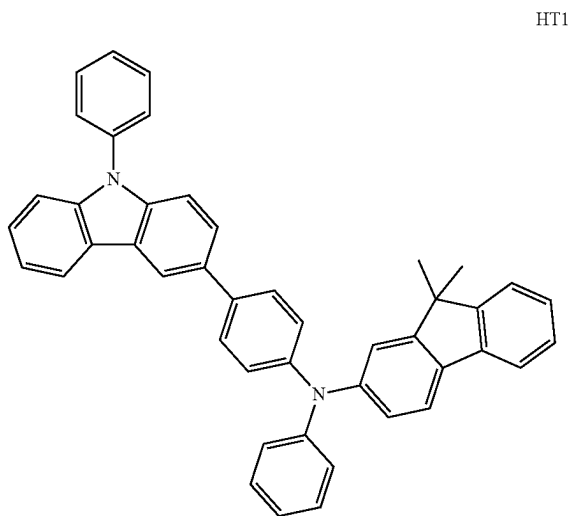
[0181] a phenyl group, a naphthyl group, an anthracenyl group, or a pyridinyl group, each substituted with at least one of deuterium, $-F$, $-Cl$, $-Br$, $-I$, $-SF_5$, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkoxy group, a C_1 - C_{20} alkylthio group, a phenyl group, a naphthyl group, an anthracenyl group, a pyridinyl group, or a combination thereof.

[0182] In one or more embodiments, the compound represented by Formula 201 may be represented by Formula 201A, but embodiments are not limited thereto:

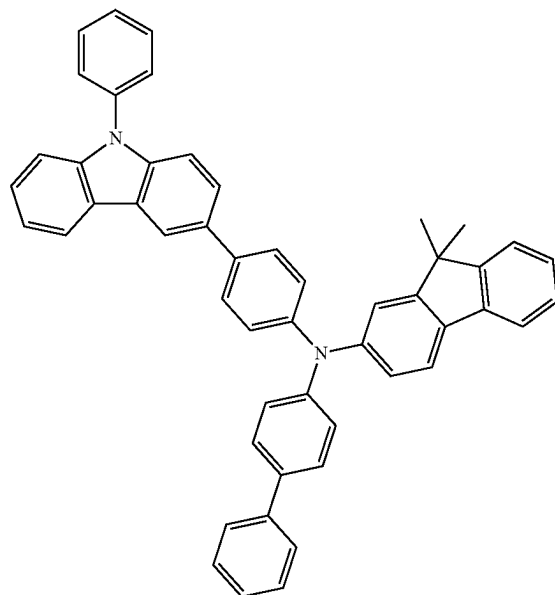
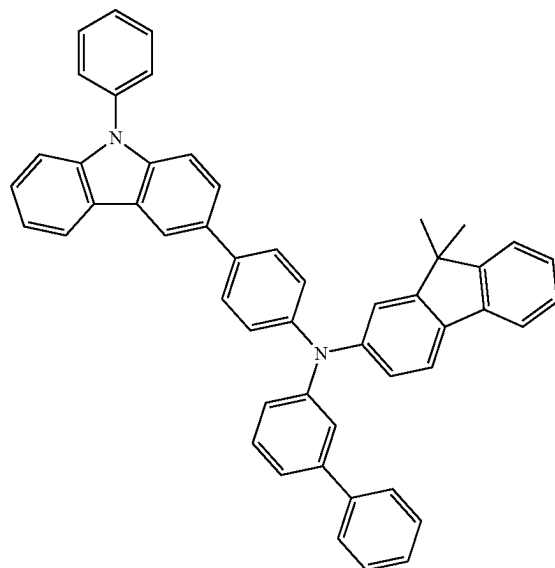


wherein, in Formula 201A, R_{101} , R_{111} , R_{112} , and R_{109} are each as described herein.

[0183] For example, the compound represented by Formula 201 and the compound represented by Formula 202 may include at least one of Compounds HT1 to HT20, but embodiments are not limited thereto:



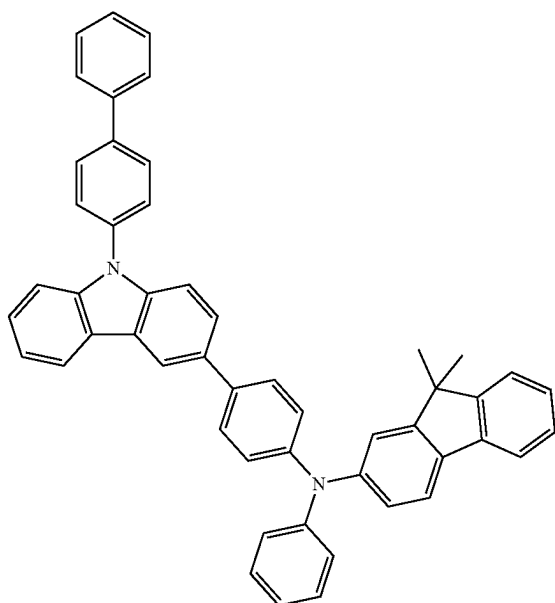
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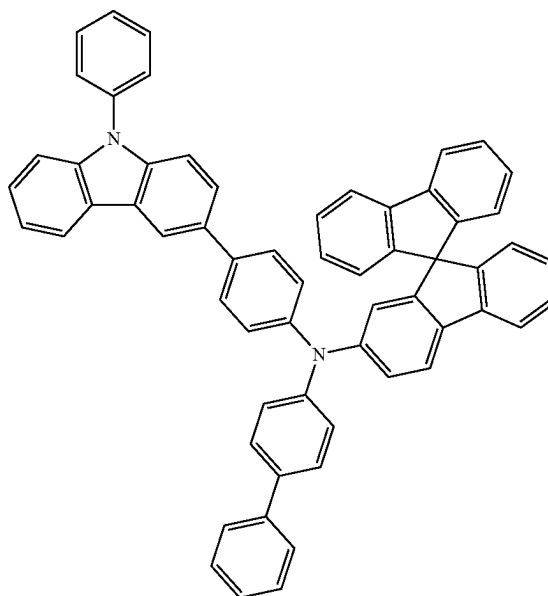
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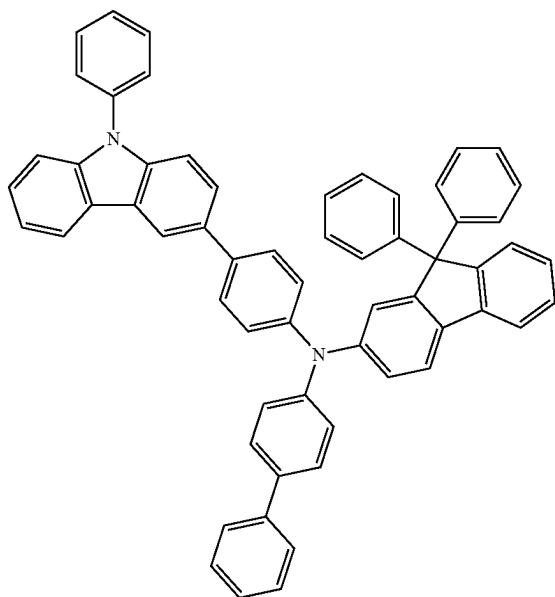
HT4



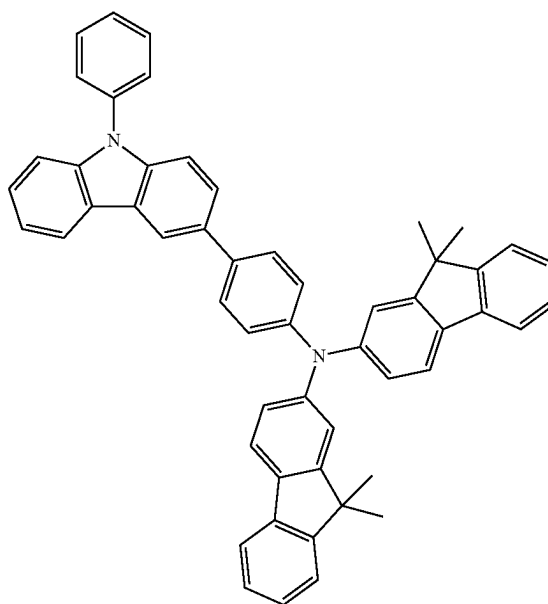
HT6



HT5

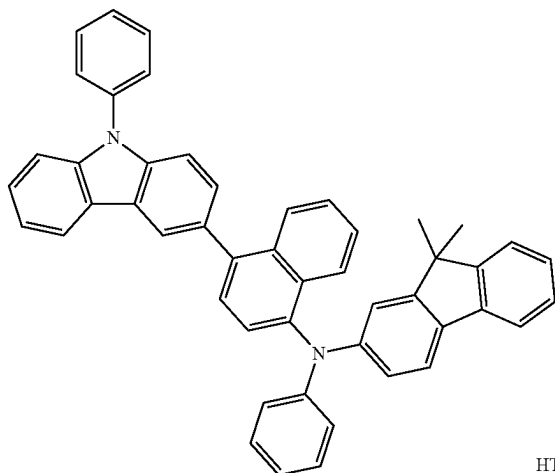


HT7

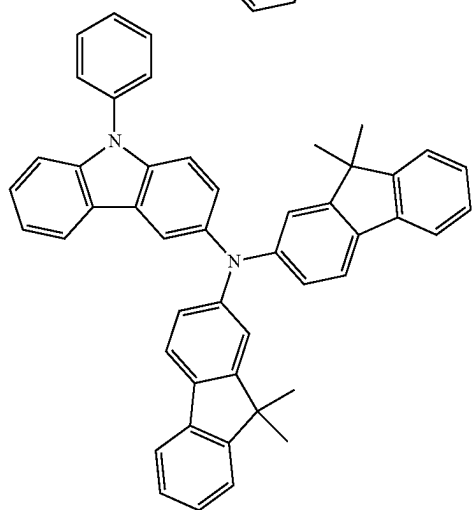


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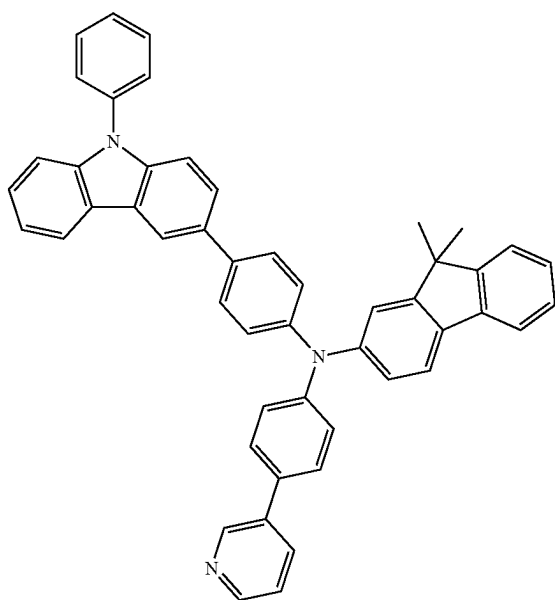
HT8



HT9

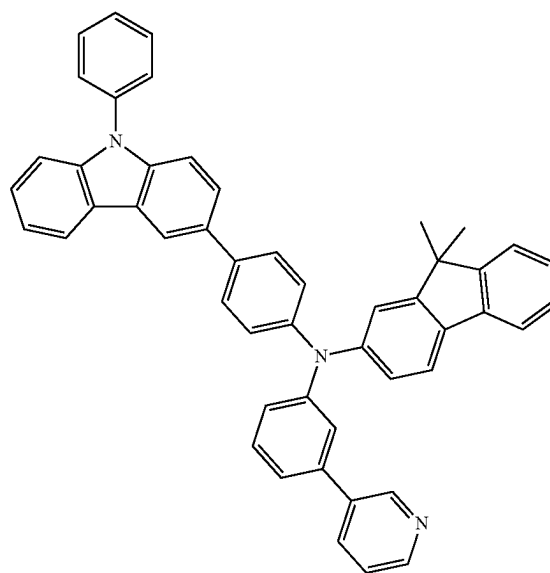


HT10

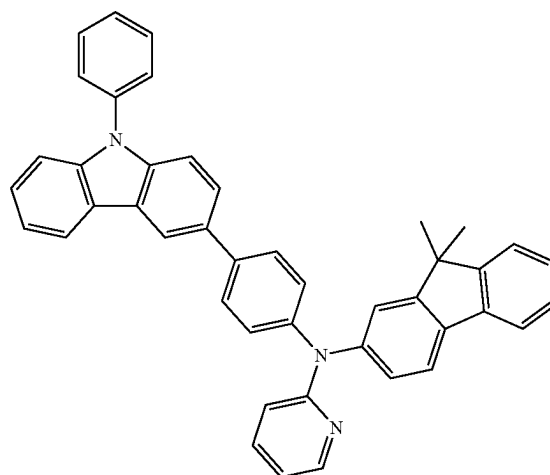


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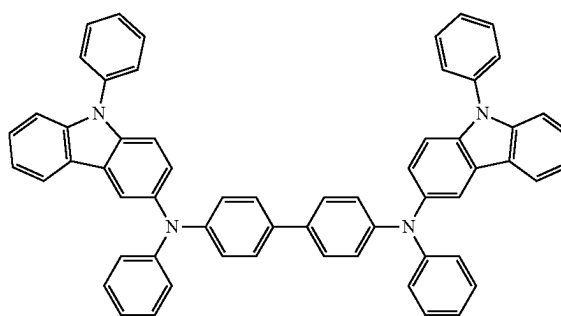
HT11



HT12

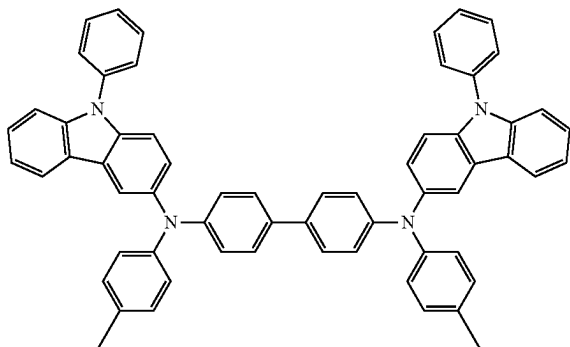


HT13

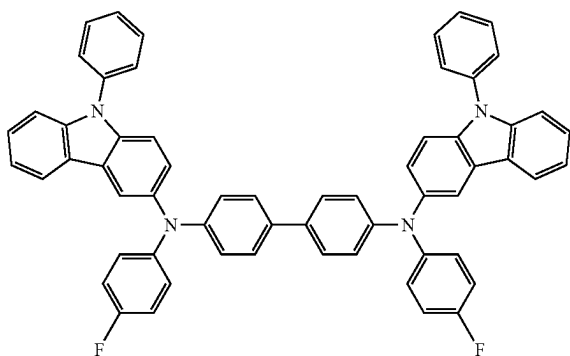


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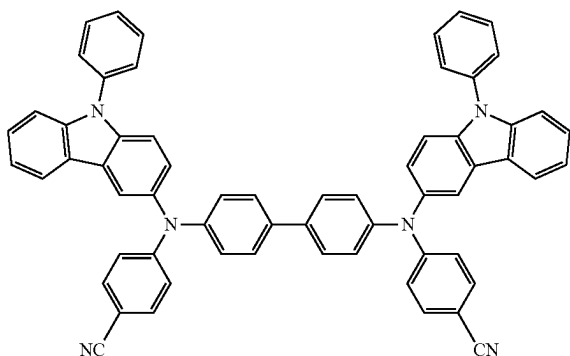
HT14



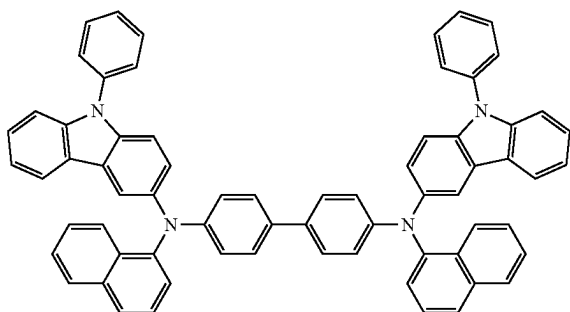
HT15



HT16

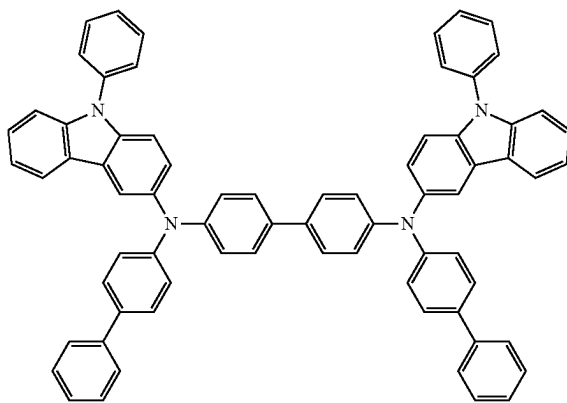


HT17

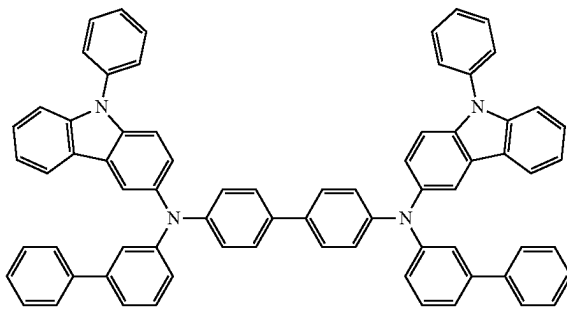


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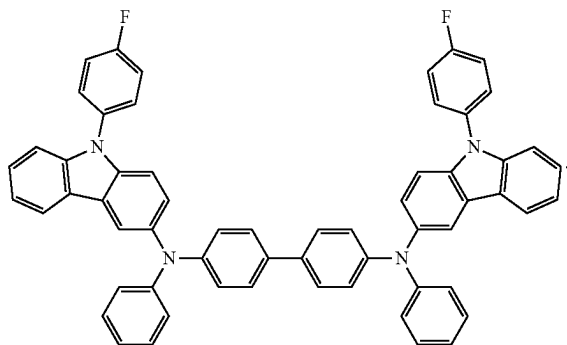
HT18



HT19



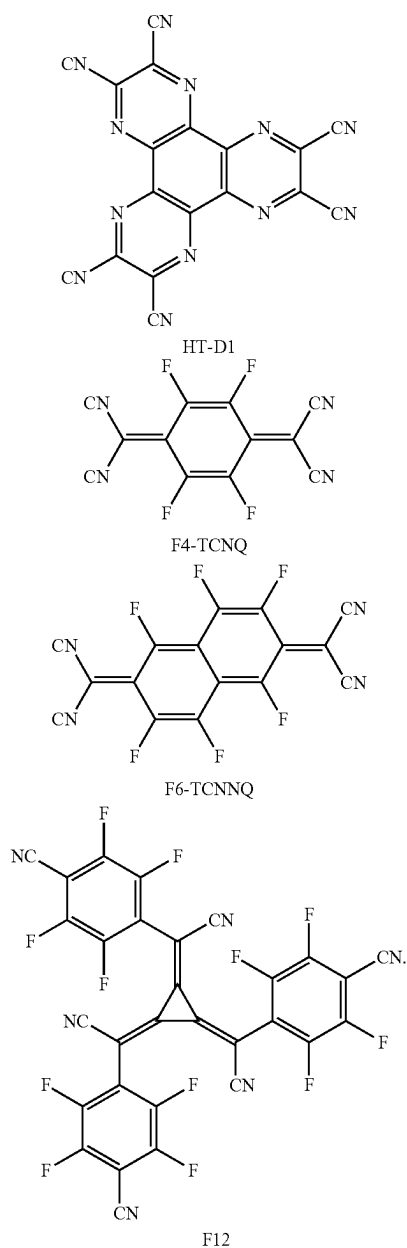
HT20



[0184] The thickness of the hole transport region may be about 100 Å to about 10,000 Å, for example, about 100 Å to about 1,000 Å. When the hole transport region includes at least one of a hole injection layer and a hole transport layer, the thickness of the hole injection layer may be about 50 Å to about 10,000 Å, for example, about 100 Å to about 1,000 Å, and the thickness of the hole transport layer may be about 50 Å to about 2,000 Å, for example, about 100 Å to about 1,500 Å. When the thicknesses of the hole transport region, the hole injection layer and the hole transport layer are within these ranges, satisfactory hole transporting characteristics may be obtained without a substantial increase in driving voltage.

[0185] The hole transport region may further include, in addition to these materials, a charge-generation material for the improvement of conductive properties. The charge-generation material may be homogeneously or non-homogeneously dispersed in the hole transport region.

[0186] The charge-generation material may be, for example, a p-dopant. The p-dopant may be one of a quinone derivative, a metal oxide, or a cyano group-containing compound, but embodiments are not limited thereto. For example, the p-dopant may include a quinone derivative, such as tetracyanoquinodimethane (TCNQ), 2,3,5,6-tetrafluoro-tetracyano-1,4-benzoquinonedimethane (F4-TCNQ), 1,3,4,5,7,8-hexafluorotetracyanonaphthoquinodimethane (F6-TCNNQ), or the like; a metal oxide, such as a tungsten oxide, a molybdenum oxide, or the like; or a cyano group-containing compound, such as one or more of Compounds HT-D1 or F12, but embodiments are not limited thereto:



[0187] The hole transport region may further include a buffer layer.

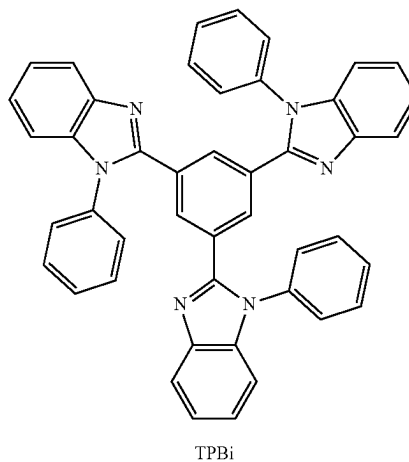
[0188] The buffer layer may compensate for an optical resonance distance according to a wavelength of light emitted from the emission layer, and thus, efficiency of a formed organic light-emitting device may be improved.

[0189] Then, an emission layer (EML) may be formed on the hole transport region by vacuum deposition, spin coating, casting, LB deposition, or the like, but embodiments are not limited thereto. When the emission layer is formed by vacuum deposition or spin coating, the deposition or coating conditions may be similar to those applied in forming the hole injection layer, though the deposition or coating conditions may vary according to a material that is used to form the emission layer.

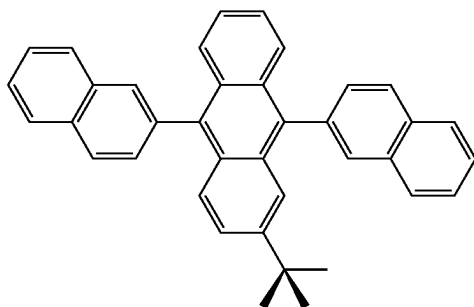
[0190] Meanwhile, when the hole transport region includes an electron blocking layer, a material for forming the electron blocking layer may be selected from materials for the hole transport region described herein and host materials as explained herein. However, the material for forming the electron blocking layer is not limited thereto. For example, when the hole transport region includes an electron blocking layer, a material for the electron blocking layer may be mCP, as described herein.

[0191] The emission layer may include a host and a dopant, and the dopant may include at least one of the organometallic compounds represented by Formula 1.

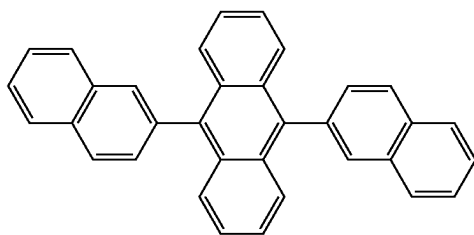
[0192] The host may include at least one of 1,3,5-tri(1-phenyl-1H-benzo[d]imidazol-2-yl)benzene (TPBi), 3-tert-butyl-9,10-di(naphth-2-yl)anthracene (TBADN), 9,10-di(naphthalene-2-yl)anthracene (ADN) (also referred to as "DNA"), 4,4'-bis(N-carbazolyl)-1,1'-biphenyl (CBP), 4,4'-bis(9-carbazolyl)-2,2'-dimethyl-biphenyl (CDBP), 1,3,5-tris(carbazole-9-yl)benzene (TCP), 1,3-bis(N-carbazolyl)benzene (mCP), Compound H50, or Compound H51, but embodiments are not limited thereto:



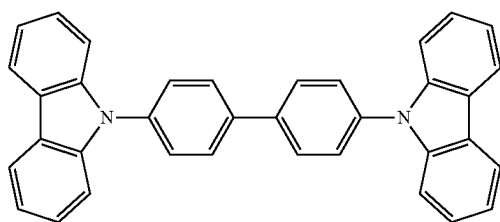
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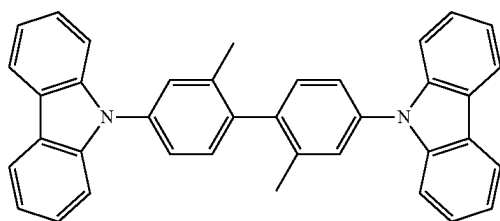
TBADN



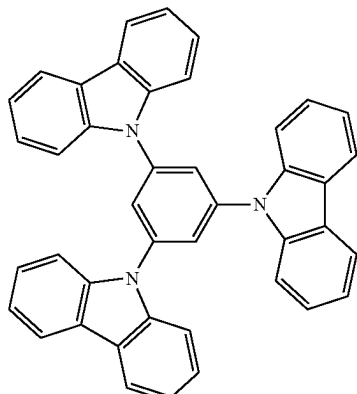
ADN



CBP

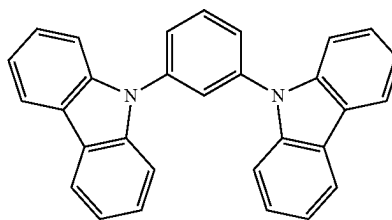


CDBP

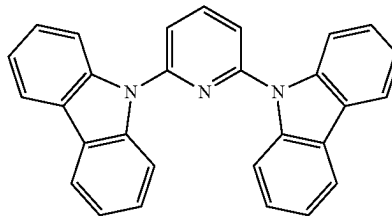


TCP

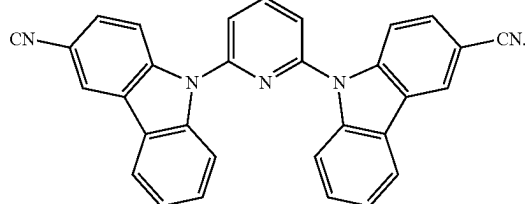
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mCP



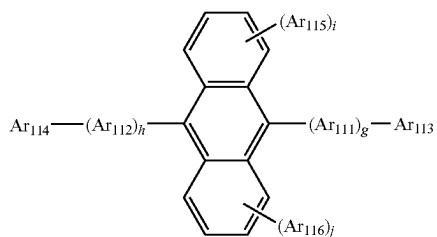
H50



H51

[0193] In one or more embodiments, the host may include a compound represented by Formula 301, but embodiments are not limited thereto:

Formula 301



wherein, in Formula 301, Ar₁₁₁ and Ar₁₁₂ may each independently be:

[0194] a phenylene group, a naphthylene group, a phenanthrenylene group, or a pyrenylene group; or

[0195] a phenylene group, a naphthylene group, a phenanthrenylene group, or a pyrenylene group, each substituted with at least one of a phenyl group, a naphthyl group, an anthracenyl group, or a combination thereof.

[0196] Ar₁₁₃ to Ar₁₁₆ in Formula 301 may each independently be:

[0197] a C₁-C₁₀ alkyl group, a phenyl group, a naphthyl group, a phenanthrenyl group, or a pyrenyl group; or

[0198] a phenyl group, a naphthyl group, a phenanthrenyl group, or a pyrenyl group, each substituted with at least one of a phenyl group, a naphthyl group, an anthracenyl group, or a combination thereof.

[0199] g, h, i, and j in Formula 301 may each independently be an integer from 0 to 4, for example, 0, 1, or 2.

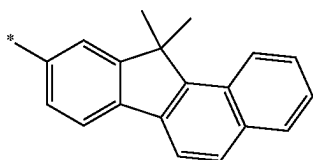
[0200] Ar_{113} to Ar_{116} in Formula 301 may each independently be:

[0201] a C_1 - C_{10} alkyl group that is substituted with at least one of a phenyl group, a naphthyl group, an anthracenyl group, or a combination thereof;

[0202] a phenyl group, a naphthyl group, an anthracenyl group, a pyrenyl group, a phenanthrenyl group, or a fluorenyl group;

[0203] a phenyl group, a naphthyl group, an anthracenyl group, a pyrenyl group, a phenanthrenyl group, or a fluorenyl group, each substituted with at least one of deuterium, $-F$, $-Cl$, $-Br$, $-I$, $-SF_5$, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1 - C_{60} alkyl group, a C_2 - C_{60} alkenyl group, a C_2 - C_{60} alkynyl group, a C_1 - C_{60} alkoxy group, a C_1 - C_{60} alkylthio group, a phenyl group, a naphthyl group, an anthracenyl group, a pyrenyl group, a phenanthrenyl group, a fluorenyl group, or a combination thereof; or

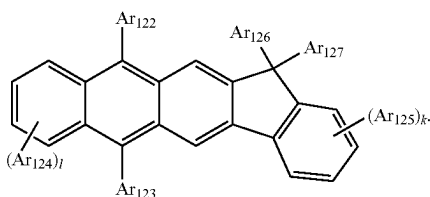
[0204] a group of the structure:



[0205] but, embodiments are not limited thereto.

[0206] In one or more embodiments, the host may include a compound represented by Formula 302, but embodiments are not limited thereto:

Formula 302



[0207] Ar_{122} to Ar_{125} in Formula 302 are each as described herein in connection with Ar_{113} in Formula 301.

[0208] Ar_{126} and Ar_{127} in Formula 302 may each independently be a C_1 - C_{10} alkyl group (for example, a methyl group, an ethyl group, a propyl group, or the like).

[0209] k and i in Formula 302 may each independently be an integer from 0 to 4. For example, k and i may be 0, 1, or 2.

[0210] When the organic light-emitting device is a full-color organic light-emitting device, the emission layer may be patterned into a red emission layer, a green emission layer, and/or a blue emission layer. In one or more embodiments, due to a stacked structure including a red emission

layer, a green emission layer, and/or a blue emission layer, the emission layer may emit white light, and various modifications may be made.

[0211] When the emission layer includes a host and a dopant, the amount of the dopant in the emission layer may be about 0.01 parts by weight to about 15 parts by weight, based on 100 parts by weight of the host, but embodiments are not limited thereto.

[0212] The thickness of the emission layer may be about 100 Å to about 1,000 Å, for example, about 200 Å to about 600 Å. When the thickness of the emission layer is within the range described above, excellent luminescence characteristics may be obtained without a substantial increase in driving voltage.

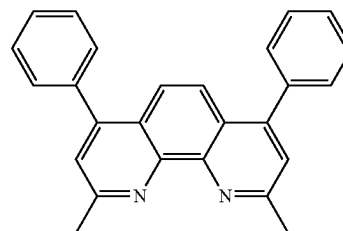
[0213] An electron transport region is disposed on the emission layer.

[0214] The electron transport region may include a hole blocking layer, an electron transport layer, an electron injection layer, or a combination thereof.

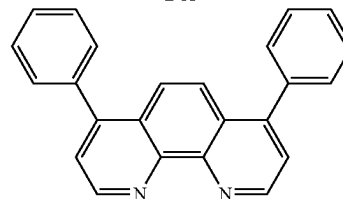
[0215] For example, the electron transport region may have a hole blocking layer/electron transport layer/electron injection layer structure, or an electron transport layer/electron injection layer structure, but the disclosure is not limited thereto. The electron transport layer may have a single-layered structure or a multilayer structure including two or more different materials.

[0216] Conditions for forming the hole blocking layer, the electron transport layer, and the electron injection layer which constitute the electron transport region may be understood by referring to the conditions for forming the hole injection layer.

[0217] When the electron transport region includes a hole blocking layer, the hole blocking layer may include, for example, at least one of 2,9-dimethyl-4,7-diphenyl-1,10-phenanthroline (BCP), 4,7-diphenyl-1,10-phenanthroline (Bphen), or bis(2-methyl-8-quinolinolato-N1,O8)-(1,1'-bi-phenyl-4-olato)aluminum (BALq), but embodiments are not limited thereto:



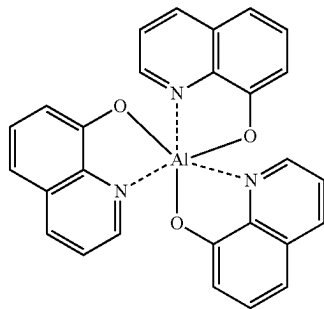
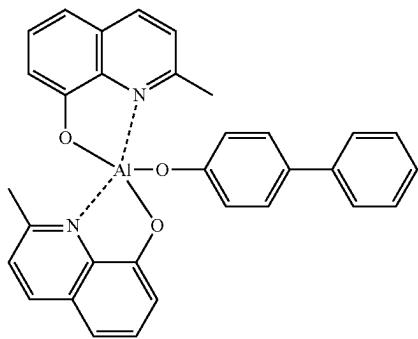
BCP



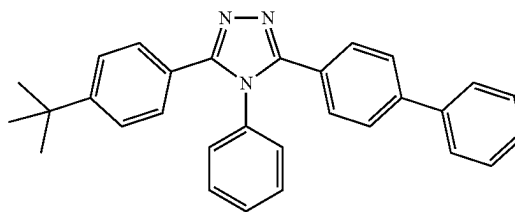
Bphen

[0218] The thickness of the hole blocking layer may be about 20 Å to about 1,000 Å, for example, about 30 Å to about 300 Å. When the thickness of the hole blocking layer is within the range as described above, excellent hole blocking characteristics may be obtained without a substantial increase in driving voltage.

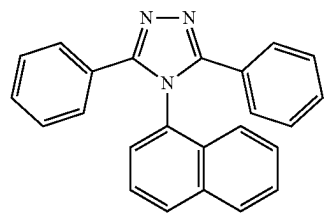
[0219] The electron transport layer may further include at least one of 2,9-dimethyl-4,7-diphenyl-1,10-phenanthroline (BCP), 4,7-diphenyl-1,10-phenanthroline (Bphen), tris(8-hydroxy-quinolinato)aluminum (Alq_3), bis(2-methyl-8-quinolinolato-N1,O8)-(1,1'-biphenyl-4-olato)aluminum (BALq), 3-(4-biphenyl)-4-phenyl-5-tert-butylphenyl-1,2,4-triazole (TAZ), or 4-(naphthalen-1-yl)-3,5-diphenyl-4H-1,2,4-triazole (NTAZ), but embodiments are not limited thereto:

 Alq_3  BALq

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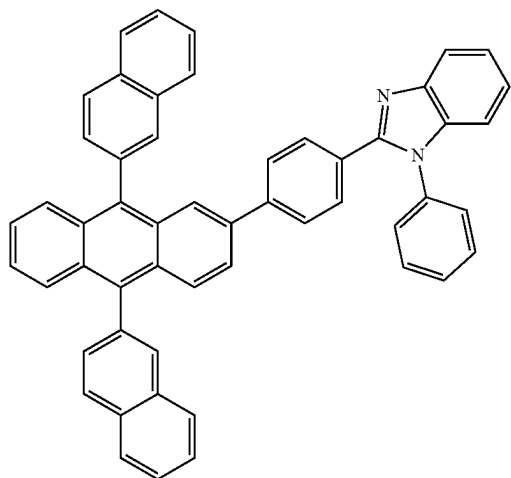


TAZ

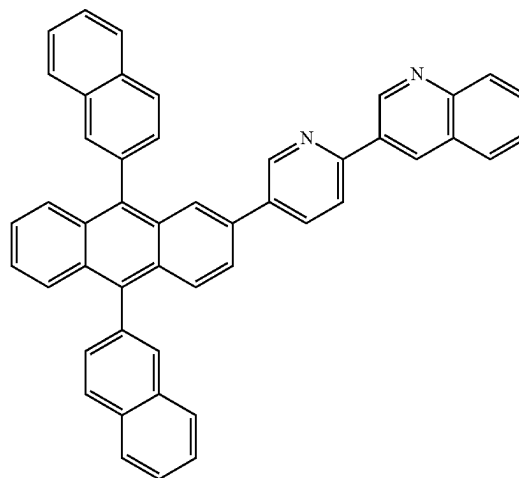


NTAZ

[0220] In one or more embodiments, the electron transport layer may include at least one of Compounds ET1 to ET25, but embodiments are not limited thereto:

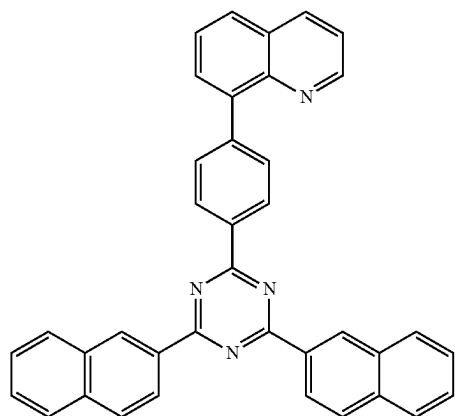


ET1

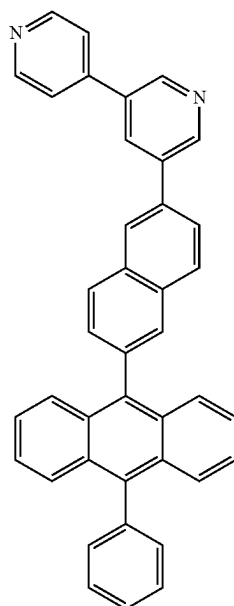


ET2

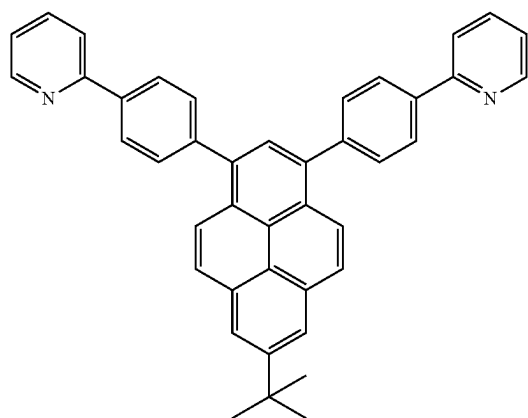
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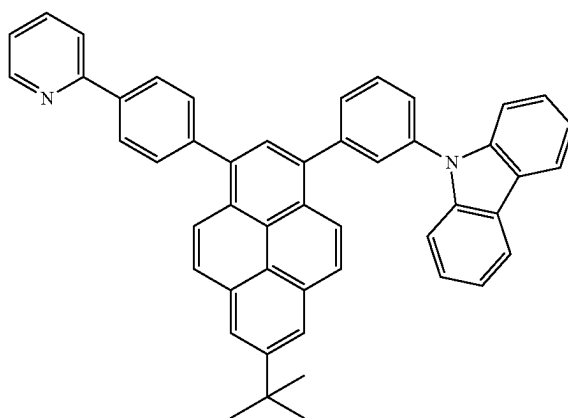
ET3



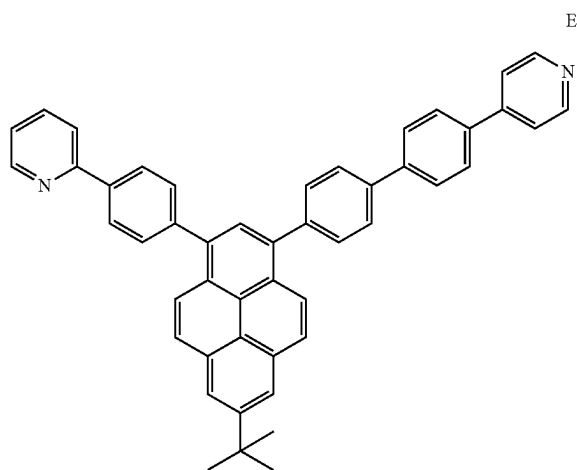
ET4



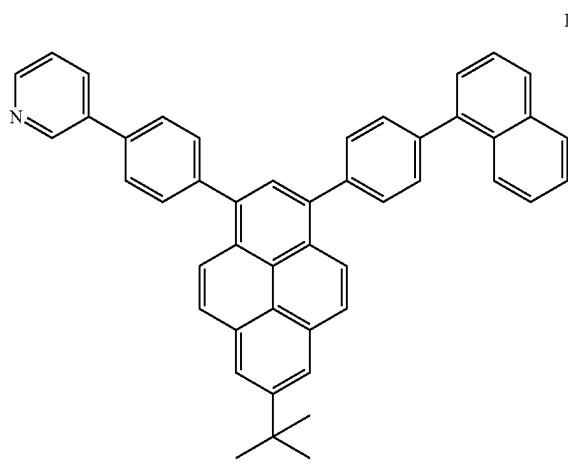
ET5



ET6

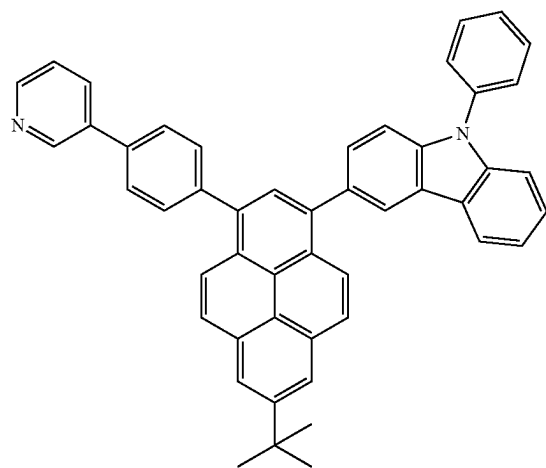


ET7

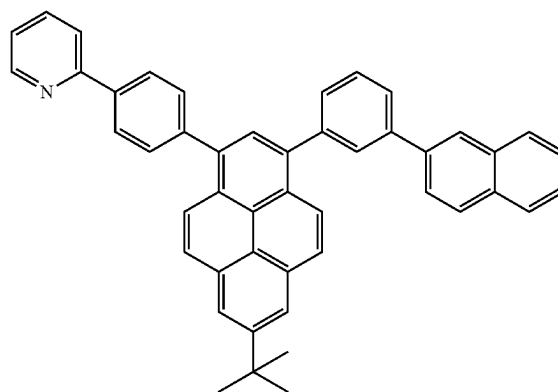


ET8

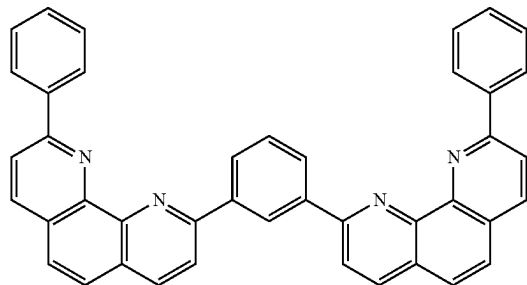
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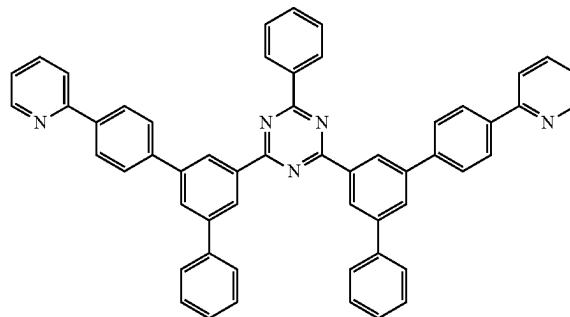
ET9



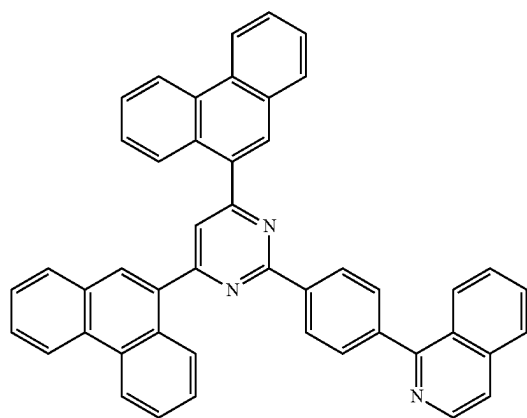
ET10



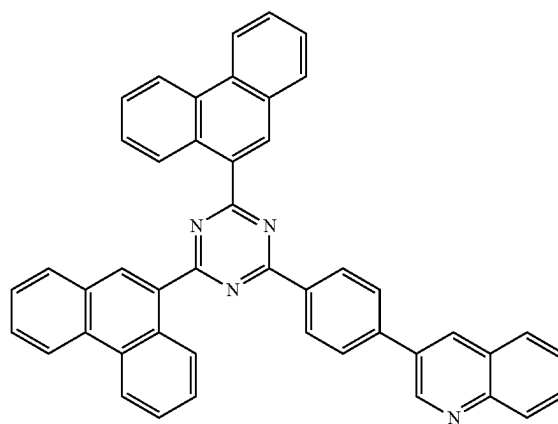
ET11



ET12



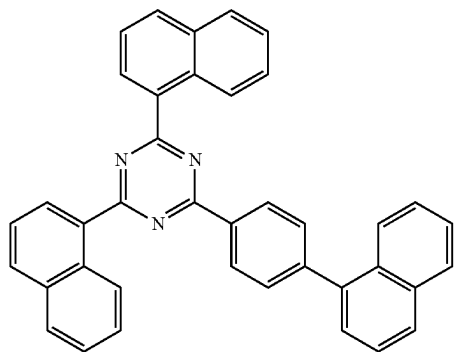
ET13



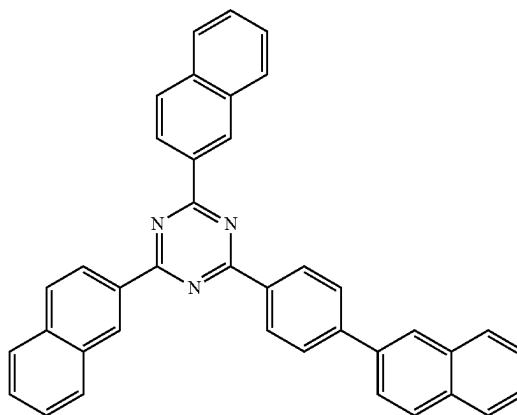
ET14

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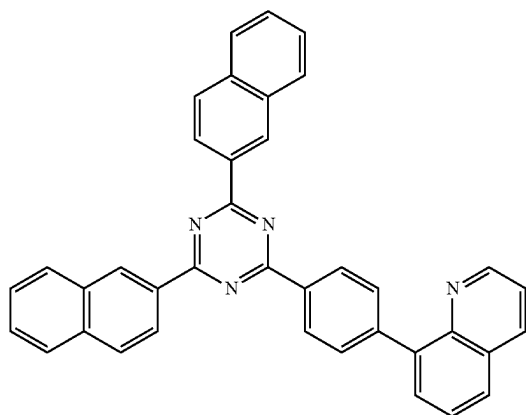
ET15



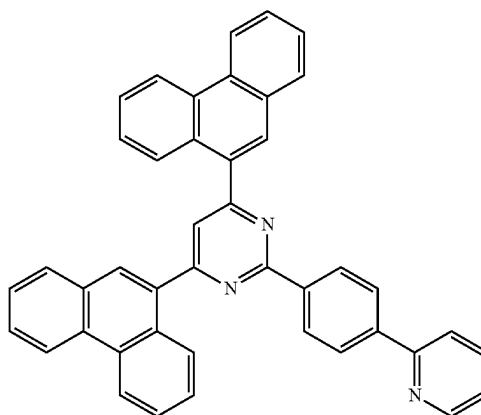
ET16



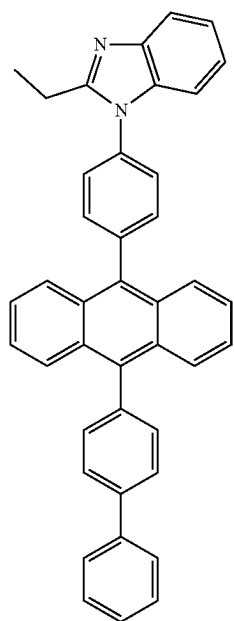
ET17



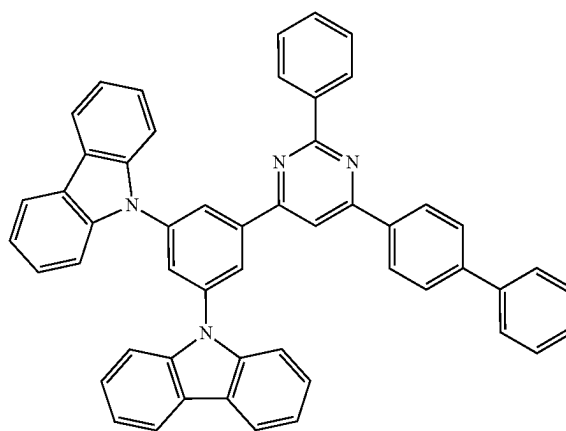
ET18



ET19

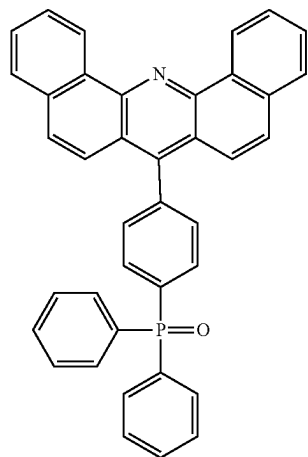


ET20

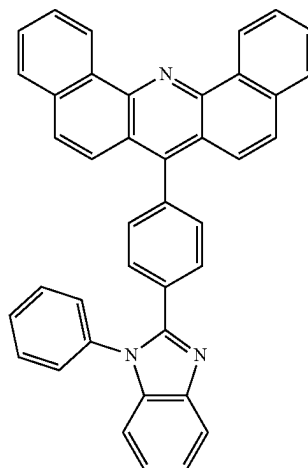


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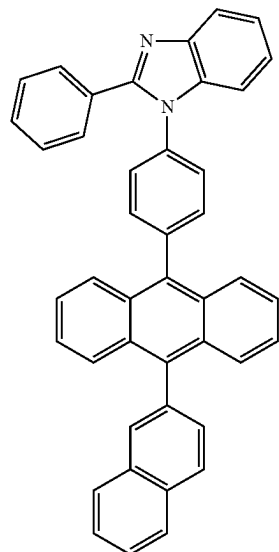
ET21



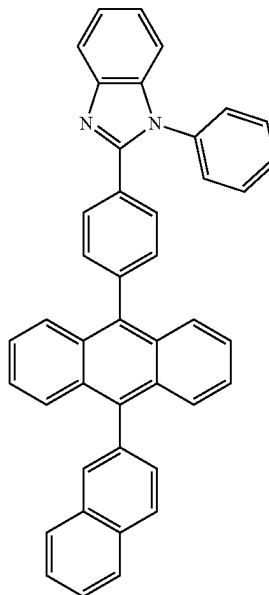
ET22



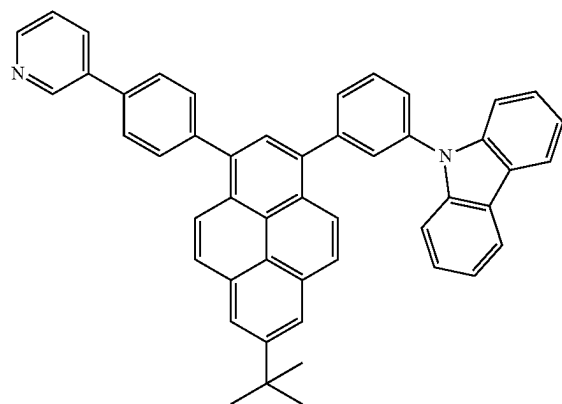
ET23



ET24



ET25

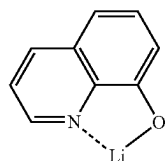


[0221] The thickness of the electron transport layer may be about 100 Å to about 1,000 Å, for example, about 150 Å to about 500 Å. When the thickness of the electron transport

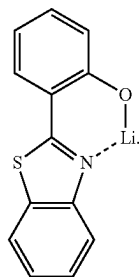
layer is within the range as described above, satisfactory electron transport characteristics may be obtained without a substantial increase in driving voltage.

[0222] Also, the electron transport layer may further include, in addition to the materials described above, a metal-containing material.

[0223] The metal-containing material may include a Li complex. The Li complex may include, for example, Compound ET-D1 (lithium quinolate, LiQ) or ET-D2, but embodiments are not limited thereto:



ET-D1



ET-D2

[0224] In addition, the electron transport region may include an electron injection layer that facilitates electron injection from the second electrode 19.

[0225] The electron injection layer may include LiF, NaCl, CsF, Li₂O, BaO, or a combination thereof.

[0226] The thickness of the electron injection layer may be about 1 Å to about 100 Å, for example, about 3 Å to about 90 Å. When the thickness of the electron injection layer is within the range as described above, satisfactory electron injection characteristics may be obtained without a substantial increase in driving voltage.

[0227] The second electrode 19 is located on the organic layer 15. The second electrode 19 may be a cathode. A material for forming the second electrode 19 may be metal, an alloy, an electrically conductive compound, or a combination thereof, which have a relatively low work function. For example, lithium (Li), magnesium (Mg), aluminum (Al), silver (Ag), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), or magnesium-silver (Mg—Ag) may be used as the material for forming the second electrode 19. In one or more embodiments, to manufacture a top-emission type light-emitting device, various modifications, such as formation of a transmissive second electrode 19 using ITO or IZO, is possible.

[0228] Hereinbefore, the organic light-emitting device has been described with reference to the FIGURE, but embodiments are not limited thereto.

[0229] According to another aspect, provided is a diagnostic composition including at least one of the organometallic compounds represented by Formula 1.

[0230] The organometallic compound represented by Formula 1 provides a high luminescence efficiency. Accordingly, a diagnostic composition including at least one of the organometallic compounds represented by Formula 1 may have high diagnostic efficiency.

[0231] The diagnostic composition may be used in various applications including a diagnosis kit, a diagnosis reagent, a biosensor, or a biomarker, but embodiments are not limited thereto.

[0232] The term “C₁-C₆₀ alkyl group” as used herein refers to a linear or branched saturated aliphatic hydrocarbon monovalent group having 1 to 60 carbon atoms, and non-limiting examples thereof include a methyl group, an ethyl group, a propyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, a pentyl group, an isoamyl group, a hexyl group, or the like. The term “C₁-C₆₀ alkylene group” as used herein refers to a divalent group having the same structure as the C₁-C₆₀ alkyl group.

[0233] The term “C₁-C₆₀ alkoxy group” used herein refers to a monovalent group represented by —OA₁₀₁ (wherein A₁₀₁ is the C₁-C₆₀ alkyl group), and non-limiting examples thereof include a methoxy group, an ethoxy group, an isopropoxy group, or the like. The term “C₁-C₆₀ alkylthio group” as used herein refers to a monovalent group represented by —SA₁₀₁ (wherein A₁₀₁ is the C₁-C₆₀ alkyl group).

[0234] The term “C₂-C₆₀ alkenyl group” as used herein has a structure including at least one carbon-carbon double bond in the middle or at the terminus of the C₂-C₆₀ alkyl group, and non-limiting examples thereof include an ethenyl group, a propenyl group, a butenyl group, or the like. The term “C₂-C₆₀ alkenylene group” as used herein refers to a divalent group having the same structure as the C₂-C₆₀ alkenyl group.

[0235] The term “C₂-C₆₀ alkynyl group” as used herein has a structure including at least one carbon-carbon triple bond in the middle or at the terminus of the C₂-C₆₀ alkyl group, and non-limiting examples thereof include an ethynyl group, a propynyl group, or the like. The term “C₂-C₆₀ alkynylene group” as used herein refers to a divalent group having the same structure as the C₂-C₆₀ alkynyl group.

[0236] The term “C₃-C₁₀ cycloalkyl group” as used herein refers to a monovalent saturated hydrocarbon ring group having 3 to 10 carbon atoms, and non-limiting examples thereof include a cyclopropyl group, a cyclobutyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, or the like. The term “C₃-C₁₀ cycloalkylene group” as used herein refers to a divalent group having the same structure as the C₃-C₁₀ cycloalkyl group.

[0237] The term “C₁-C₁₀ heterocycloalkyl group” as used herein refers to a monovalent ring group having at least one heteroatom selected from B, N, O, P, Si, S, Se, or Ge as a ring-forming atom and 1 to 10 carbon atoms as ring-forming atom(s), and non-limiting examples thereof include a tetrahydrofuran group, a tetrahydrothiophene group (tetrahydrothiophene group), or the like. The term “C₁-C₁₀ heterocycloalkylene group” as used herein refers to a divalent group having the same structure as the C₁-C₁₀ heterocycloalkyl group.

[0238] The term “C₃-C₁₀ cycloalkenyl group” as used herein refers to a monovalent ring group that includes 3 to 10 carbon atoms and at least one carbon-carbon double bond in the ring thereof and has no aromaticity, and non-limiting examples thereof include a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, or the like. The term “C₃-C₁₀ cycloalkenylene group” as used herein refers to a divalent group having the same structure as the C₃-C₁₀ cycloalkenyl group.

[0239] The term “C₁-C₁₀ heterocycloalkenyl group” as used herein refers to a monovalent ring group that has at

least one heteroatom selected from B, N, O, P, Si, S, Se, and Ge as a ring-forming atom, 1 to 10 carbon atoms as ring-forming atom(s), and at least one double bond in the ring thereof. Non-limiting examples of the C_1 - C_{10} heterocycloalkenyl group include a 2,3-dihydrofuran group, a 2,3-dihydrothiophenyl group, or the like. The term " C_1 - C_{10} heterocycloalkenylene group" as used herein refers to a divalent group having the same structure as the C_1 - C_{10} heterocycloalkenyl group.

[0240] The term " C_6 - C_{60} aryl group" as used herein refers to a monovalent group having a carbocyclic aromatic ring system having 6 to 60 carbon atoms, and the term " C_6 - C_{60} arylene group" as used herein refers to a divalent group having a carbocyclic aromatic ring system having 6 to 60 carbon atoms. Non-limiting examples of the C_6 - C_{60} aryl group include a phenyl group, a naphthyl group, an anthracenyl group, a phenanthrenyl group, a pyrenyl group, a chrysenyl group, or the like. When the C_6 - C_{60} aryl group and the C_6 - C_{60} arylene group each include two or more rings, the two or more rings may be fused to each other.

[0241] The term " C_7 - C_{60} alkyl aryl group" as used herein refers to a C_6 - C_{60} aryl group that is substituted with at least one C_1 - C_{60} alkyl group. The term " C_7 - C_{60} aryl alkyl group" as used herein refers to a C_1 - C_{60} alkyl group that is substituted with at least one C_6 - C_{60} aryl group.

[0242] The term " C_1 - C_{60} heteroaryl group" as used herein refers to a monovalent group having a heteroaromatic ring system having at least one heteroatom selected from B, N, O, P, Si, S, Se, and Ge as a ring-forming atom and 1 to 60 carbon atoms as ring-forming atom(s). The term " C_1 - C_{60} heteroarylene group" as used herein refers to a divalent group having a heteroaromatic ring system having at least one heteroatom selected from B, N, O, P, Si, S, Se, and Ge as a ring-forming atom and 1 to 60 carbon atoms as ring-forming atom(s). Non-limiting examples of the C_1 - C_{60} heteroaryl group include a pyridinyl group, a pyrimidinyl group, a pyrazinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, or the like. When the C_1 - C_{60} heteroaryl group and the C_1 - C_{60} heteroarylene group each include two or more rings, the two or more rings may be fused to each other.

[0243] The term " C_2 - C_{60} alkyl heteroaryl group" as used herein refers to a C_1 - C_{60} heteroaryl group that is substituted with at least one C_1 - C_{60} alkyl group. The term " C_2 - C_{60} heteroaryl alkyl group" as used herein refers to a C_1 - C_{60} alkyl group that is substituted with at least one C_1 - C_{60} heteroaryl group.

[0244] The term " C_6 - C_{60} aryloxy group" as used herein refers to $-OA_{102}$ (wherein A_{102} is the C_6 - C_{60} aryl group), and the term " C_6 - C_{60} arylthio group" as used herein refers to $-SA_{103}$ (wherein A_{103} is the C_6 - C_{60} aryl group).

[0245] The term " C_1 - C_{60} heteroaryloxy group" as used herein refers to $-OA_{104}$ (wherein A_{104} is the C_1 - C_{60} heteroaryl group), and the term " C_1 - C_{60} heteroarylthio group" as used herein refers to $-SA_{105}$ (wherein A_{105} is the C_1 - C_{60} heteroaryl group).

[0246] The term "monovalent non-aromatic condensed polycyclic group" as used herein refers to a monovalent group (for example, having 8 to 60 carbon atoms) having two or more rings condensed to each other, only carbon atoms as ring-forming atoms, and no aromaticity in its molecular structure when considered as a whole. Non-

limiting examples of the monovalent non-aromatic condensed polycyclic group include a fluorenyl group or the like.

[0247] The term "divalent non-aromatic condensed polycyclic group" as used herein refers to a divalent group having the same structure as the monovalent non-aromatic condensed polycyclic group.

[0248] The term "monovalent non-aromatic condensed heteropolycyclic group" as used herein refers to a monovalent group having two or more rings condensed with each other, a heteroatom selected from B, N, O, P, Si, S, Se, and Ge, other than carbon atoms (for example, having 1 to 60 carbon atoms), as a ring-forming atom, and no aromaticity in its molecular structure when considered as a whole. Non-limiting examples of the monovalent non-aromatic condensed heteropolycyclic group include a carbazolyl group or the like. The term "divalent non-aromatic condensed heteropolycyclic group" as used herein refers to a divalent group having the same structure as the monovalent non-aromatic condensed heteropolycyclic group.

[0249] The term " C_5 - C_{30} carbocyclic group" as used herein refers to a saturated or unsaturated ring group having, as a ring-forming atom, 5 to 30 carbon atoms only. The C_5 - C_{30} carbocyclic group may be a monocyclic group or a polycyclic group.

[0250] The term " C_1 - C_{30} heterocyclic group" as used herein refers to a saturated or unsaturated ring group having 1 to 30 carbon atoms as a ring-forming atom and at least one heteroatom selected from B, N, O, P, Si, S, Se, and Ge. The C_1 - C_{30} heterocyclic group may be a monocyclic group or a polycyclic group.

[0251] At least one substituent of the substituted C_5 - C_{30} carbocyclic group, the substituted C_1 - C_{30} heterocyclic group, the substituted C_1 - C_{60} alkyl group, the substituted C_2 - C_{60} alkenyl group, the substituted C_2 - C_{60} alkynyl group, the substituted C_1 - C_{60} alkoxy group, the substituted C_1 - C_{60} alkylthio group, the substituted C_1 - C_{60} alkylthio group, the substituted C_3 - C_{10} cycloalkyl group, the substituted C_1 - C_{10} heterocycloalkyl group, the substituted C_3 - C_{10} cycloalkenyl group, the substituted C_1 - C_{10} heterocycloalkenyl group, the substituted C_6 - C_{60} aryl group, the substituted C_7 - C_{60} alkyl aryl group, the substituted C_7 - C_{60} aryl alkyl group, the substituted C_6 - C_{60} aryloxy group, the substituted C_6 - C_{60} arylthio group, the substituted C_1 - C_{60} heteroaryl group, the substituted C_2 - C_{60} alkyl heteroaryl group, the substituted C_2 - C_{60} heteroaryl alkyl group, the substituted C_1 - C_{60} heteroaryloxy group, the substituted C_1 - C_{60} heteroarylthio group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed heteropolycyclic group may be:

[0252] deuterium, $-F$, $-Cl$, $-Br$, $-I$, $-SF_5$, $-CD_3$, $-CD_2H$, $-CDH_2$, $-CF_3$, $-CF_2H$, $-CFH_2$, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazide group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1 - C_{60} alkyl group, a C_2 - C_{60} alkenyl group, a C_2 - C_{60} alkynyl group, a C_1 - C_{60} alkoxy group, or a C_1 - C_{60} alkylthio group;

[0253] a C_1 - C_{60} alkyl group, a C_2 - C_{60} alkenyl group, a C_2 - C_{60} alkynyl group, a C_1 - C_{60} alkoxy group, or a C_1 - C_{60} alkylthio group, each substituted with at least one of deuterium, $-F$, $-Cl$, $-Br$, $-I$, $-SF_5$, $-CD_3$, $-CD_2H$, $-CDH_2$, $-CF_3$, $-CF_2H$, $-CFH_2$, a

hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₇-C₆₀ alkyl aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a C₂-C₆₀ alkyl heteroaryl group, a C₁-C₆₀ heteroaryloxy group, a C₁-C₆₀ heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q₁₁)(Q₁₂)(Q₁₃), —Ge(Q₁₁)(Q₁₂)(Q₁₃), —N(Q₁₄)(Q₁₅),

[0254] —B(Q₁₆)(Q₁₇), —P(Q₁₈)(Q₁₉), —P(=O)(Q₁₈)(Q₁₉), or a combination thereof;

[0255] a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₇-C₆₀ alkyl aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a C₂-C₆₀ alkyl heteroaryl group, a C₁-C₆₀ heteroaryloxy group, a C₁-C₆₀ heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, or a monovalent non-aromatic condensed heteropolycyclic group;

[0256] a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₇-C₆₀ alkyl aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a C₂-C₆₀ alkyl heteroaryl group, a C₁-C₆₀ heteroaryloxy group, a C₁-C₆₀ heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, or a monovalent non-aromatic condensed heteropolycyclic group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF₅, —CD₃, —CD₂H, —CDH₂, —CF₃, —CF₂H, —CFH₂, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-Co alkoxy group, a C₁-C₆₀ alkylthio group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₇-C₆₀ alkyl aryl group, a C₇-C₆₀ aryl alkyl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a C₂-C₆₀ alkyl heteroaryl group, a C₁-C₆₀ heteroaryloxy group, a C₁-C₆₀ heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q₂₁)(Q₂₂)(Q₂₃),

[0257] —Ge(Q₂₁)(Q₂₂)(Q₂₃), —N(Q₂₄)(Q₂₅), —B(Q₂₆)(Q₂₇), —P(Q₂₈)(Q₂₉), —P(=O)(Q₂₈)(Q₂₉), or a combination thereof; or

[0258] —Si(Q₃₁)(Q₃₂)(Q₃₃), —Ge(Q₃₁)(Q₃₂)(Q₃₃), —N(Q₃₄)(Q₃₅), —B(Q₃₆)(Q₃₇), —P(Q₃₈)(Q₃₉), or

[0259] —P(=O)(Q₃₈)(Q₃₉), and

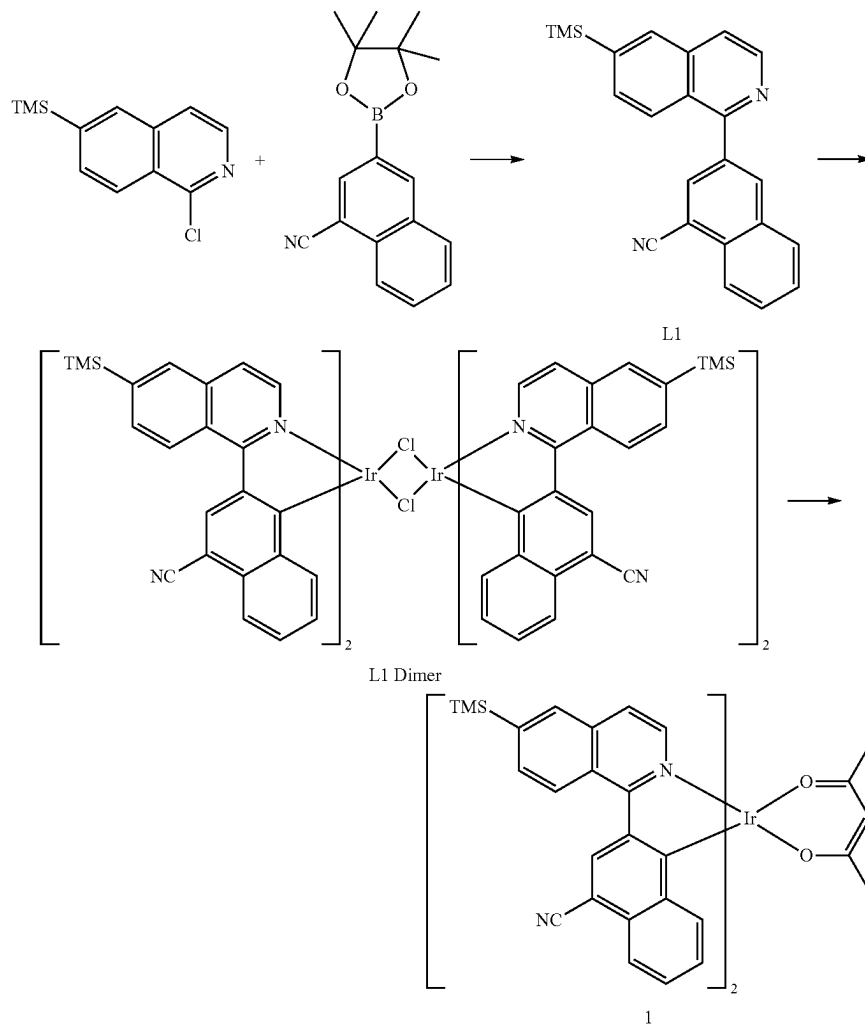
[0260] Q₁ to Q₉, Q₁₁ to Q₁₉, Q₂₁ to Q₂₉, and Q₃₁ to Q₃₉ may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, —SF₅, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C₁-C₆₀ alkyl group, a substituted or unsubstituted C₂-C₆₀ alkenyl group, a substituted or unsubstituted C₂-C₆₀ alkynyl group, a substituted or unsubstituted C₁-C₆₀ alkoxy group, a substituted or unsubstituted C₁-C₆₀ alkylthio group, a substituted or unsubstituted C₃-C₁₀ cycloalkyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkyl group, a substituted or unsubstituted C₃-C₁₀ cycloalkenyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkenyl group, a substituted or unsubstituted C₆-C₆₀ aryl group, a substituted or unsubstituted C₇-C₆₀ alkyl aryl group, a substituted or unsubstituted C₇-C₆₀ aryl alkyl group, a substituted or unsubstituted C₆-C₆₀ aryloxy group, a substituted or unsubstituted C₆-C₆₀ arylthio group, a substituted or unsubstituted C₁-Co heteroaryl group, a substituted or unsubstituted C₂-C₆₀ alkyl heteroaryl group, a substituted or unsubstituted C₂-C₆₀ heteroaryl alkyl group, a substituted or unsubstituted C₁-C₆₀ heteroaryloxy group, a substituted or unsubstituted C₁-C₆₀ heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, or a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group.

[0261] For example, Q₁ to Q₉, Q₁₁ to Q₁₉, Q₂₁ to Q₂₉, and Q₃₁ to Q₃₉ may each independently be a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, a C₁-C₆₀ alkylthio group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₇-C₆₀ alkyl aryl group, a C₇-C₆₀ aryl alkyl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a C₂-C₆₀ alkyl heteroaryl group, a C₂-C₆₀ heteroaryl alkyl group, a C₁-C₆₀ heteroaryloxy group, a C₁-C₆₀ heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, or a monovalent non-aromatic condensed heteropolycyclic group, each unsubstituted or substituted with at least one of hydrogen, deuterium, —F, —Cl, —Br, —I, —SF₅, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof; deuterium, a C₁-C₆₀ alkyl group, a C₃-C₁₀ cycloalkyl group, a C₆-C₆₀ aryl group, or a combination thereof.

[0262] Hereinafter, exemplary organometallic compounds represented by Formula 1 and organic light-emitting devices according to exemplary embodiments are described in further detail with reference to Synthesis Examples and Examples. However, embodiments are not limited to Synthesis Examples and Examples. The wording “B was used instead of A” used in describing Synthesis Examples means that an amount of A used was identical to an amount of B used, in terms of a molar equivalent.

EXAMPLES

Synthesis Example 1: Synthesis of Compound 1



Synthesis of Intermediate L1

[0263] 2.0 grams (g) (8.48 millimoles (mmol)) 1-chloro-6-(trimethylsilyl)isoquinoline was mixed with 60 milliliters (mL) of tetrahydrofuran (THF) and 15 mL of water, and then 2.8 g (10.17 mmol) of 3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1-naphthonitrile, 0.7 g (0.59 mmol) of (tetrakis(triphenylphosphine)palladium(0), Pd(PPh₃)₄), and 2.9 g (21.1 mmol) of K₂CO₃ were added thereto. The resulting mixture was then heated under reflux for 24 hours. The resulting product obtained therefrom was subjected to an extraction process by using ethyl acetate and water to obtain an organic layer. The organic layer was then dried with magnesium sulfate, the solvent was removed under a reduced pressure, and the product was purified by liquid chromatography, to thereby obtain 2.2 g (yield of 75%) of Intermediate L1.

[0264] Liquid chromatography-mass spectrometry (LC-MS) m/z=353 (M+H)⁺.

Synthesis of L1 dimer

[0265] 1.0 g (2.8 mmol) of Intermediate L1 and 0.47 g (1.3 mmol) of iridium chloride hydrate were mixed with 60 mL of ethoxyethanol and 15 mL of deionized (DI) water and then heated under reflux for 24 hours. The temperature of the resulting product obtained therefrom was lowered to room temperature, and the resulting solids were filtered and then sufficiently washed with water/methanol/hexane in the stated order. The solids obtained without separate filtration were dried in a vacuum oven, to thereby obtain 0.9 g of L1 dimer.

Synthesis of Compound 1

[0266] 0.9 g (0.48 mmol) of L1 dimer, 0.12 g (1.2 mmol) of acetylacetone, and 0.17 g (1.2 mmol) of K₂CO₃ were mixed with 30 mL of ethoxyethanol and then stirred at 110° C. for 18 hours. After the reaction was completed, the temperature was lowered to room temperature, and the

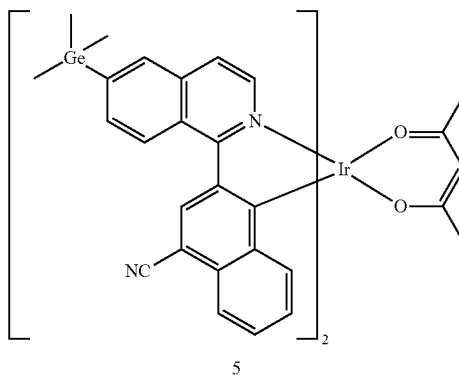
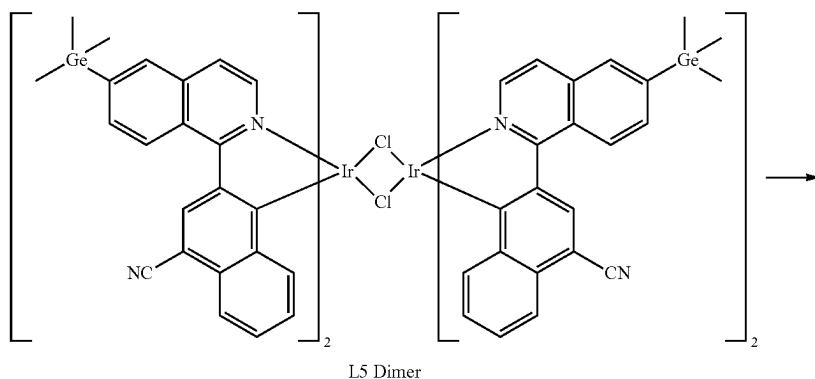
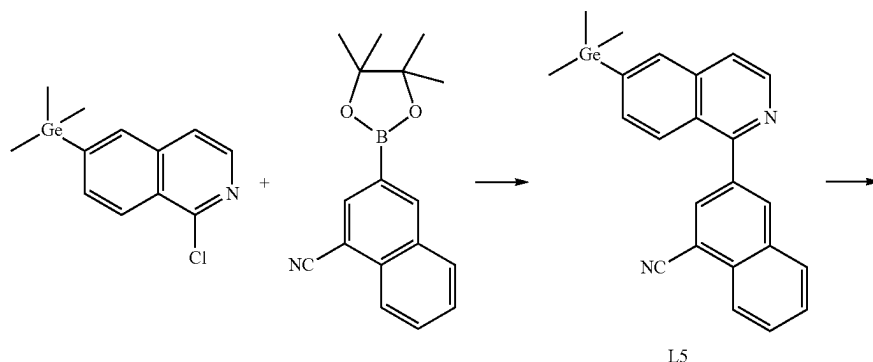
resulting solids were filtered. The product was purified by liquid chromatography, to thereby obtain 0.5 g (yield of 50%) of Compound 1.

[0267] LC-MS $m/z=995$ (M+H)⁺.

Synthesis Example 2: Synthesis of Compound 5

Synthesis of L5 Dimer

[0270] The L5 dimer was synthesized in a similar manner as used to synthesize L1 dimer of Synthesis Example 1, except that Intermediate L5 was used instead of Intermediate L1.



Synthesis of Intermediate L5

[0268] 2.2 g (yield of 78%) of Intermediate L9 was synthesized in a similar manner as used to synthesize Intermediate L2 of Synthesis Example 1, except that 1-chloro-6-(trimethylgermyl)isoquinoline was used instead of 1-chloro-6-(trimethylsilyl)isoquinoline.

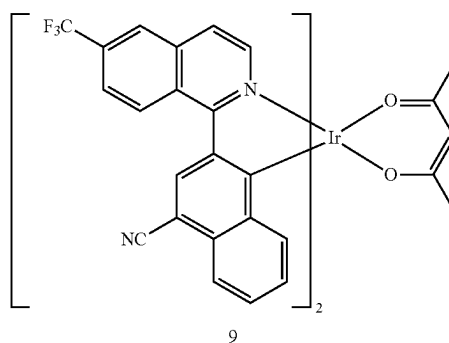
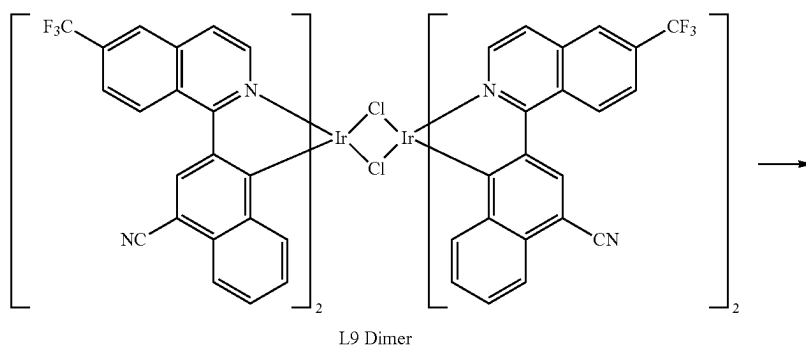
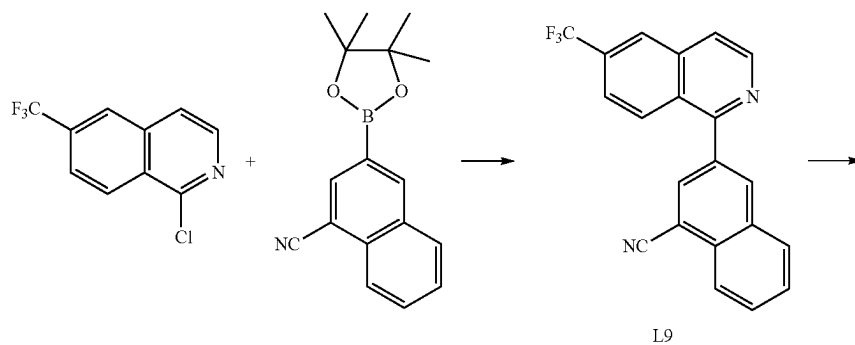
[0269] LC-MS $m/z=399$ (M+H)⁺.

Synthesis of Compound 5

[0271] 5 g (yield of 48%) of Compound 5 was synthesized in a similar manner as used to synthesize Compound 1 of Synthesis Example 1, except that L5 dimer was used instead of L1 dimer.

[0272] LC-MS $m/z=1087$ (M+H)⁺.

Synthesis Example 3: Synthesis of Compound 9



Synthesis of Intermediate L9

[0273] 1.3 g (yield of 90%) of Intermediate L9 was synthesized in a similar manner as used to synthesize Intermediate L1 of Synthesis Example 1, except that 1-chloro-6-(trifluoromethyl)isoquinoline was used instead of 1-chloro-6-(trimethylsilyl)isoquinoline, and ethanol was used instead of THE as a solvent.

[0274] LC-MS $m/z=349$ ($M+H$)⁺.

Synthesis of L9 Dimer

[0275] The L9 dimer was synthesized in a similar manner as used to synthesize L1 dimer of Synthesis Example 1,

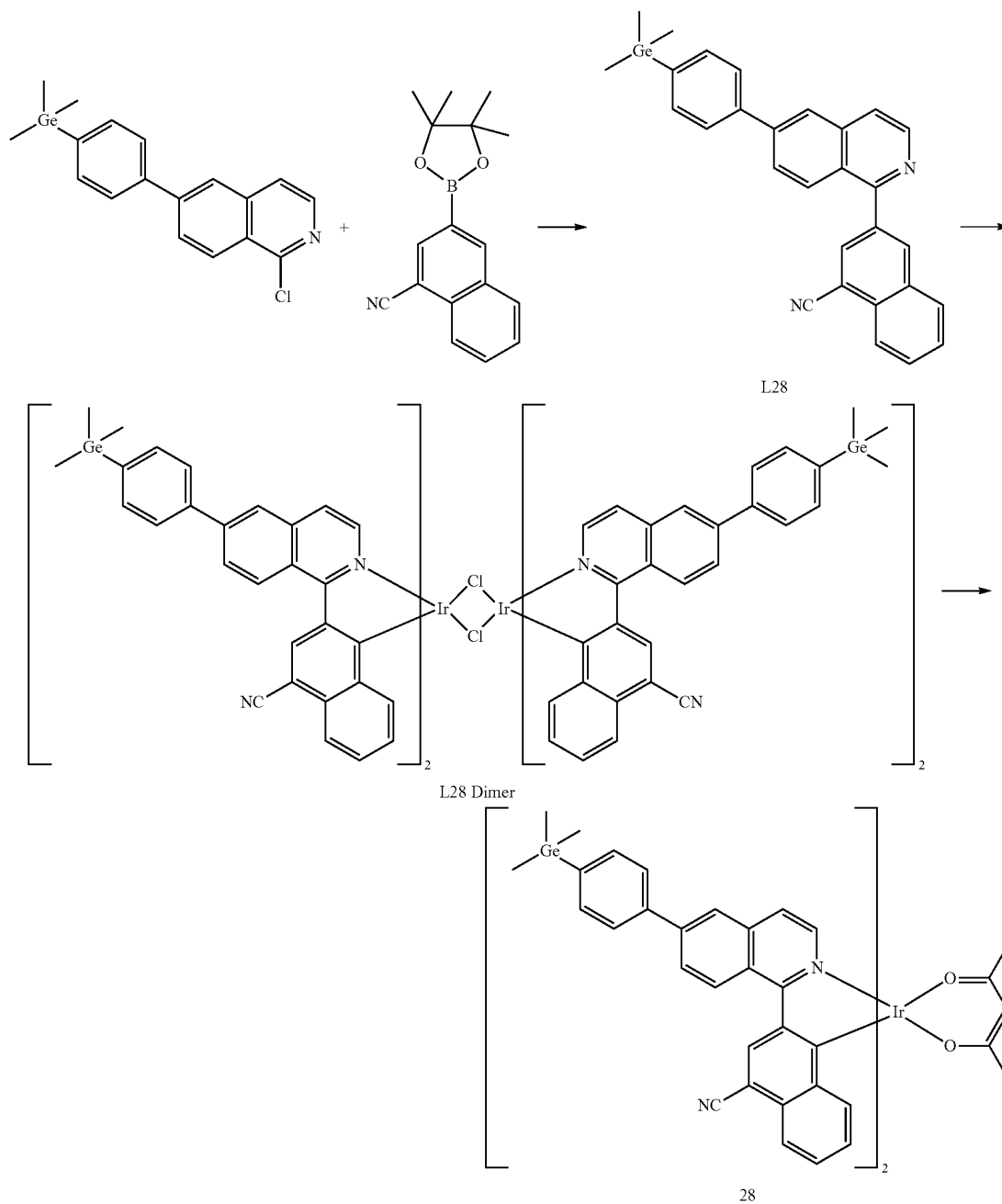
except that Intermediate L9 was used instead of Intermediate L1.

Synthesis of Compound 9

[0276] 0.3 g (yield of 35%) of Compound 9 was synthesized in a similar manner as used to synthesize Compound 1 of Synthesis Example 1, except that the L9 dimer was used instead of the L1 dimer.

[0277] LC-MS $m/z=987$ ($M+H$)⁺.

Synthesis Example 4: Synthesis of Compound 28



Synthesis of Intermediate L28

[0278] 0.9 g (yield of 75%) of Intermediate L28 was synthesized in a similar manner as used to synthesize Intermediate L1 of Synthesis Example 1, except that 1-chloro-6-(4-(trimethylgermyl)phenyl)isoquinoline was used instead of 4-chloro-7-methyl benzo[c][2,7]naphthylidene.

[0279] LC-MS $m/z=474$ ($M+H$)⁺.

Synthesis of L28 Dimer

[0280] The L28 dimer was synthesized in a similar manner as used to synthesize L1 dimer of Synthesis Example 1, except that Intermediate L28 was used instead of Intermediate L1.

Synthesis of Compound 28

[0281] 0.2 g (yield of 30%) of Compound 28 was synthesized in a similar manner as used to synthesize Compound

1 of Synthesis Example 1, except that the L28 dimer was used instead of the L1 dimer.

[0282] LC-MS $m/z=1237$ (M+H)⁺.

Example 1

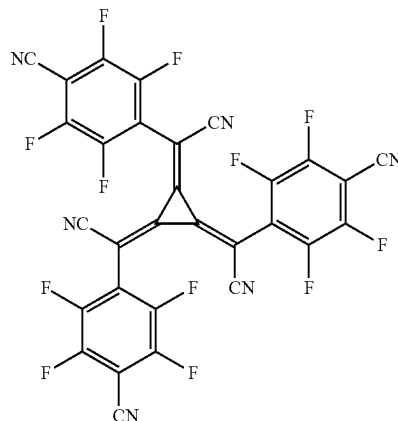
[0283] As an anode, an ITO-patterned glass substrate was cut to a size of 50 millimeters (mm)×50 mm×0.5 mm, sonicated with isopropyl alcohol and DI water, each for 5 minutes, and then cleaned by irradiation of ultraviolet rays and exposure of ozone thereto for 30 minutes. The resultant glass substrate was loaded onto a vacuum deposition apparatus.

[0284] Compound HT3 and F12-P-Dopant were vacuum co-deposited on the anode at a weight ratio of 98:2 to form a hole injection layer having a thickness of 100 Å, and Compound HT3 was vacuum deposited on the hole injection layer to form a hole transport layer having a thickness of 1,600 Å.

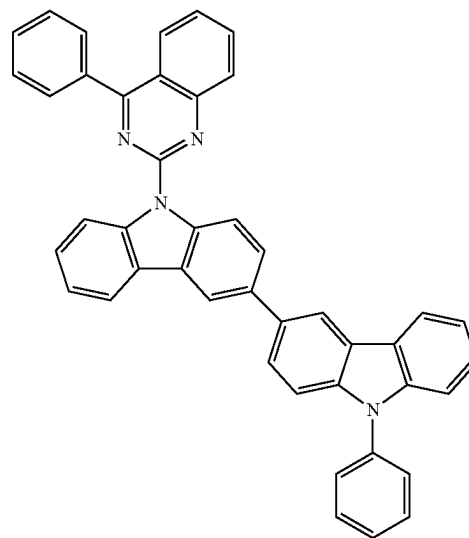
[0285] Next, Compound RH3 (host) and Compound 1 (dopant) were co-deposited on the hole transport layer at a weight ratio of 97:3 to form an emission layer having a thickness of 400 Å.

[0286] Afterwards, Compound ETL and Liq-N-Dopant were co-deposited on the emission layer at a volume ratio of 50:50 to form an electron transport layer having a thickness of 350 Å, Liq-N-Dopant was vacuum deposited on the electron transport layer to form an electron injection layer having a thickness of 10 Å, and Al was vacuum deposited on the electron injection layer to form a cathode having a thickness of 1,000 Å, thereby completing manufacture of an organic light-emitting device.

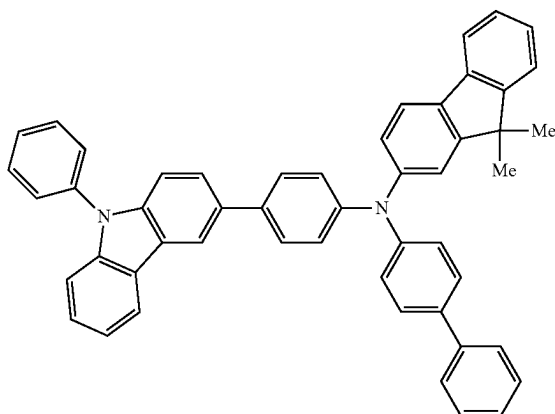
-continued



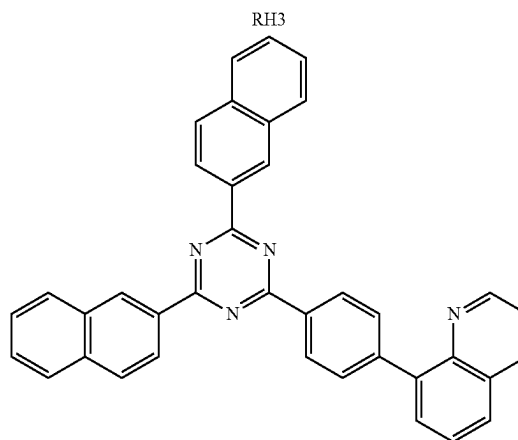
F12-P-Dopant



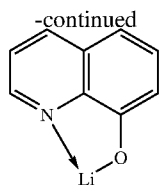
RH3



HT3



ETL



Liq-N-Dopant

Examples 2 to 4 and Comparative Examples 1 to 4
[0287] Organic light-emitting devices were manufactured in a similar manner as in Example 1, except that, in forming the emission layer, each of compounds shown in Table 1 was used instead of Compound 1.

Evaluation Example 1: Evaluation of
 Characteristics of Organic Light-Emitting Devices

[0288] For the organic light-emitting devices of Examples 1 to 4 and Comparative Examples 1 to 4, driving voltage (volts, V), maximum value of external quantum efficiency (Max EQE, %), maximum emission wavelength (nm), full width at half maximum (FWHM, nm), and roll-off ratio (%) were evaluated, and the results thereof are shown in Table 1. A current voltmeter (Keithley 2400) and a luminance meter (Minolta Cs-1000A) were used as evaluation apparatuses. The roll-off ratio was calculated according to Equation 20.

$$\text{Roll-off ratio} = \{1 - (\text{efficiency (at } 3,500 \text{ cd/m}^2)/\text{maximum luminescence efficiency})\} \times 100\% \quad \text{Equation 20}$$

TABLE 1

	Dopant compound	Driving voltage (V)	Max EQE (%)	Maximum emission wavelength (nm)	FWHM (nm)	Roll-Off (%)
Example 1	1	4.5	25	619	36	10
Example 2	5	4.5	25	618	36	10
Example 3	9	4.5	25	640	37	11
Example 4	28	4.6	27	622	39	12
Comparative Example 1	CE1	4.8	23	610	35	13
Comparative Example 2	CE2	4.9	23	650	36	14
Comparative Example 3	CE3	4.8	24	656	37	13
Comparative Example 4	CE4	4.8	20	651	38	15

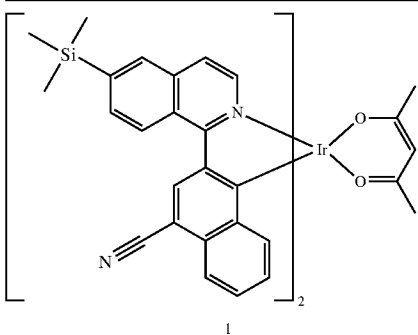
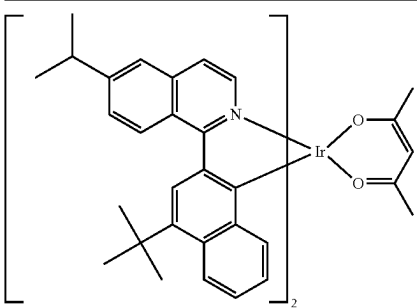
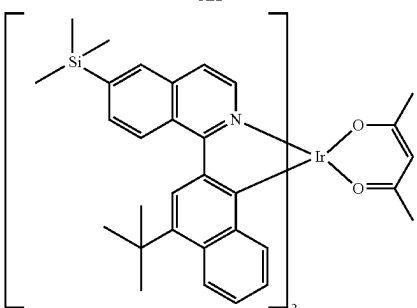
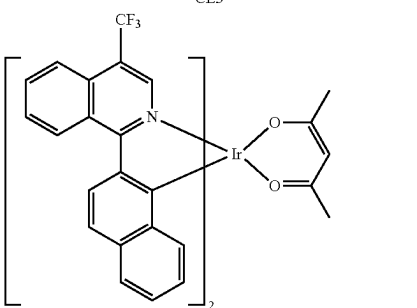


TABLE 1-continued

Dopant compound	Driving voltage (V)	Max EQE (%)	Maximum emission wavelength (nm)	FWHM (nm)	Roll-Off (%)
5					
9					
28					
CE1					

TABLE 1-continued

Dopant compound	Driving voltage (V)	Max EQE (%)	Maximum emission wavelength (nm)	FWHM (nm)	Roll-Off (%)
					
CE2					
					
CE3					
					
CE4					

[0289] From Table 1, it was confirmed that each of the organic light-emitting devices of Examples 1 to 4 had a low driving voltage, a high maximum external quantum efficiency, a narrow FWHM, and a low roll-off ratio.

[0290] In addition, it was confirmed that the organic light-emitting devices of Examples 1 to 4 exhibited a lower driving voltage, an equal or greater maximum external quantum efficiency, and a lower roll-off ratio, compared to the organic light-emitting devices of Comparative Examples 1 to 4.

[0291] The organometallic compound represented by Formula 1 has excellent electrical characteristics, and thus, an electronic device, e.g., an organic light-emitting device, including at least one of the organometallic compounds represented by Formula 1 may have a low driving voltage, a high maximum external quantum efficiency, a narrow FWHM, and a low roll-off ratio. Therefore, a high-quality organic light-emitting device may be embodied by using the organometallic compound represented by Formula 1.

[0292] It should be understood that embodiments described herein should be considered in a descriptive sense only and not for purposes of limitation. Descriptions of features or aspects within each embodiment should typically be considered as available for other similar features or aspects in other embodiments. While one or more embodiments have been described with reference to the figure, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope as defined by the following claims.

What is claimed is:

1. An organometallic compound represented by Formula 1:



wherein, in Formula 1,

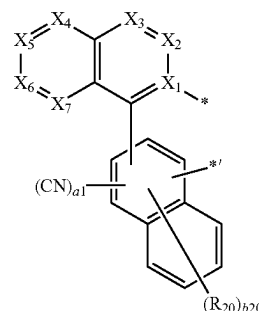
M_1 is a transition metal,

L_1 is a ligand represented by Formula 1A,

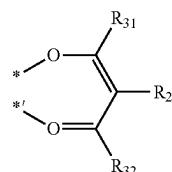
L_2 is a ligand represented by Formula 1B, and

$n1$ and $n2$ are each independently 1 or 2,

Formula 1A



Formula 1B



wherein, in Formulae 1A and 1B,

X_1 is C or N,

X_2 is C(R_2) or N, X_3 is C(R_3) or N, X_4 is C(R_4) or N, X_5 is C(R_5) or N, X_6 is C(R_6) or N, and X_7 is C(R_7) or N, wherein at least one of X_2 to X_7 is not N,

$a1$ is 1 or 2,

R_2 to R_7 , R_{20} , and R_{31} to R_{33} are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, —SF₅, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C₁-C₆₀ alkyl group, a substituted or unsubstituted C₂-C₆₀ alkenyl group, a substituted or unsubstituted C₂-C₆₀ alkynyl group, a substituted or unsubstituted C₁-C₆₀ alkoxy group, a substituted or unsubstituted C₁-C₆₀ alkylthio group, a substituted or unsubstituted C₃-C₁₀ cycloalkyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkyl group, a substituted or unsubstituted C₃-C₁₀ cycloalkenyl group, a substituted or unsubstituted

C₁-C₁₀ heterocycloalkenyl group, a substituted or unsubstituted C₆-C₆₀ aryl group, a substituted or unsubstituted C₇-C₆₀ alkyl aryl group, a substituted or unsubstituted C₇-C₆₀ aryl alkyl group, a substituted or unsubstituted C₆-C₆₀ aryloxy group, a substituted or unsubstituted C₆-C₆₀ arylthio group, a substituted or unsubstituted C₁-C₆₀ heteroaryl group, a substituted or unsubstituted C₂-C₆₀ alkyl heteroaryl group, a substituted or unsubstituted C₂-C₆₀ heteroaryl alkyl group, a substituted or unsubstituted C₁-C₆₀ heteroaryloxy group, a substituted or unsubstituted C₁-C₆₀ heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q₁)(Q₂)(Q₃), —Ge(Q₁)(Q₂)(Q₃), —N(Q₄)(Q₅), —B(Q₆)(Q₇), —P(Q₈)(Q₉), or —P(=O)(Q₈)(Q₉), at least one of R₂ to R₇ is:

—F, —Si(Q₁)(Q₂)(Q₃), or —Ge(Q₁)(Q₂)(Q₃); or

a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₆-C₆₀ aryl group, or a C₁-C₆₀ heteroaryl group, each substituted with at least one of —F,

—Si(Q₃₁)(Q₃₂)(Q₃₃), —Ge(Q₃₁)(Q₃₂)(Q₃₃), or a combination thereof,

b20 is an integer from 1 to 5,

two or more of R₂ to R₇ are optionally linked to each other to form a C₅-C₃₀ carbocyclic group that is unsubstituted or substituted with at least one R_{10a}, or two or more of R₂ to R₇ are optionally linked to each other to form a C₁-C₃₀ heterocyclic group that is unsubstituted or substituted with at least one R_{10a},

two or more of a plurality of R₂₀ are optionally linked to each other to form a C₅-C₃₀ carbocyclic group that is unsubstituted or substituted with at least one R_{10a}, or two or more of the plurality of R₂₀ are optionally linked to each other to form a C₁-C₃₀ heterocyclic group that is unsubstituted or substituted with at least one R_{10a},

R_{10a} is as defined for R₁,

* and * each indicate a binding site to M₁,

a substituent of the substituted C₅-C₃₀ carbocyclic group, the substituted C₁-C₃₀ heterocyclic group, the substituted C₁-C₆₀ alkyl group, the substituted C₂-C₆₀ alkenyl group, the substituted C₂-C₆₀ alkynyl group, the substituted C₁-C₆₀ alkoxy group, the substituted C₁-C₆₀ alkylthio group, the substituted C₃-C₁₀ cycloalkyl group, the substituted C₁-C₁₀ heterocycloalkyl group, the substituted C₃-C₁₀ cycloalkenyl group, the substituted C₁-C₁ heterocycloalkenyl group, the substituted C₆-C₆₀ aryl group, the substituted C₇-C₆₀ alkyl aryl group, the substituted C₇-C₆₀ aryl alkyl group, the substituted C₆-C₆₀ aryloxy group, the substituted C₆-C₆₀ arylthio group, the substituted C₁-C₆₀ heteroaryl group, the substituted C₂-C₆₀ alkyl heteroaryl group, the substituted C₂-C₆₀ heteroaryl alkyl group, the substituted C₁-C₆₀ heteroaryloxy group, the substituted C₁-C₆₀ heteroarylthio group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed heteropolycyclic group is each independently:

deuterium, —F, —Cl, —Br, —I, —SF₅, —CD₃, —CD₂H, —CDH₂, —CF₃, —CF₂H, —CFH₂, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, or a C₁-C₆₀ alkylthio group;

a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, or a C₁-C₆₀ alkylthio group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF₅, —CD₃, —CD₂H, —CDH₂, —CF₃, —CF₂H, —CFH₂, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₇-C₆₀ alkyl aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a C₂-C₆₀ alkyl heteroaryl group, a C₁-C₆₀ heteroaryloxy group, a C₁-C₆₀ heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q₁₁)(Q₁₂)(Q₁₃), —Ge(Q₁₁)(Q₁₂)(Q₁₃), —N(Q₁₄)(Q₁₅), —B(Q₁₆)(Q₁₇), —P(Q₁₈)(Q₁₉), —P(=O)(Q₁₈)(Q₁₉), or a combination thereof;

a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₇-C₆₀ alkyl aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a C₂-C₆₀ alkyl heteroaryl group, a C₁-C₆₀ heteroaryloxy group, a C₁-C₆₀ heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, or a monovalent non-aromatic condensed heteropolycyclic group;

a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₇-C₆₀ alkyl aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a C₂-C₆₀ alkyl heteroaryl group, a C₁-C₆₀ heteroaryloxy group, a C₁-C₆₀ heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, or a monovalent non-aromatic condensed heteropolycyclic group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF₅, —CD₃, —CD₂H, —CDH₂, —CF₃, —CF₂H, —CFH₂, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, a C₁-C₆₀ alkylthio group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl

group, a C₇-C₆₀ alkyl aryl group, a C₇-C₆₀ aryl alkyl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a C₂-C₆₀ alkyl heteroaryl group, a C₂-C₆₀ heteroaryl alkyl group, a C₁-C₆₀ heteroaryloxy group, a C₁-C₆₀ heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q₂₁)(Q₂₂)(Q₂₃), —Ge(Q₂₁)(Q₂₂)(Q₂₃), —N(Q₂₄)(Q₂₅), —B(Q₂₆)(Q₂₇), —P(Q₂₈)(Q₂₉), —P(=O)(Q₂₈)(Q₂₉), or a combination thereof; or

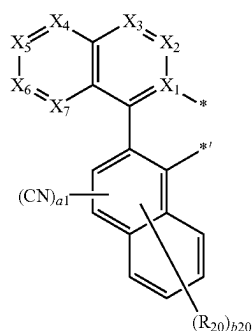
—Si(Q₃₁)(Q₃₂)(Q₃₃), —Ge(Q₃₁)(Q₃₂)(Q₃₃), —N(Q₃₄)(Q₃₅), —B(Q₃₆)(Q₃₇), —P(Q₃₈)(Q₃₉), or —P(=O)(Q₃₈)(Q₃₉), and

Q₁ to Q₉, Q₁₁ to Q₁₉, Q₂₁ to Q₂₉, and Q₃₁ to Q₃₉ are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, —SF₅, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C₁-C₆₀ alkyl group, a substituted or unsubstituted C₂-C₆₀ alkenyl group, a substituted or unsubstituted C₂-C₆₀ alkynyl group, a substituted or unsubstituted C₁-C₆₀ alkoxy group, a substituted or unsubstituted C₁-C₆₀ alkylthio group, a substituted or unsubstituted C₃-C₁₀ cycloalkyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkyl group, a substituted or unsubstituted C₃-C₁₀ cycloalkenyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkenyl group, a substituted or unsubstituted C₆-C₆₀ aryl group, a substituted or unsubstituted C₇-C₆₀ alkyl aryl group, a substituted or unsubstituted C₇-C₆₀ aryl alkyl group, a substituted or unsubstituted C₆-C₆₀ aryloxy group, a substituted or unsubstituted C₆-C₆₀ arylthio group, a substituted or unsubstituted C₁-C₆₀ heteroaryl group, a substituted or unsubstituted C₂-C₆₀ alkyl heteroaryl group, a substituted or unsubstituted C₂-C₆₀ heteroaryl alkyl group, a substituted or unsubstituted C₁-C₆₀ heteroaryloxy group, a substituted or unsubstituted C₁-C₆₀ heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, or a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group.

2. The organometallic compound of claim 1, wherein M₁ is Ir, and

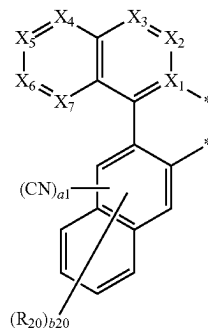
a sum of n1 and n2 is 3.

3. The organometallic compound of claim 1, wherein L₁ is a ligand represented by one of Formulae 1A-1 to 1A-3:

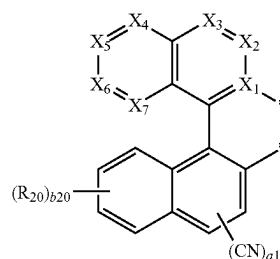


1A-1

-continued



1A-2



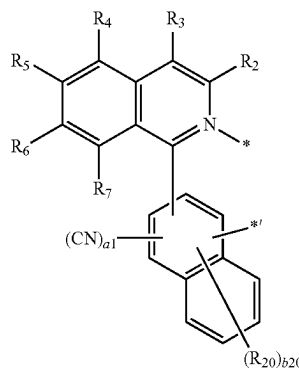
1A-3

wherein, in Formulae 1A-1 to 1A-3,

X₁ to X₇, a1, R₂₀, and b20 are each as described in claim 1, and

* and *' each indicate a binding site to M₁.

4. The organometallic compound of claim 1, wherein L₁ is a ligand represented by Formula 1A(1):



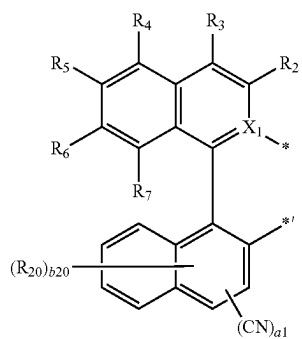
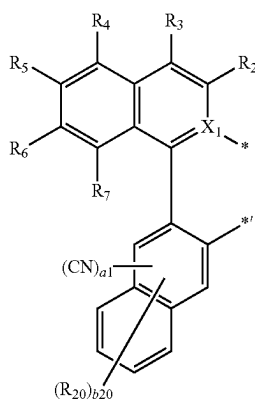
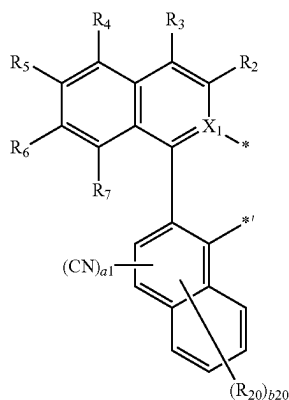
Formula 1A(1)

wherein, in Formula 1A(1),

R₂ to R₇, a1, R₂₀, and b20 are each as described in claim 1, and

* and *' each indicate a binding site to M₁.

5. The organometallic compound of claim 1, wherein L_1 is a ligand represented by one of Formulae 1A(1)-1 to 1A(1)-3:



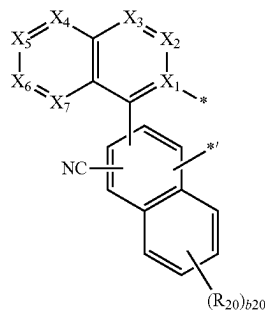
wherein, in Formulae 1A(1)-1 to 1A(1)-3,

X_1 , R_2 to R_7 , a_1 , R_{20} , and b_{20} are each as described in claim 1, and

* and *' each indicate a binding site to M_1 .

6. The organometallic compound of claim 1, wherein L_1 is a ligand represented by Formula 1A(2):

Formula 1A(2)

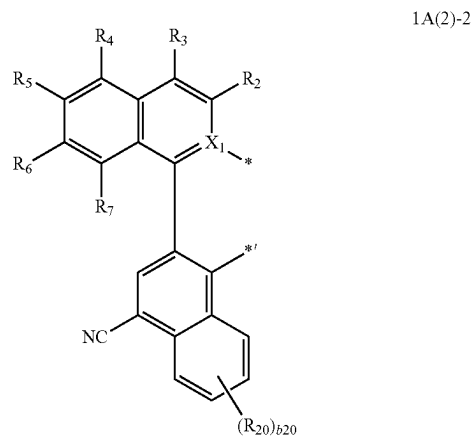
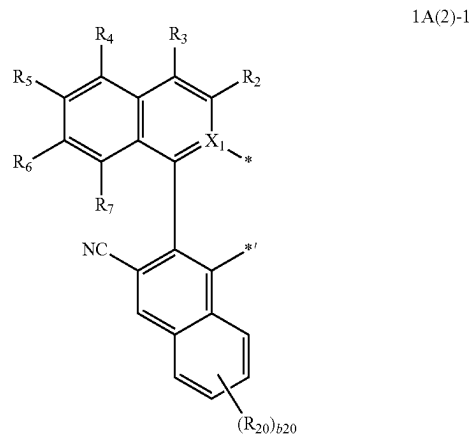


wherein, in Formula 1A(2),

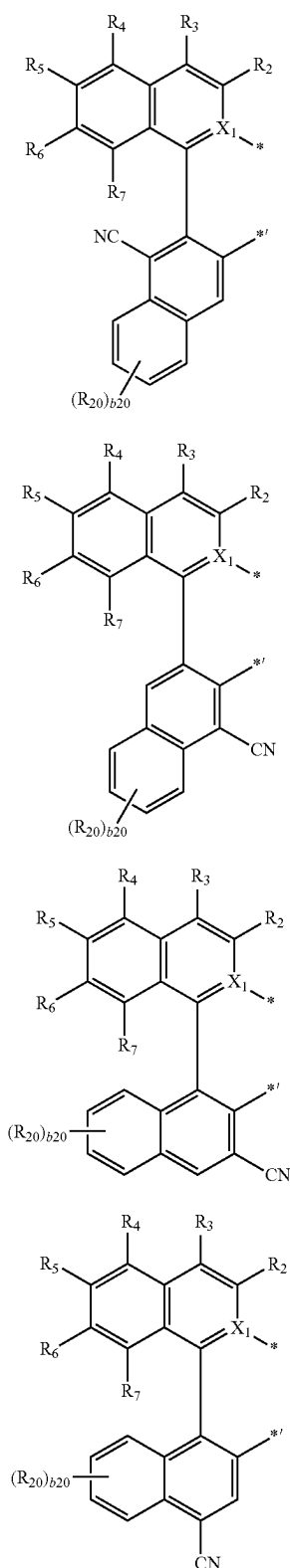
X_1 to X_7 , R_{20} , and b_{20} are each as described in claim 1, and

* and *' each indicate a binding site to M_1 .

7. The organometallic compound of claim 1, wherein L_1 is a ligand represented by one of Formulae 1A(2)-1 to 1A(2)-6:



-continued



wherein, in Formulae 1A(2)-1 to 1A(2)-6,

X₁, R₂ to R₇, a₁, R₂₀, and b₂₀ are each as described in claim 1, and* and *' each indicate a binding site to M₁.8. The organometallic compound of claim 1, wherein R₂ to R₇, R₂₀, and R₃₁ to R₃₃ are each independently:

hydrogen, deuterium, —F, —Cl, —Br, —I, —SF₅, —CD₃, —CD₂H, —CDH₂, —CF₃, —CF₂H, —CFH₂, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, or a C₁-C₂₀ alkylthio group;

a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, or a C₁-C₂₀ alkylthio group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF₅, —CD₃, —CD₂H, —CDH₂, —CF₃, —CF₂H, —CFH₂, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C₁-C₁₀ alkyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a pyridinyl group, a pyrimidinyl group, or a combination thereof;

a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazoyl group, a phenanthrolinyl group, a benzoimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazoyl group, a dibenzocarbazoyl group, an imidazopyridinyl group, or an imidazopyrimidinyl group;

a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a

pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzoimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, or an imidazopyrimidinyl group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF₅, —CD₃, —CD₂H, —CDH₂, —CF₃, —CF₂H, —CFH₂, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a C₁-C₂₀ alkylthio group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzoimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, or a combination thereof; or —Si(Q₁)(Q₂)(Q₃), —Ge(Q₁)(Q₂)(Q₃), —N(Q₄)(Q₅), —B(Q₆)(Q₇), —P(Q₈)(Q₉), or —P(=O)(Q₈)(Q₉), and

Q₁ to Q₉ are each independently:

—CH₃, —CD₃, —CD₂H, —CDH₂, —CH₂CH₃, —CH₂CD₃, —CH₂CD₂H, —CH₂CDH₂, —CHDCH₃, —CHDCD₂H, —CHDCDH₂, —CHDCD₃, —CD₂CD₃, —CD₂CD₂H, or —CD₂CDH₂;

an n-propyl group, an iso-propyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, an n-pentyl group, an isopentyl group, a sec-pentyl group, a tert-pentyl group, a phenyl group, or a naphthyl group; or

an n-propyl group, an iso-propyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group,

an n-pentyl group, an isopentyl group, a sec-pentyl group, a tert-pentyl group, a phenyl group, or a naphthyl group, each substituted with at least one of deuterium, a C₁-C₁₀ alkyl group, a phenyl group, or a combination thereof.

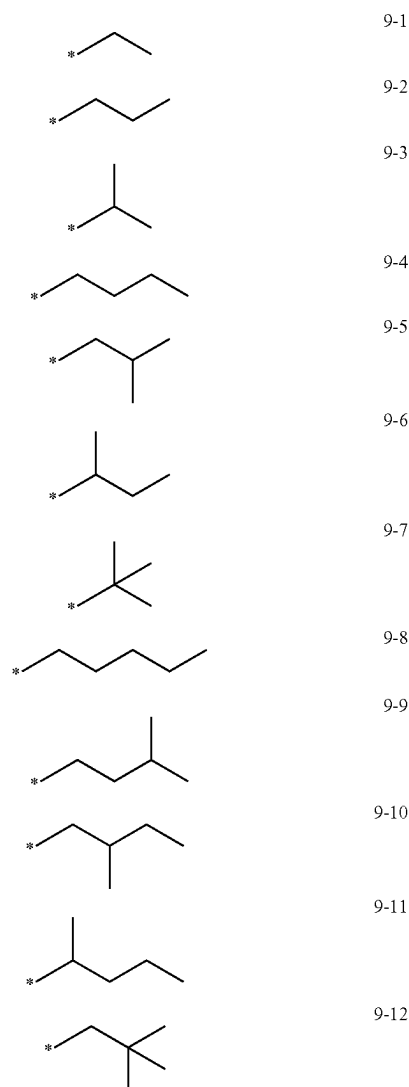
9. The organometallic compound of claim 1, wherein R₂ to R₇, R₂₀, and R₃₁ to R₃₃ are each independently:

hydrogen, deuterium, —F, —Cl, —Br, —I, —SF₅, —CD₃, —CD₂H, —CDH₂, —CF₃, —CF₂H, —CFH₂, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, or a C₁-C₆₀ alkylthio group; or

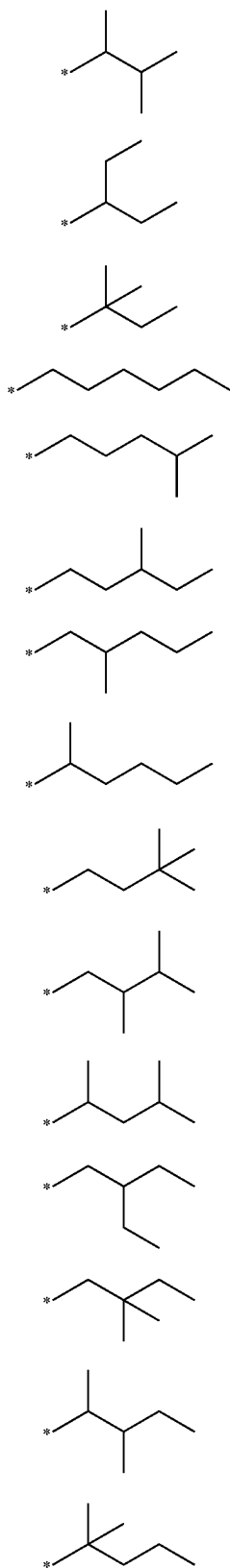
a group represented by any one of Formulae 9-1 to 9-39, 9-44 to 9-61, 9-201 to 9-237, 10-1 to 10-129, or 10-201 to 10-353; or

—Si(Q₁)(Q₂)(Q₃), —Ge(Q₁)(Q₂)(Q₃), or —N(Q₄)(Q₅), and

Q₁ to Q₅ are each as described in claim 1:



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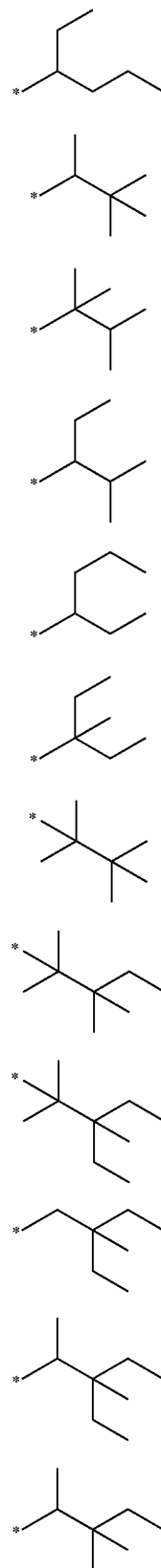
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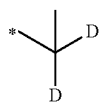
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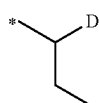
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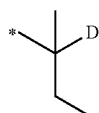
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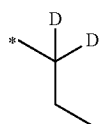
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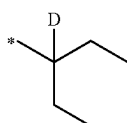
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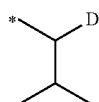
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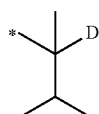
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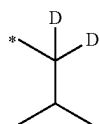
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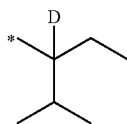
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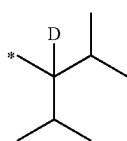
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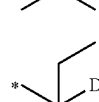
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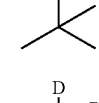
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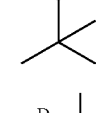
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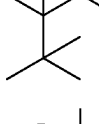
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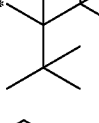
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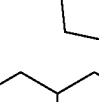
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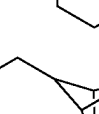
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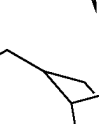
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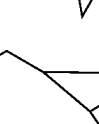
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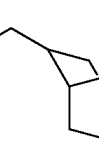
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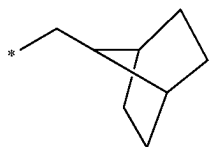


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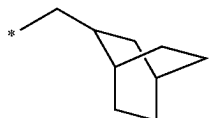


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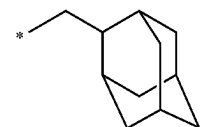
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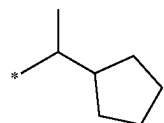
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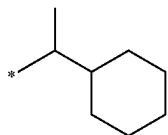
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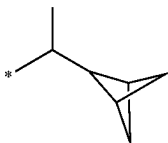
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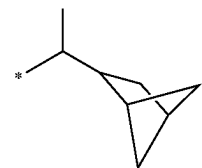
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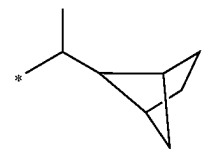
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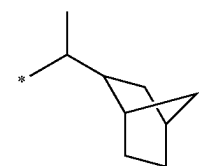
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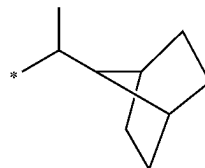


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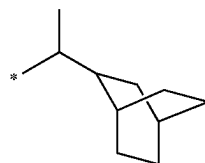


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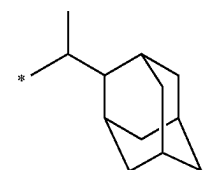
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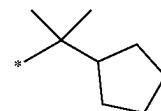
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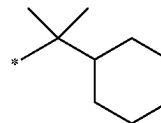
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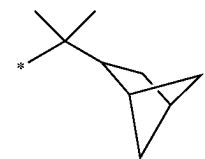
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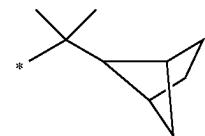
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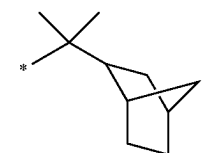
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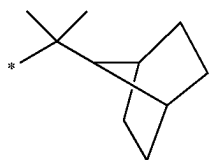


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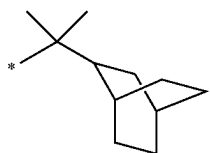


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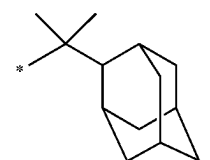
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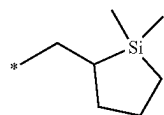
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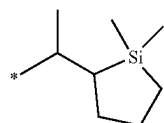
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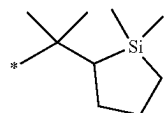
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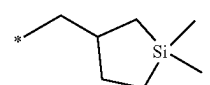
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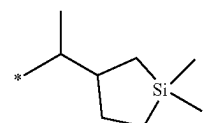
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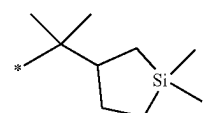
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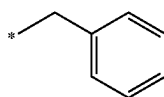
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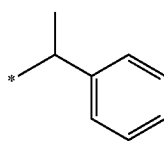
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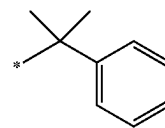


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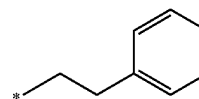


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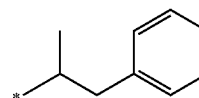
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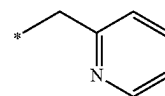
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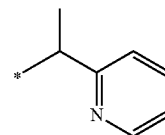
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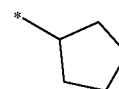
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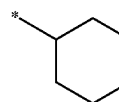
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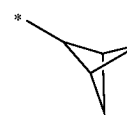
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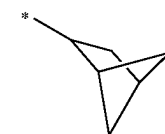
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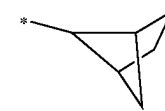
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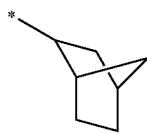
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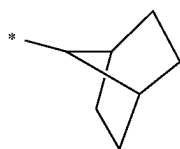


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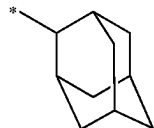
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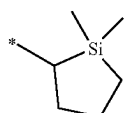
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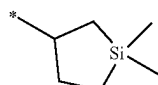
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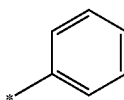
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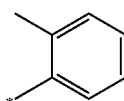
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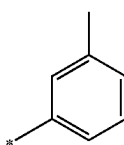
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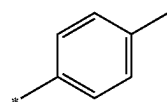
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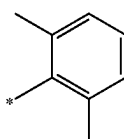
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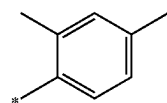
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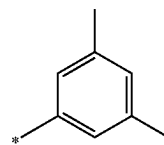


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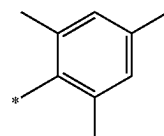


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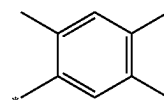
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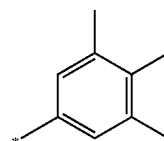
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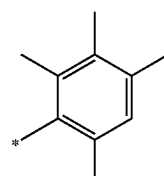
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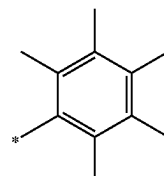
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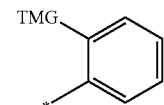
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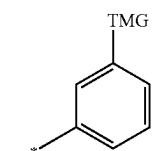
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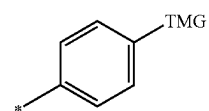
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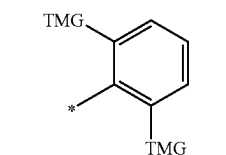


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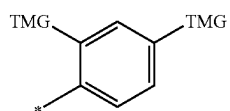


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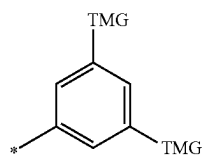
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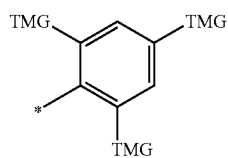
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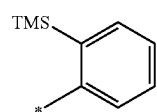
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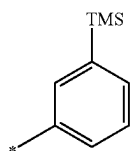
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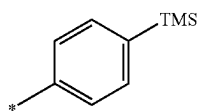
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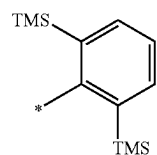
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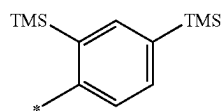
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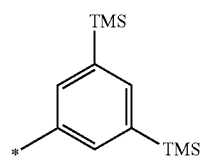
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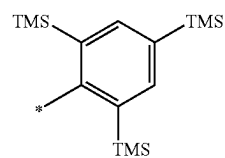


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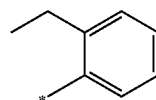


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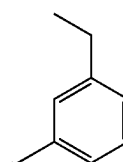
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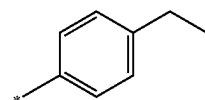
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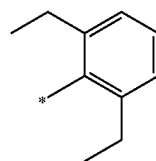
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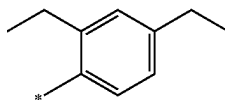
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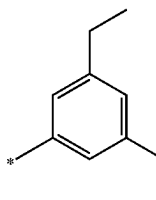
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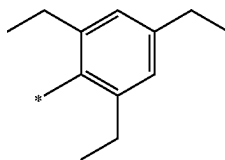
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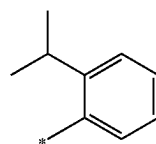
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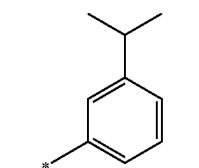


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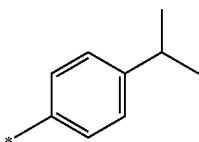


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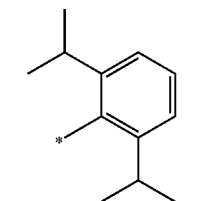
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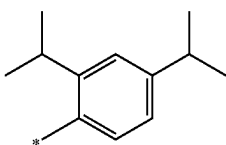
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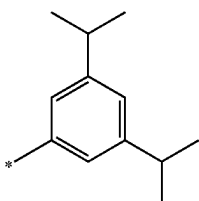
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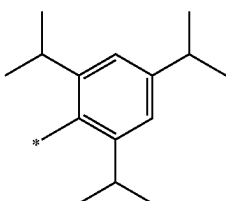
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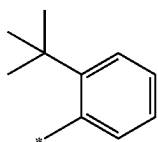
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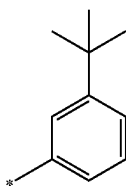
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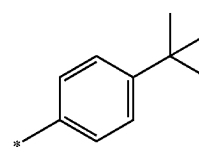


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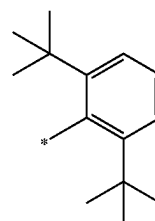


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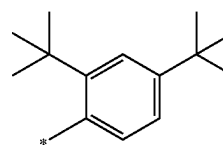
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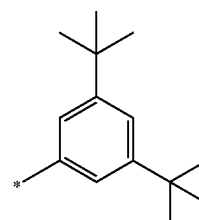
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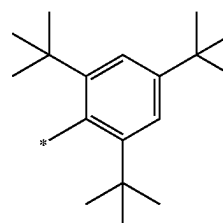
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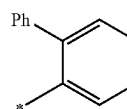
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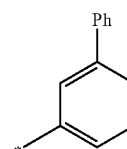
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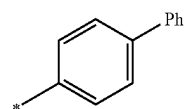
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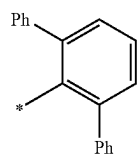


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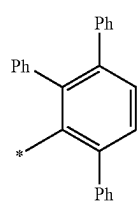


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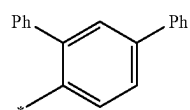
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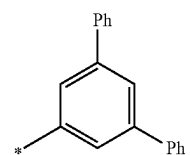
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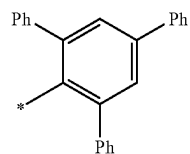
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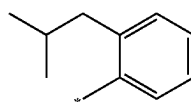
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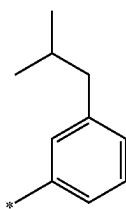
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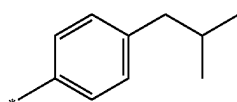
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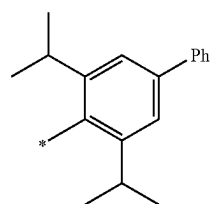
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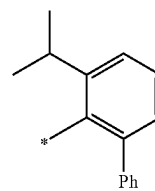


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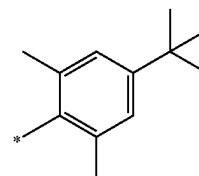


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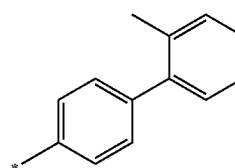
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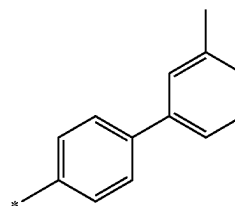
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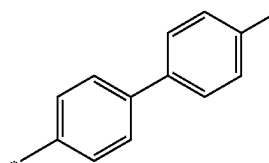
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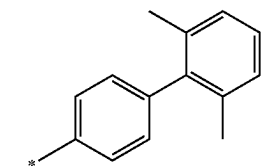
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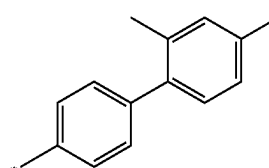
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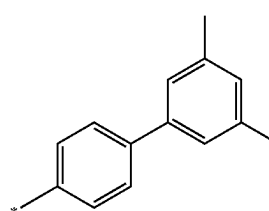
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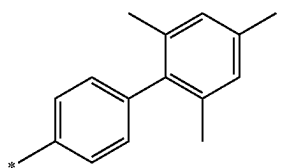


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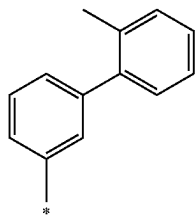


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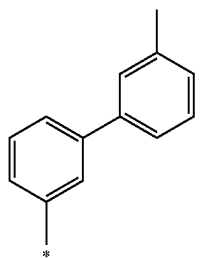
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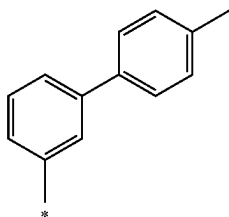
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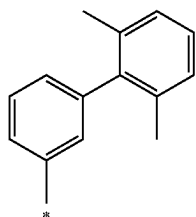
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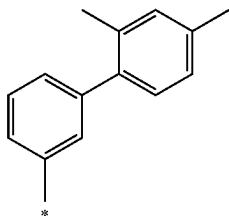
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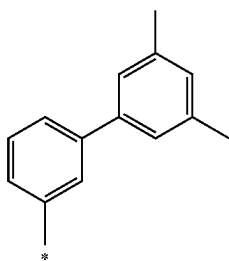
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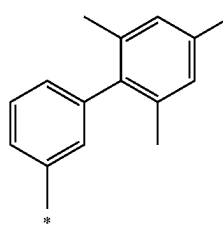


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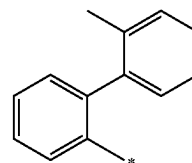


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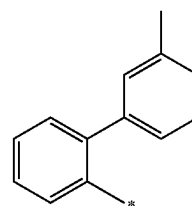
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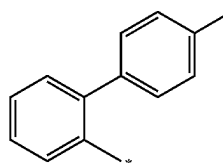
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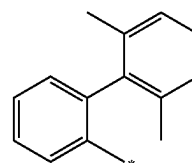
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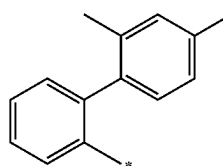
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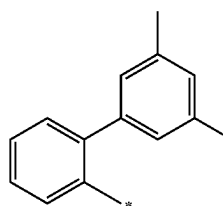
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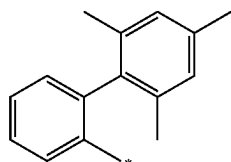


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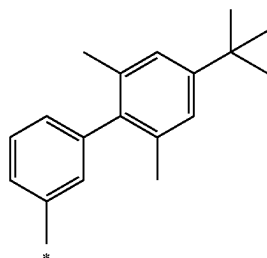


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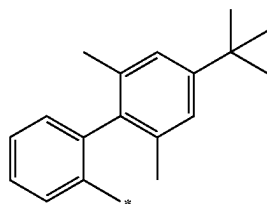
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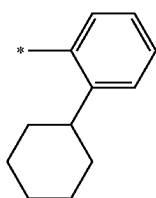
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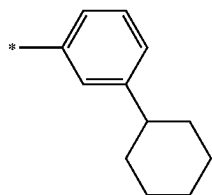
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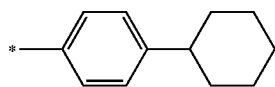
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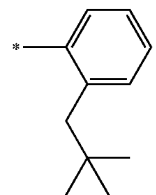
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10-97

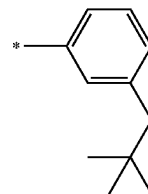


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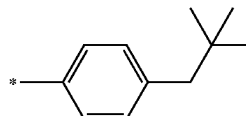


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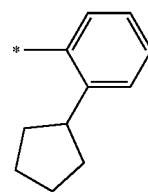
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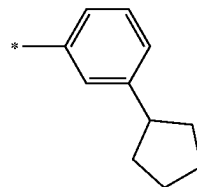
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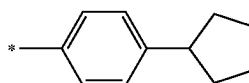
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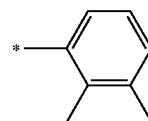
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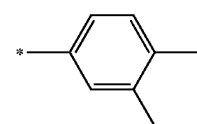
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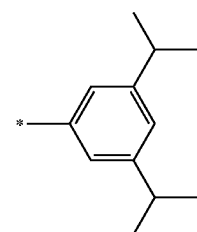
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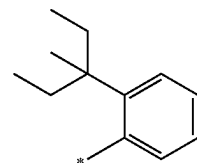
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10-106

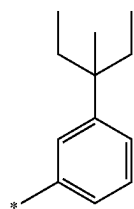


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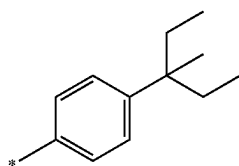


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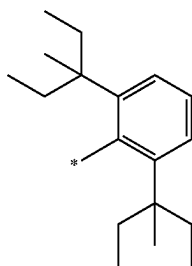
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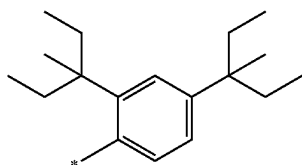
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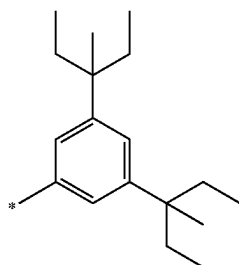
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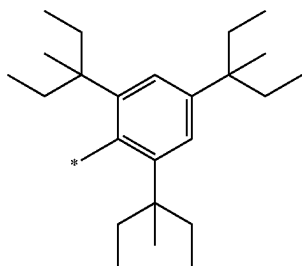
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10-112

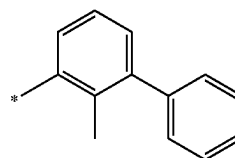


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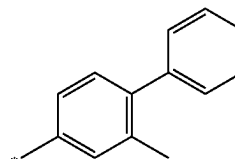


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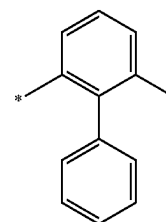
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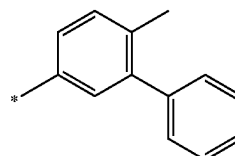
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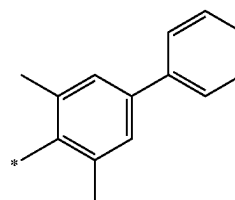
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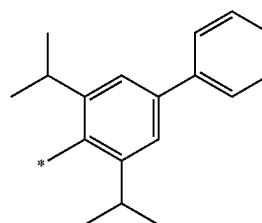
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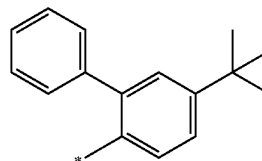
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10-119

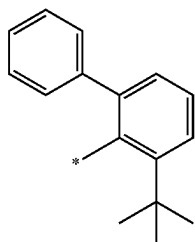


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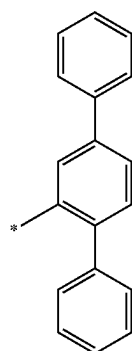


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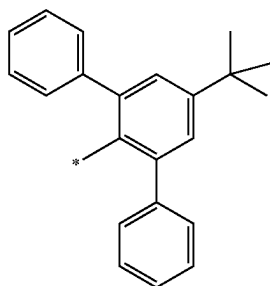
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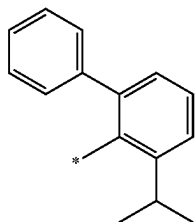
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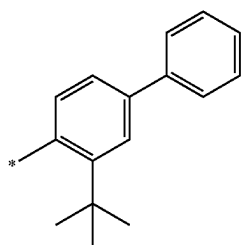
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10-124

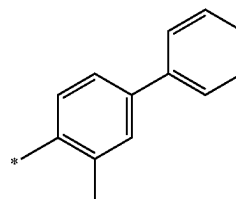


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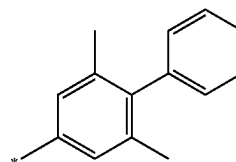


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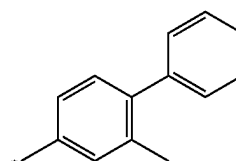
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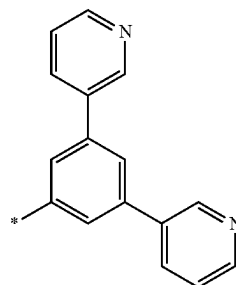
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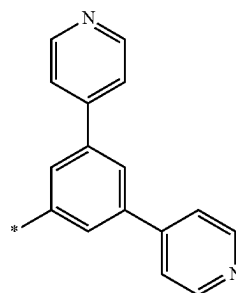
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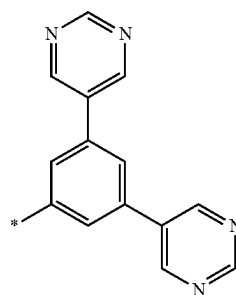
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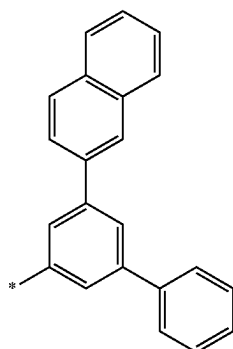


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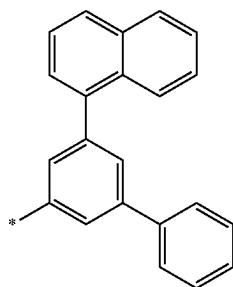


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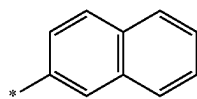
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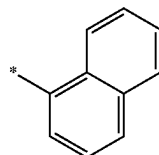
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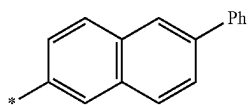
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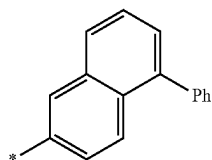
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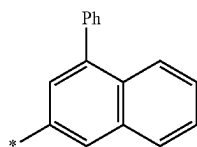
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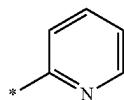
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10-209

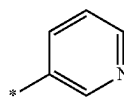


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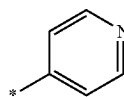


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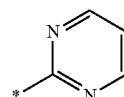
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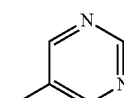
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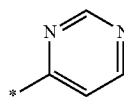
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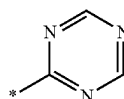
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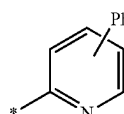
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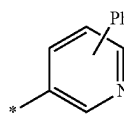
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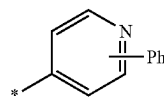
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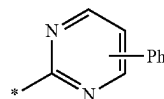
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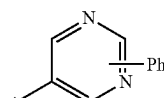
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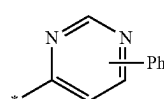
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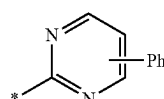
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10-222

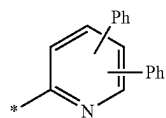


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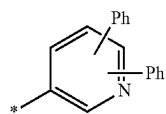


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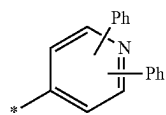
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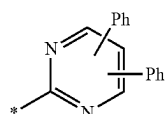
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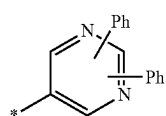
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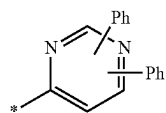
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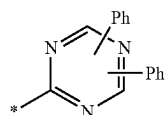
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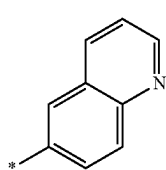
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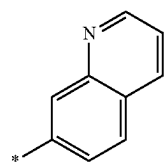
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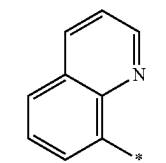
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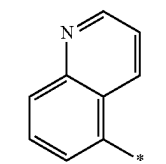
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10-233

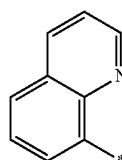


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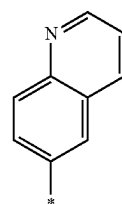


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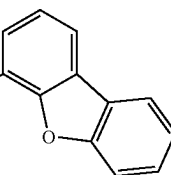
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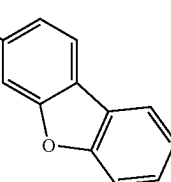
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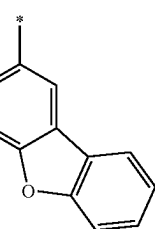
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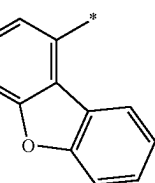
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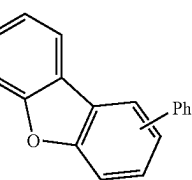
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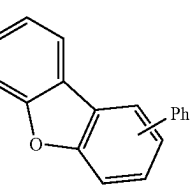
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10-241

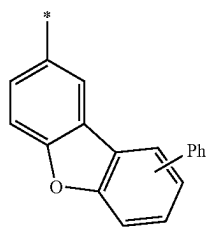


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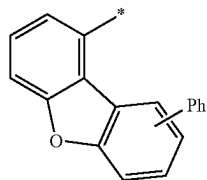


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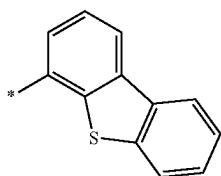
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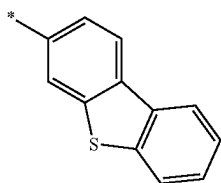
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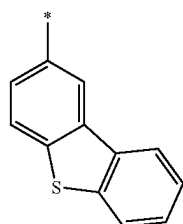
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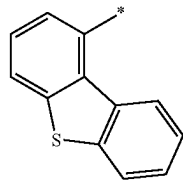
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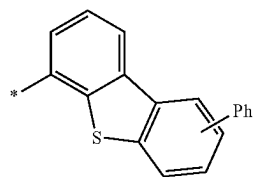
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10-248

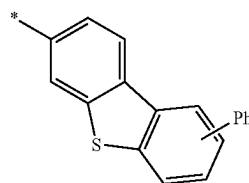


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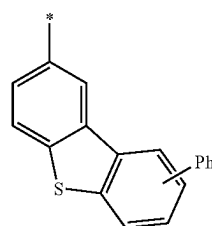


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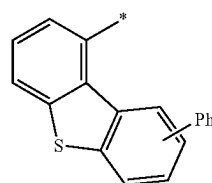
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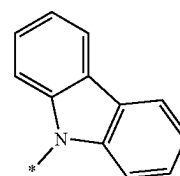
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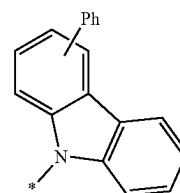
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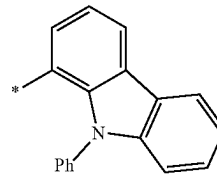
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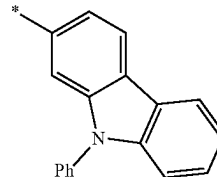
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10-255

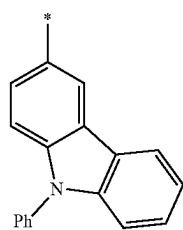


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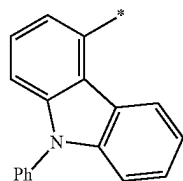


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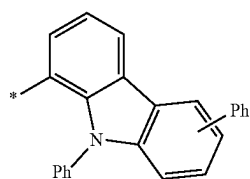
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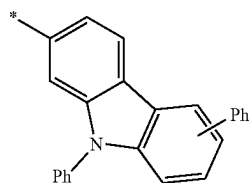
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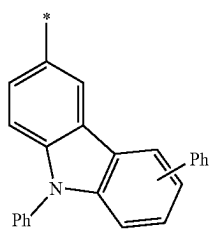
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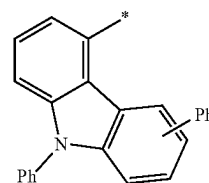
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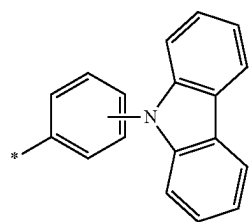
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10-262

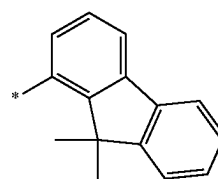


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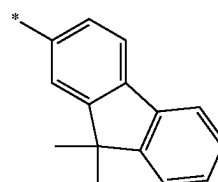


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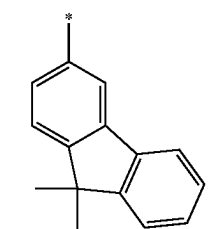
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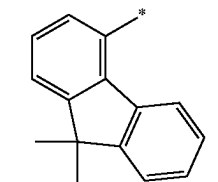
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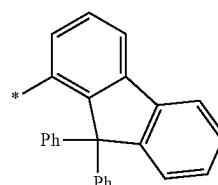
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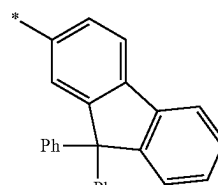
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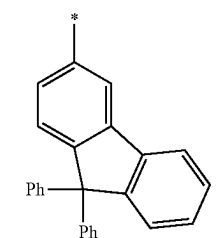
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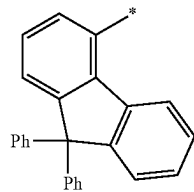


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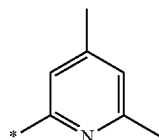


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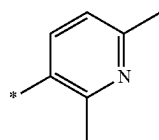
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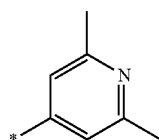
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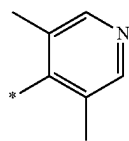
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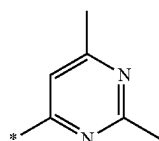
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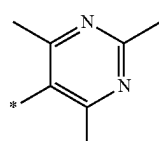
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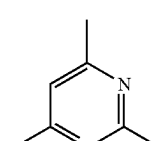
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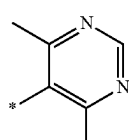
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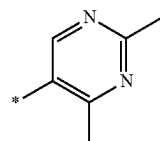


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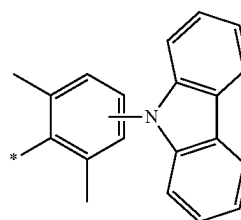


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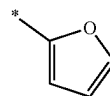
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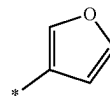
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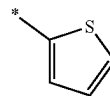
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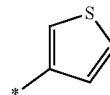
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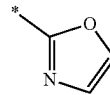
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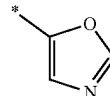
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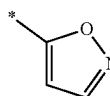
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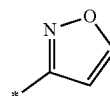
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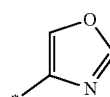
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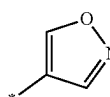
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10-290

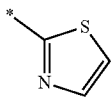


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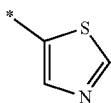


10-292

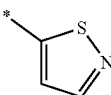
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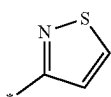
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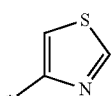
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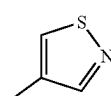
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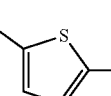
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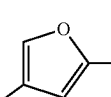
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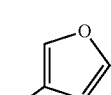
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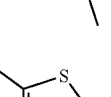
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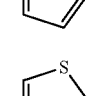
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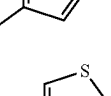
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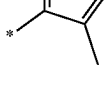
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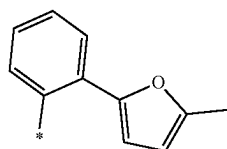
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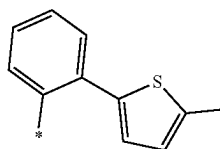
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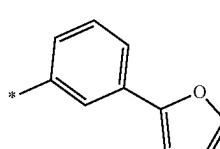
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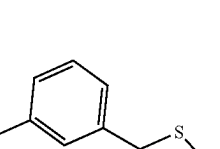
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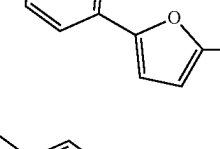
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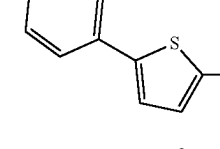
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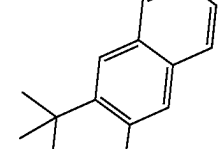
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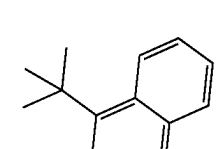
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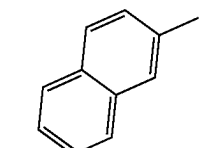
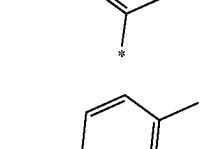
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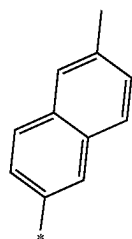
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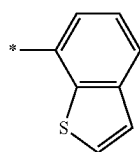
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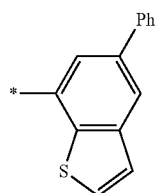
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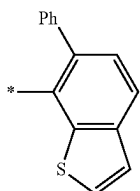
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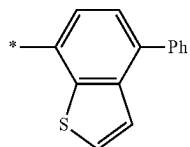
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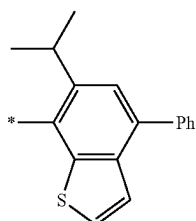
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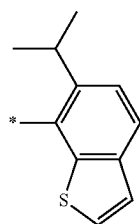
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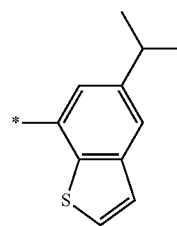


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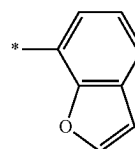


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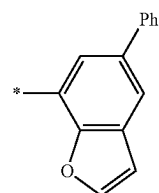
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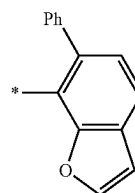
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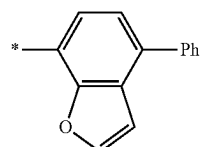
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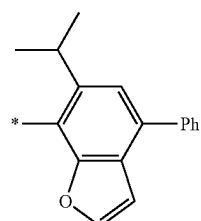
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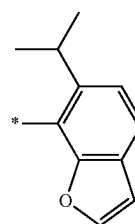
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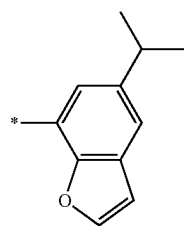


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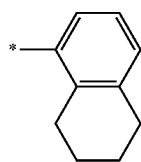


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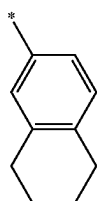
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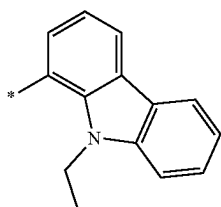
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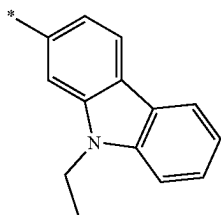
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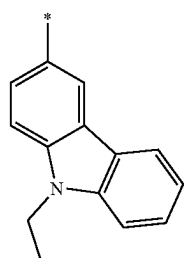
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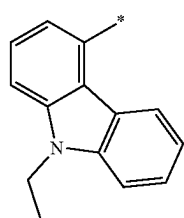
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10-332

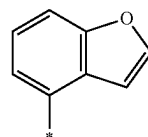


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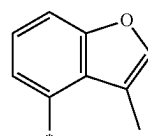


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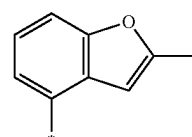
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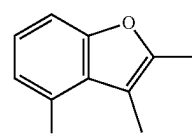
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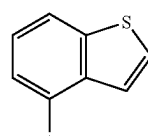
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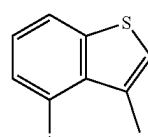
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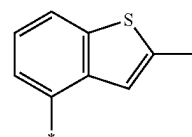
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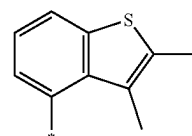
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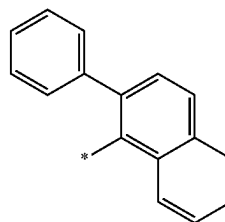
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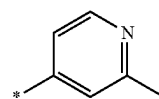
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10-342

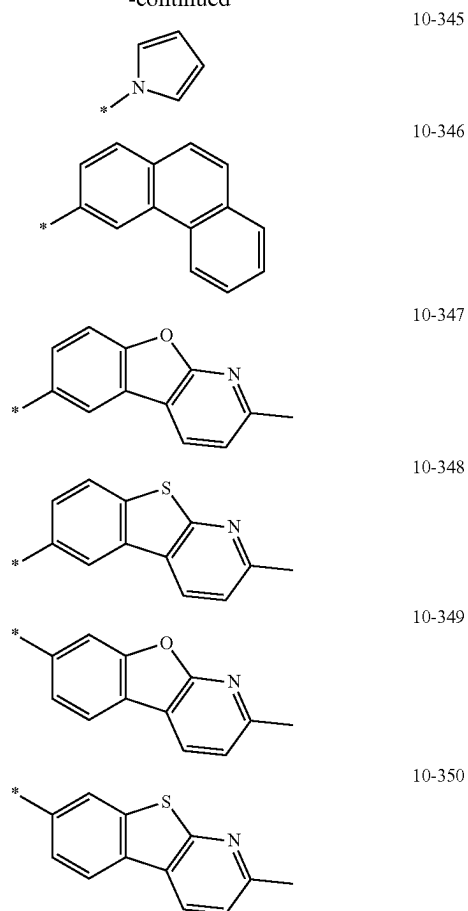


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10-344

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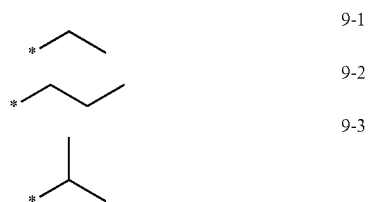
wherein, in Formulae 9-1 to 9-39, 9-44 to 9-61, 9-201 to 9-240, 10-1 to 10-129, and 10-201 to 10-350, * indicates a binding site to a neighboring atom, “Ph” indicates a phenyl group, “TMS” indicates a trimethylsilyl group, and “TMG” indicates a trimethylgermyl group.

10. The organometallic compound of claim 1, wherein at least one of R_2 to R_7 is:

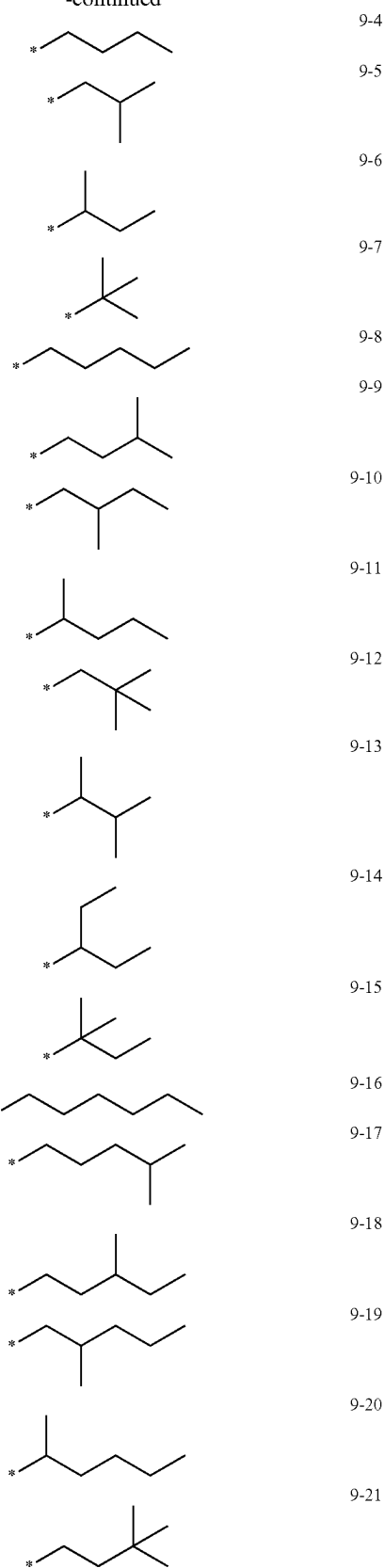
—F, —CF₃, —CF₂H, —CFH₂, —Si(Q₁)(Q₂)(Q₃), or —Ge(Q₁)(Q₂)(Q₃);

a group represented by any one of Formulae 9-1 to 9-39 in which at least one hydrogen atom is substituted with —F, —Si(Q₁)(Q₂)(Q₃), or —Ge(Q₁)(Q₂)(Q₃); or

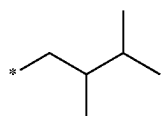
a group represented by any one of Formulae 10-1 to 10-129 or 10-201 to 10-350 in which at least one hydrogen atom is substituted with —F, —Si(Q₁)(Q₂)(Q₃), or —Ge(Q₁)(Q₂)(Q₃):



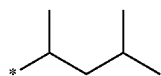
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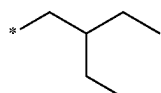
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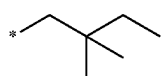
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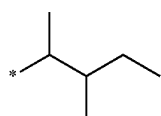
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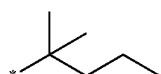
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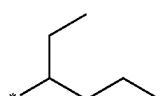
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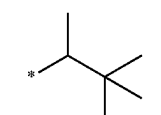
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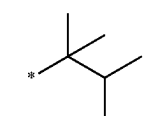
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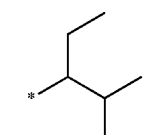
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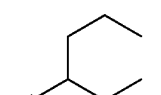
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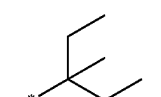
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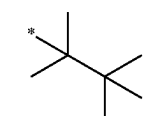
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9-32

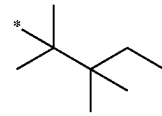


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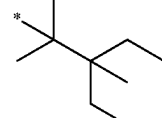


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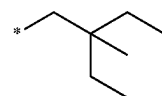
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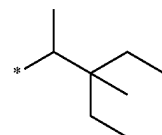
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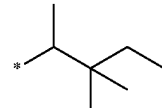
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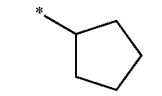
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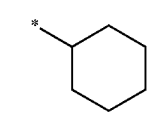
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9-39



10-1



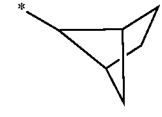
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10-3



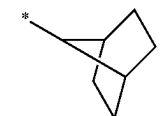
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10-5

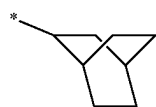


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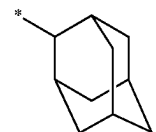


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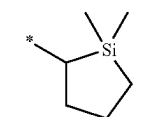
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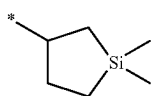
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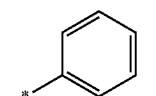
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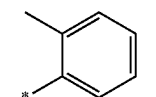
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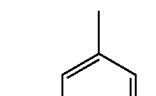
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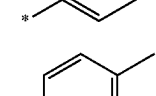
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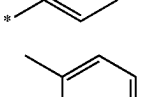
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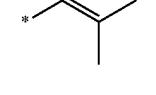
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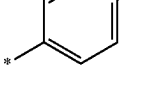
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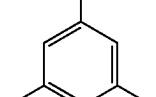
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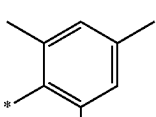
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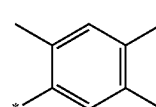
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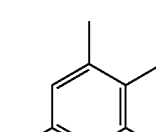
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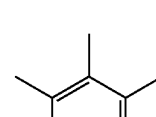
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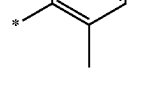
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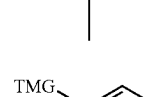
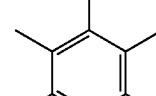
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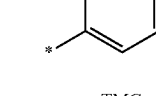
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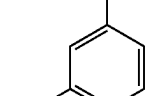
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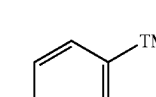
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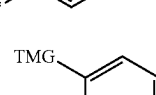
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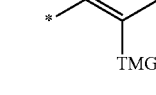
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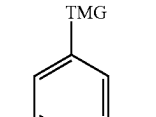
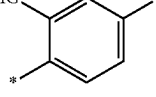
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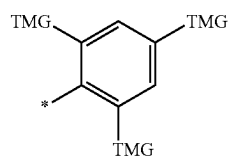
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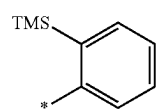
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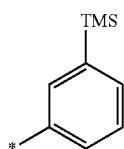
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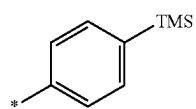
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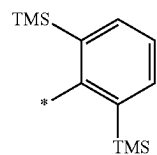
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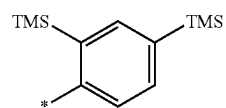
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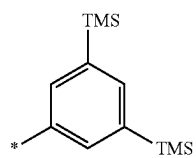
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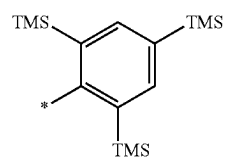
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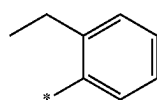
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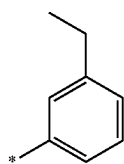
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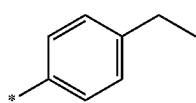
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10-38

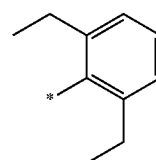


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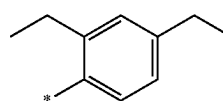


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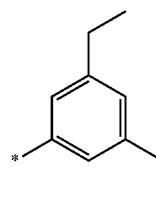
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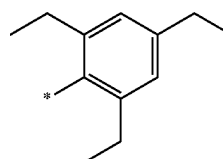
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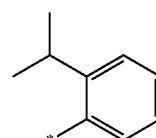
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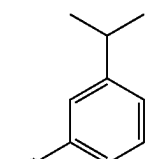
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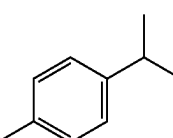
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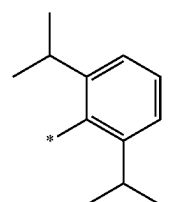
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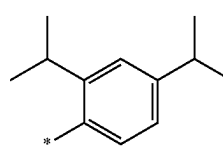
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10-47

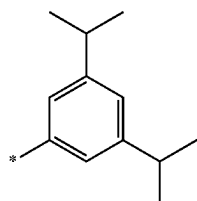


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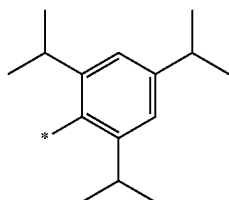


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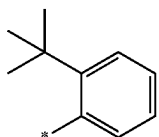
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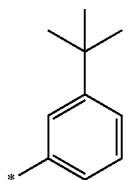
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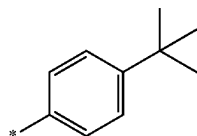
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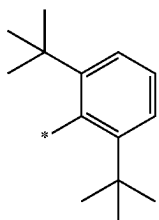
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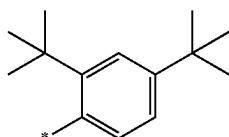
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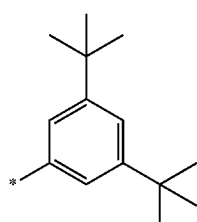
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10-55

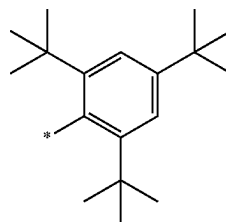


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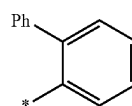


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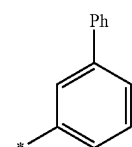
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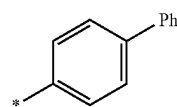
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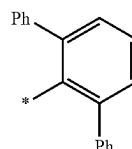
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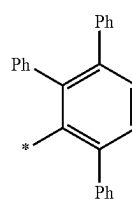
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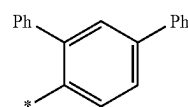
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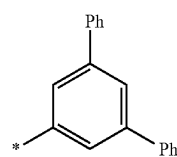
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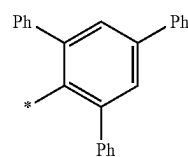
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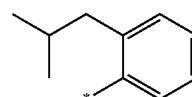
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10-65

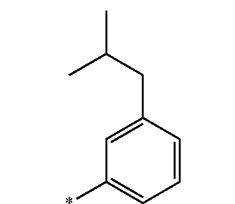


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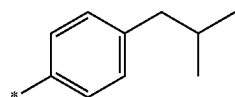


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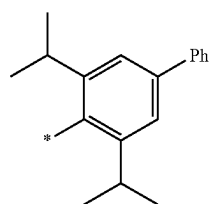
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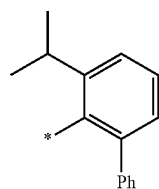
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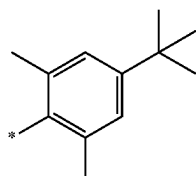
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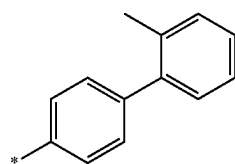
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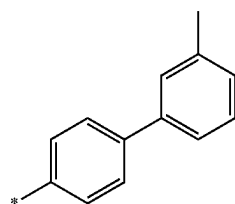
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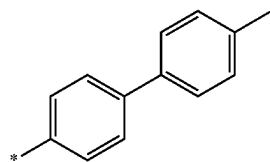
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10-73

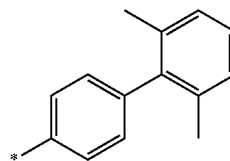


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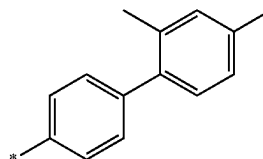


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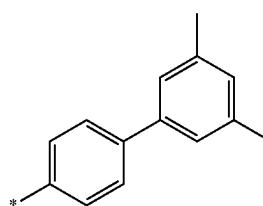
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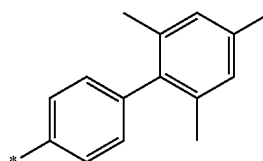
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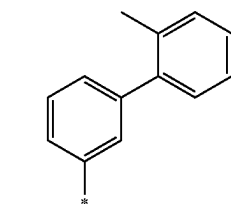
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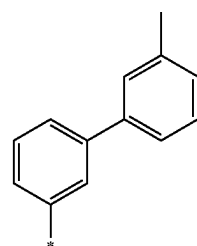
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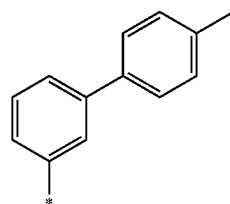
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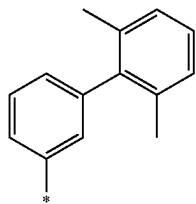


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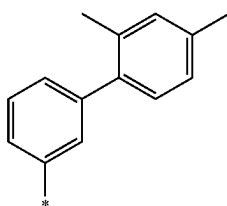


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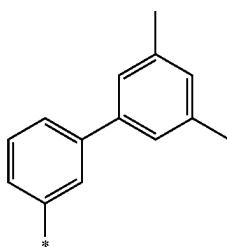
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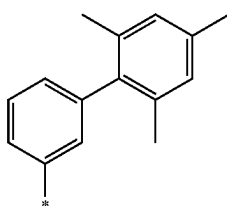
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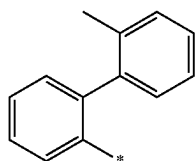
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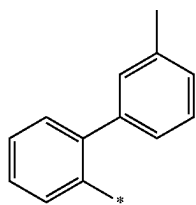
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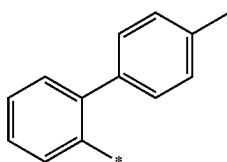
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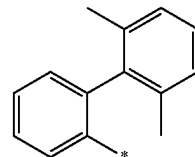


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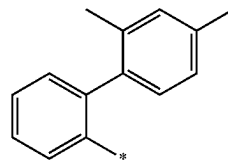


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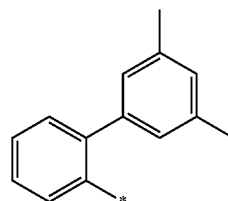
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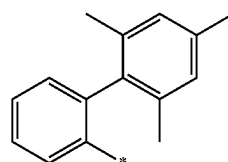
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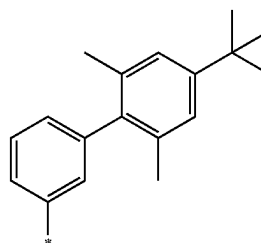
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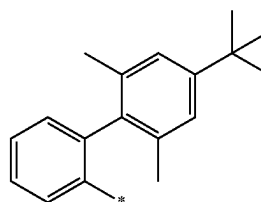
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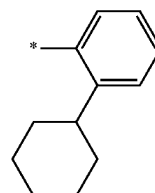
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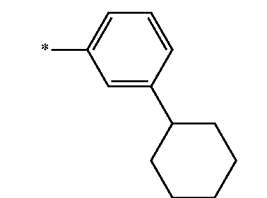


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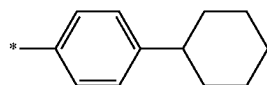


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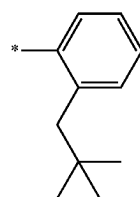
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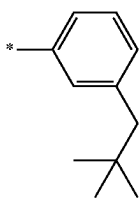
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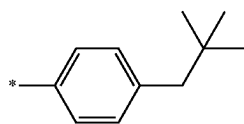
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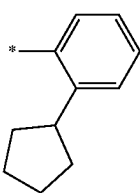
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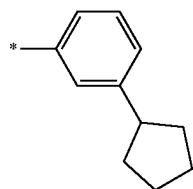
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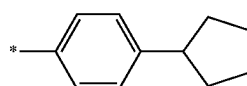
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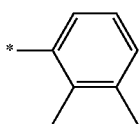
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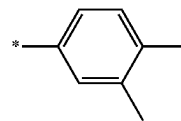


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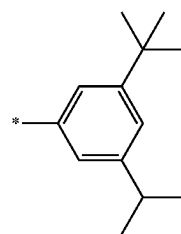


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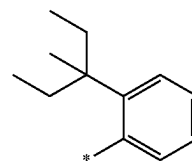
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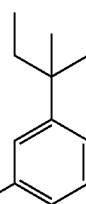
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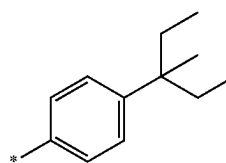
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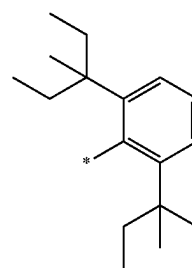
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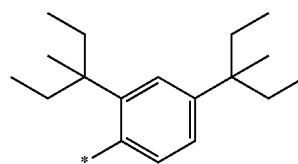
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10-110

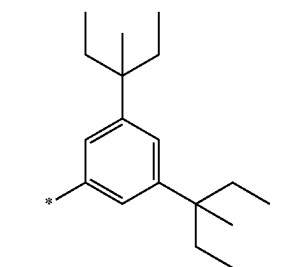


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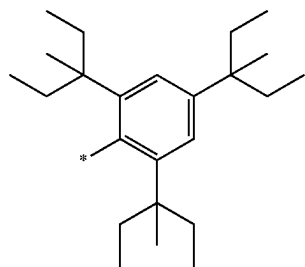


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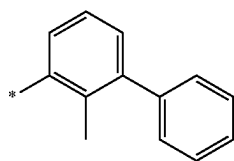
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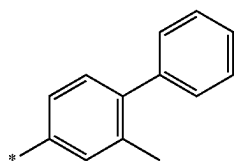
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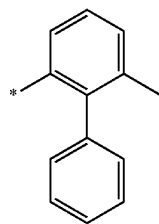
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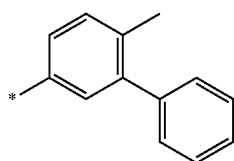
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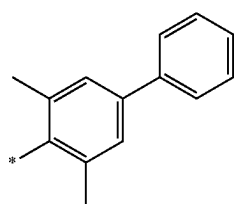
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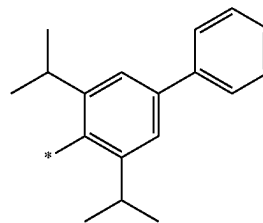


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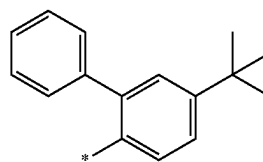


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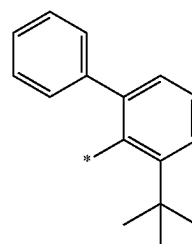
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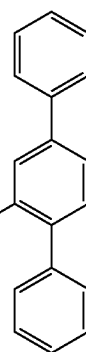
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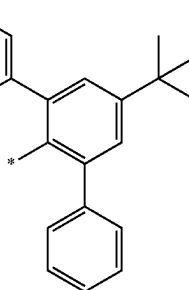
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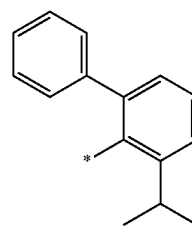
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10-123

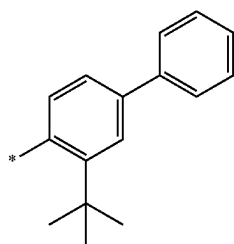


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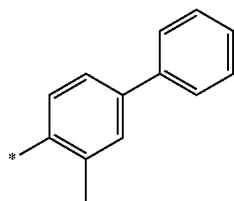


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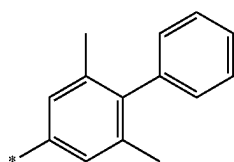
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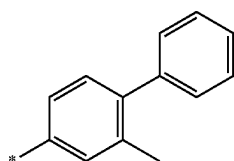
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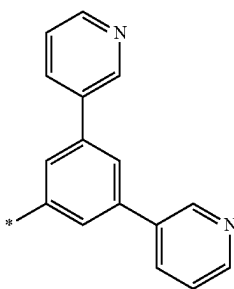
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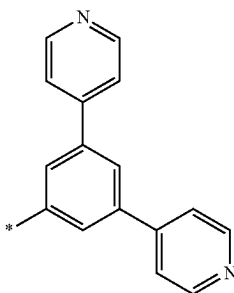
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10-129

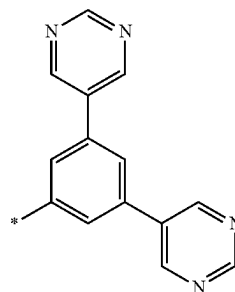


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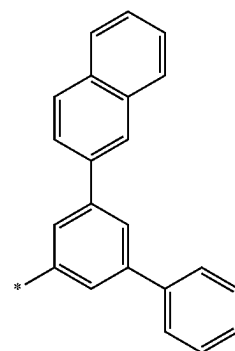


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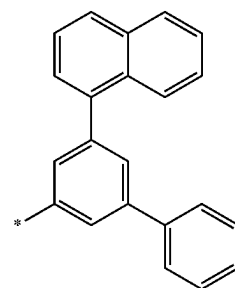
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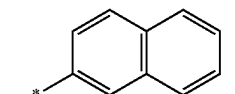
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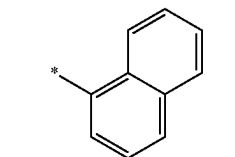
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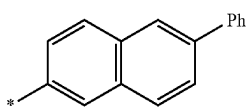
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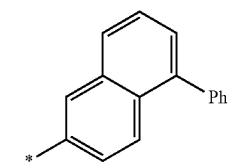
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10-207

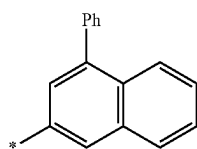


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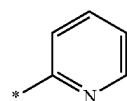


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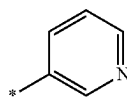
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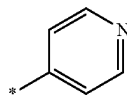
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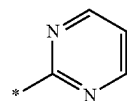
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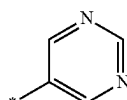
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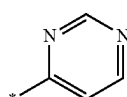
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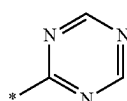
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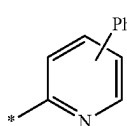
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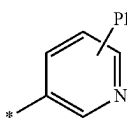
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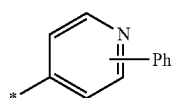
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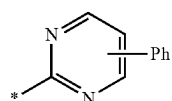
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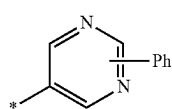
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10-220

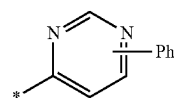


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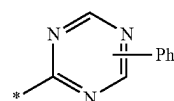


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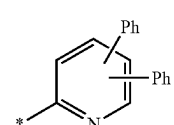
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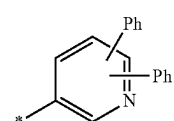
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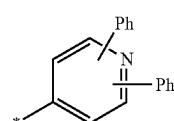
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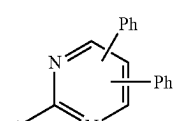
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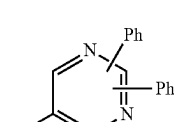
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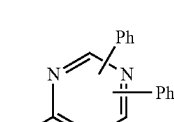
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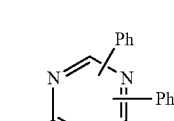
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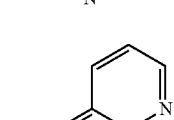
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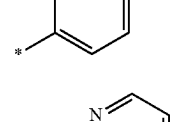
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10-231

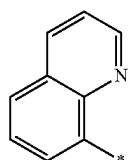


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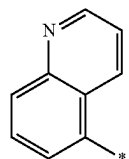


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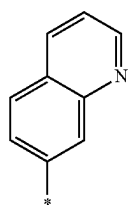
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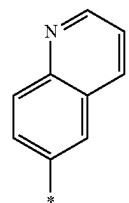
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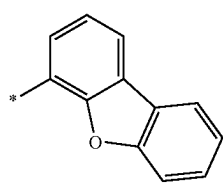
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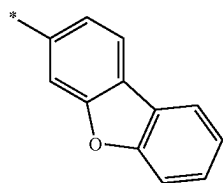
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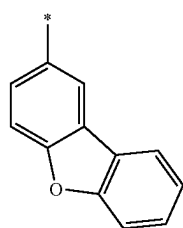
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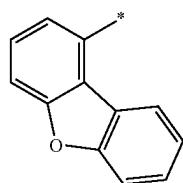
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10-239

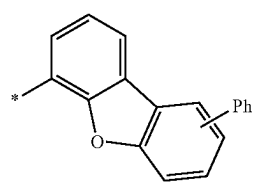


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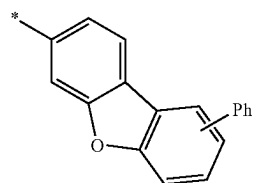


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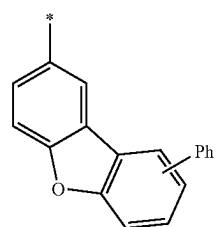
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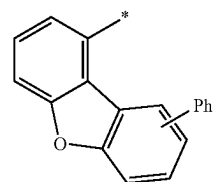
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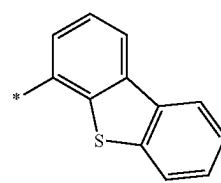
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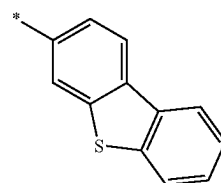
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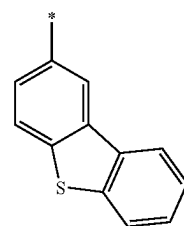
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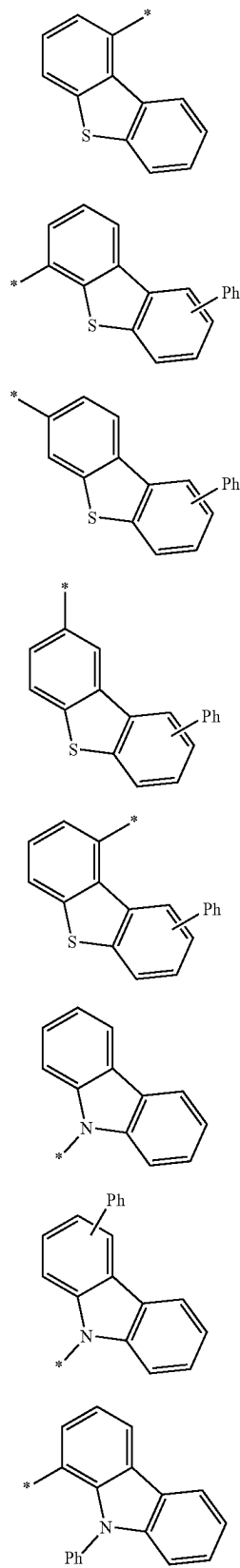


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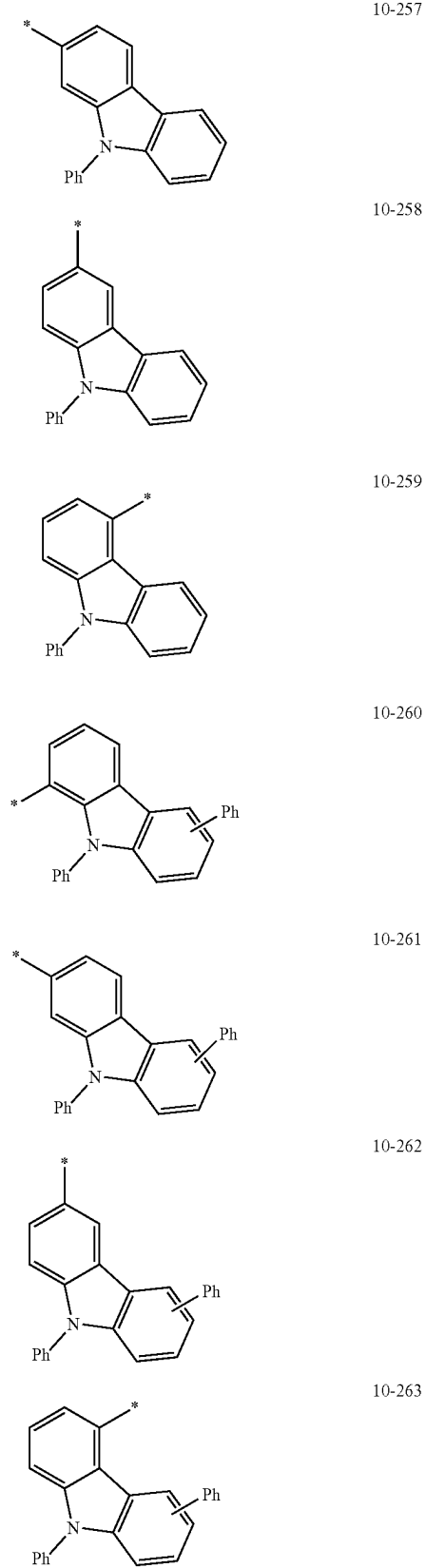


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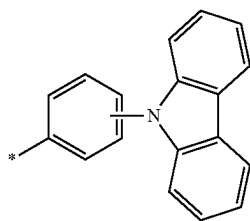
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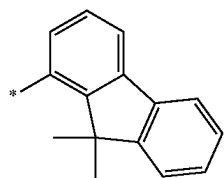
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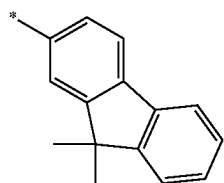
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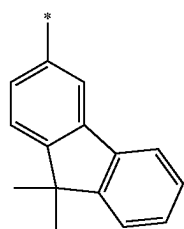
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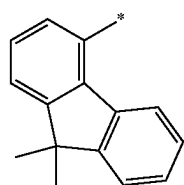
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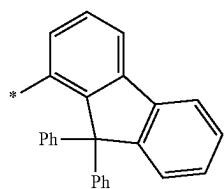
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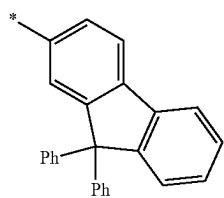
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10-268

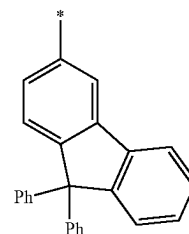


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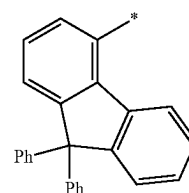


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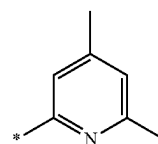
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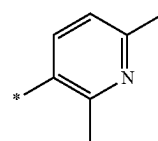
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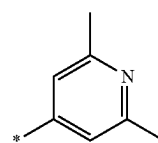
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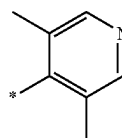
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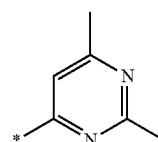
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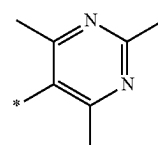
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10-276

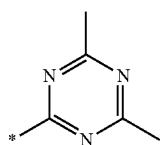


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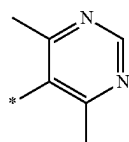


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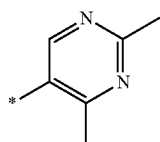
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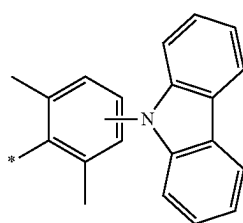
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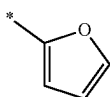
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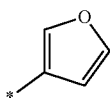
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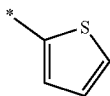
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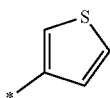
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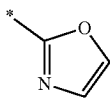
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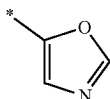
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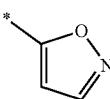
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10-287

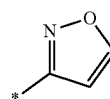


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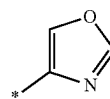


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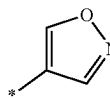
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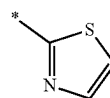
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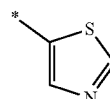
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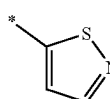
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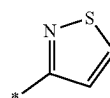
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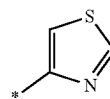
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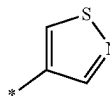
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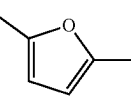
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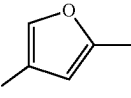
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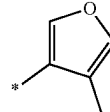
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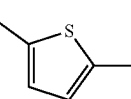
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10-300

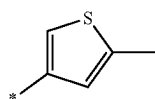


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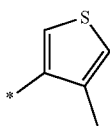


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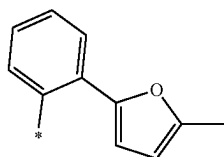
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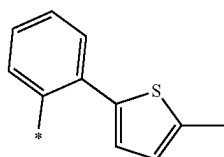
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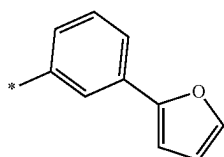
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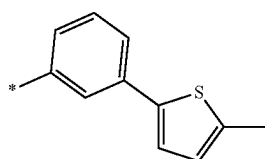
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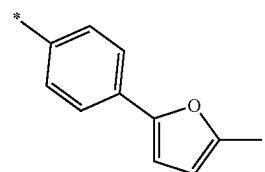
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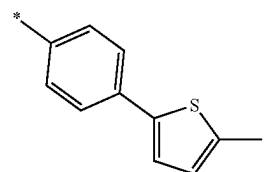
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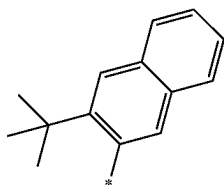
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10-309

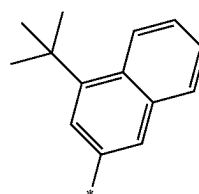


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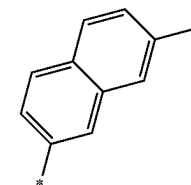


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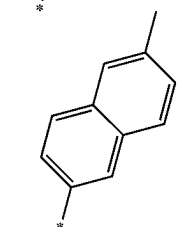
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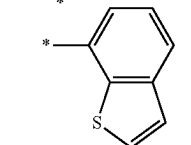
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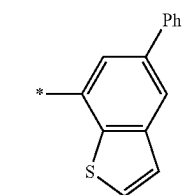
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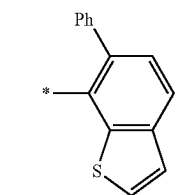
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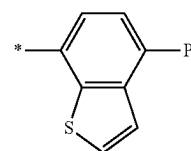
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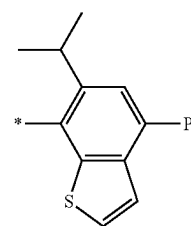
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10-317

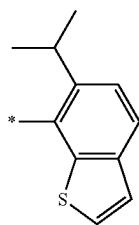


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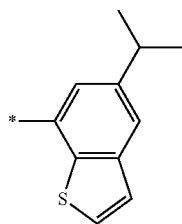


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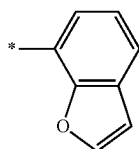
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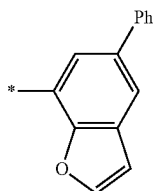
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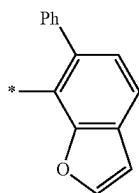
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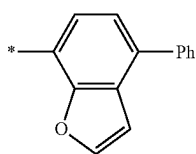
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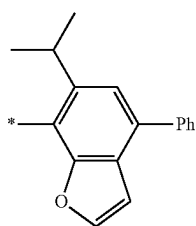
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10-324

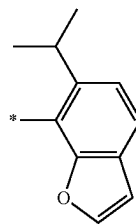


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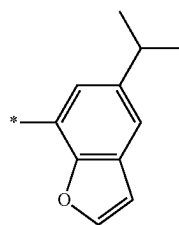


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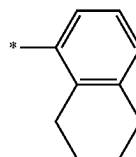
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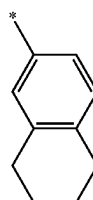
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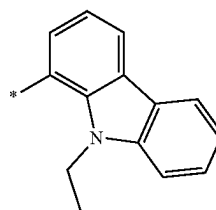
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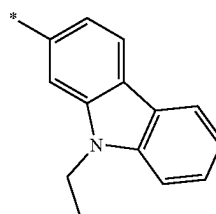
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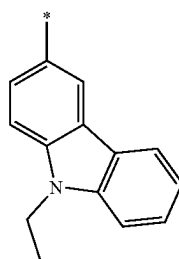
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10-331

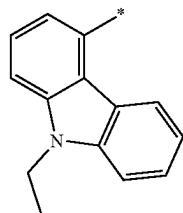


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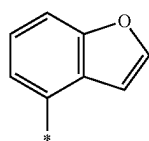


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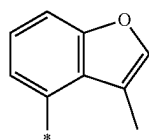
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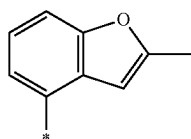
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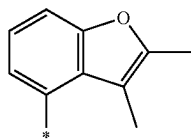
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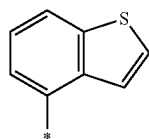
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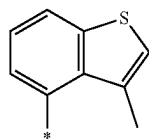
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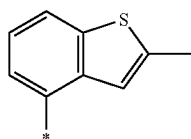
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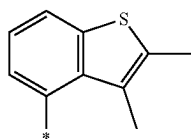
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10-340

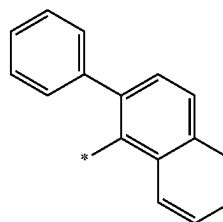


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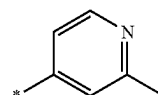


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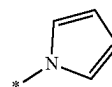
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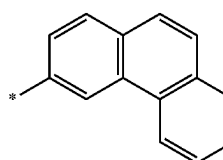
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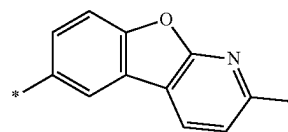
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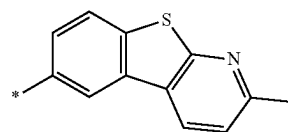
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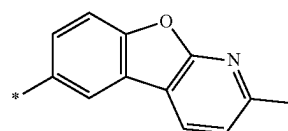
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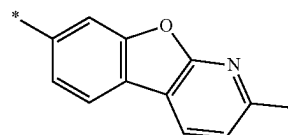
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10-348



10-349



10-350

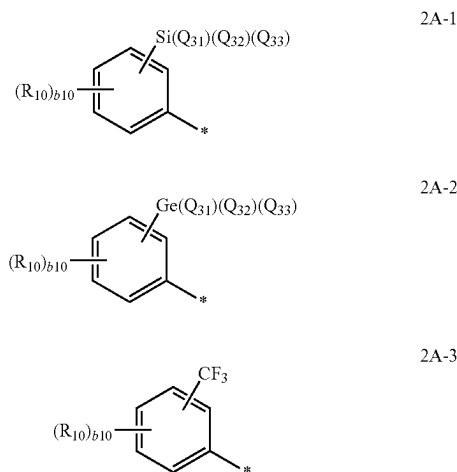
wherein, in Formulae 9-1 to 9-39, 10-1 to 10-129, and 10-201 to 10-350, * indicates a binding site to a neighboring atom, "Ph" indicates a phenyl group, "TMS" indicates a trimethylsilyl group, and "TMG" indicates a trimethylgermyl group.

11. The organometallic compound of claim 1, wherein at least one of R_2 to R_7 is:

—F, —CF₃, —CF₂H, —CFH₂, —Si(Q₁)(Q₂)(Q₃), or —Ge(Q₁)(Q₂)(Q₃);

a C₁-C₂₀ alkyl group that is substituted with at least one —F; or

a group represented by one of Formulae 2A-1 to 2A-3:



wherein, in Formulae 2A-1 to 2A-3,

R₁₀ is hydrogen, deuterium, —F, —Cl, —Br, —I, —SF₅, a hydroxyl group, a cyano group, a nitro group, an amino group, —CD₃, —CD₂H, —CDH₂, —CF₃, —CF₂H, —CFH₂, a C₁-C₁₀ alkyl group, a C₂-C₁₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₁-C₃₀ heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q₁)(Q₂)(Q₃), —Ge(Q₁)(Q₂)(Q₃), —N(Q₄)(Q₅), —B(Q₆)(Q₇), —P(Q₈)(Q₉), or —P(=O)(Q₈)(Q₉),

b10 is an integer from 1 to 4, and

Q₁ to Q₉ and Q₃₁ to Q₃₃ are each independently:

—CH₃, —CD₃, —CD₂H, —CDH₂, —CH₂CH₃, —CH₂CD₃, —CH₂CD₂H, —CH₂CDH₂, —CHDCH₃, —CHDCD₂H, —CHDCDH₂, —CHDCD₃, —CD₂CD₃, —CD₂CD₂H, or —CD₂CDH₂;

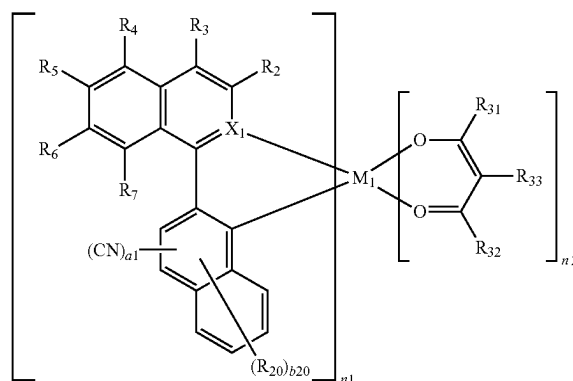
an n-propyl group, an iso-propyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, an n-pentyl group, an isopentyl group, a sec-pentyl group, a tert-pentyl group, a phenyl group, or a naphthyl group; or

an n-propyl group, an iso-propyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, an n-pentyl group, an isopentyl group, a sec-pentyl group, a tert-pentyl group, a phenyl group, or a naphthyl group, each substituted with at least one of deuterium, a C₁-C₁₀ alkyl group, a phenyl group, or a combination thereof.

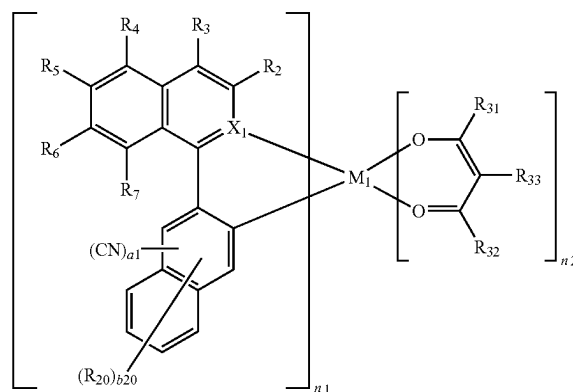
12. The organometallic compound of claim 1, wherein R₃₃ is hydrogen or deuterium.

13. The organometallic compound of claim 1, wherein the organometallic compound is represented by any one of Compounds 5-1 to 5-3:

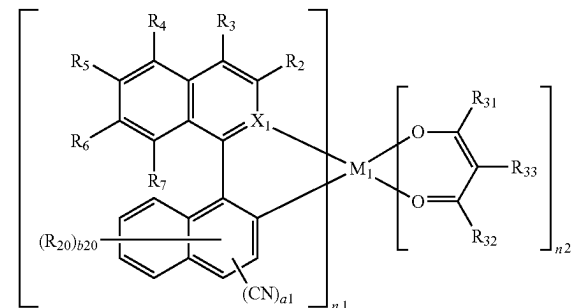
Formula 5-1



Formula 5-2



Formula 5-3

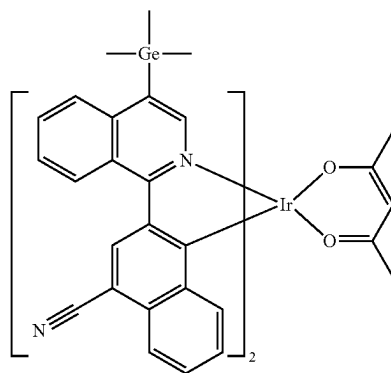
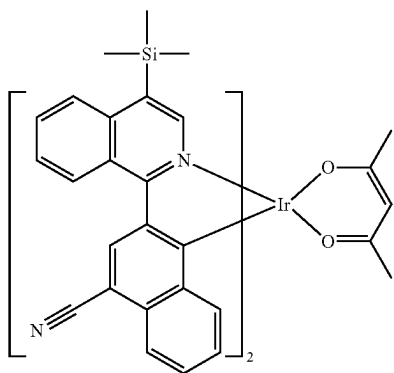
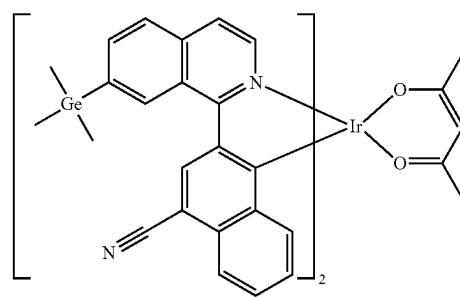
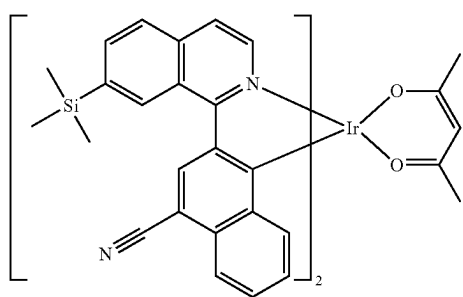
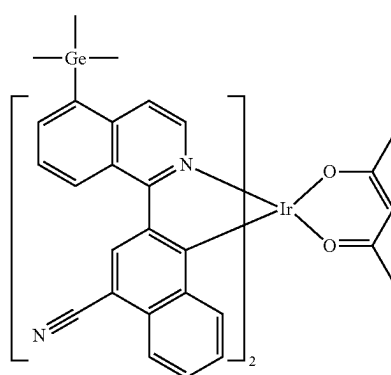
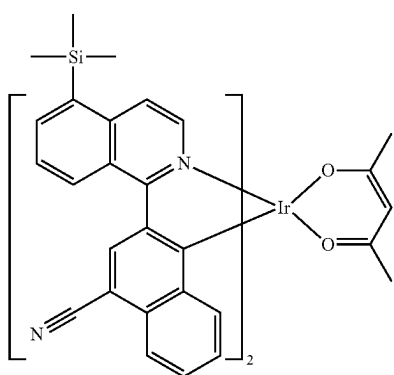
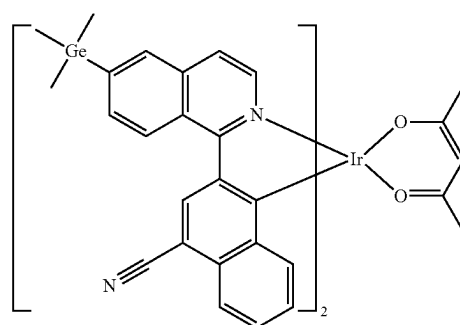
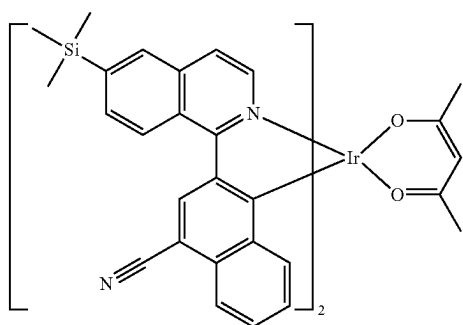


wherein, in Formulae 5-1 to 5-3,

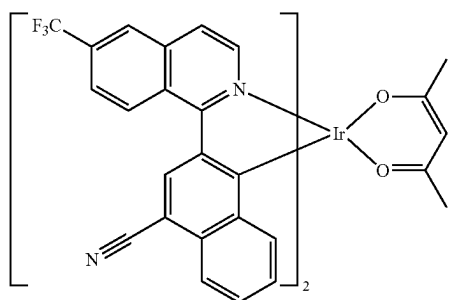
M₁, n1, n2, X₁, R₂ to R₇, a1, R₂₀, b20, and R₃₁ to R₃₃ are each as described in claim 1.

14. The organometallic compound of claim 1, wherein the organometallic compound is at least one of Compounds 1 to 72:

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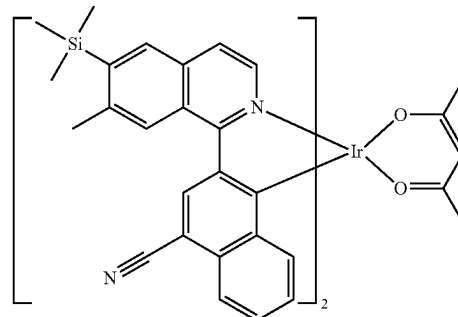


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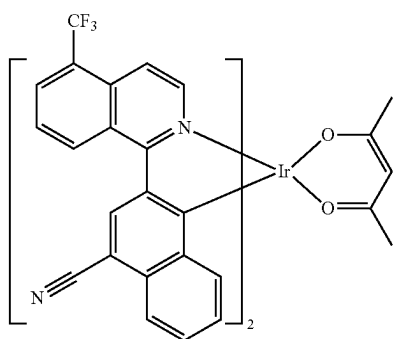
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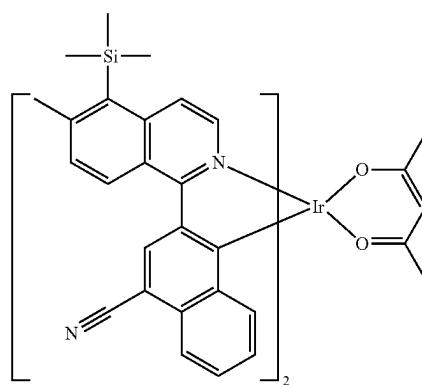


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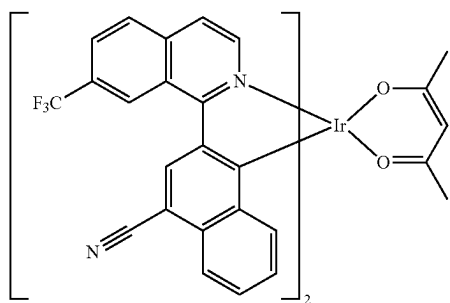
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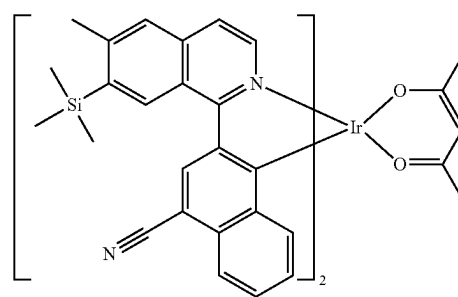
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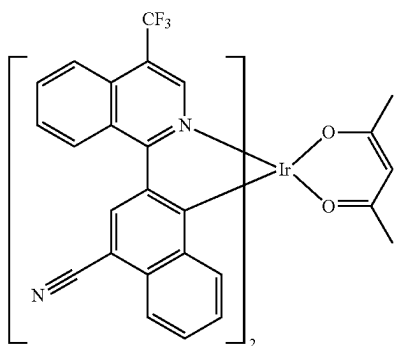
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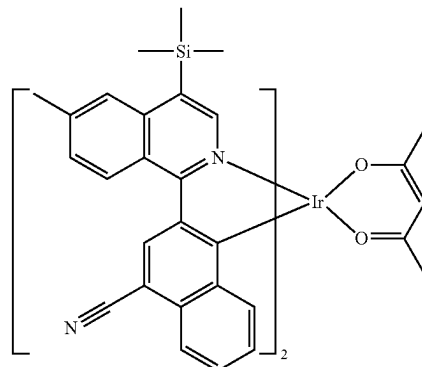
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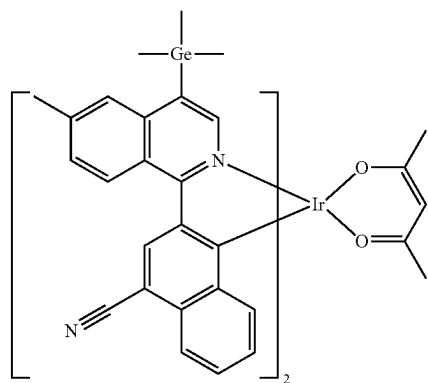
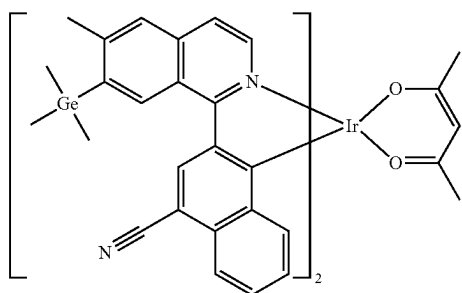
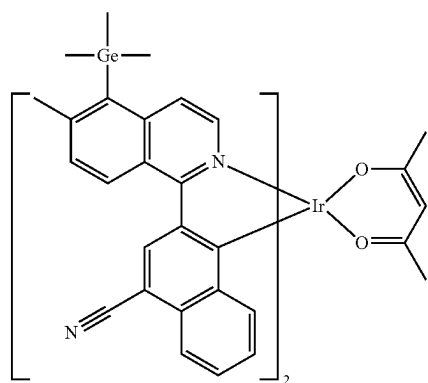
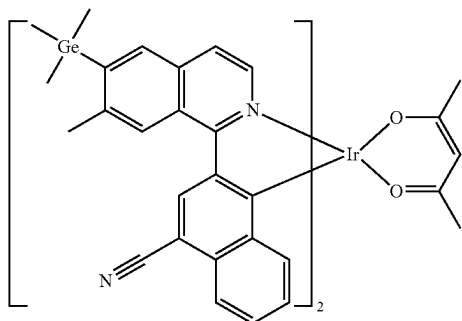
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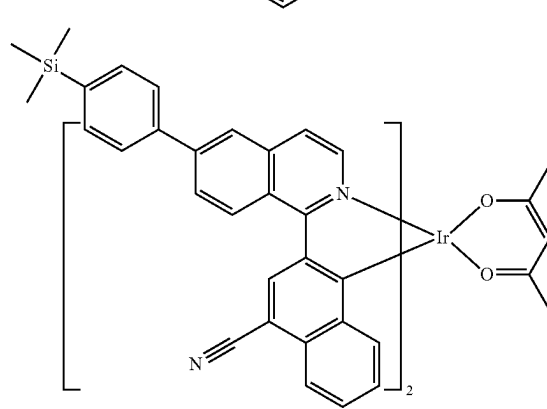
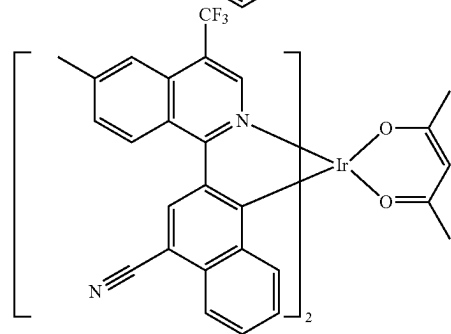
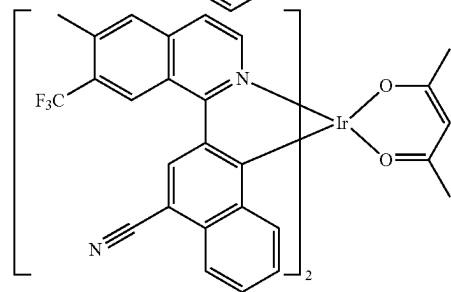
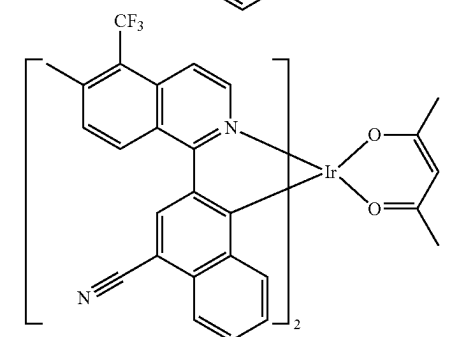
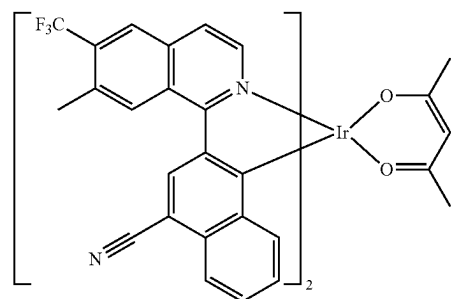
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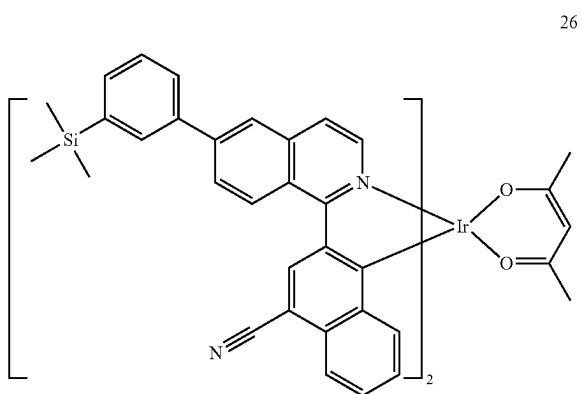
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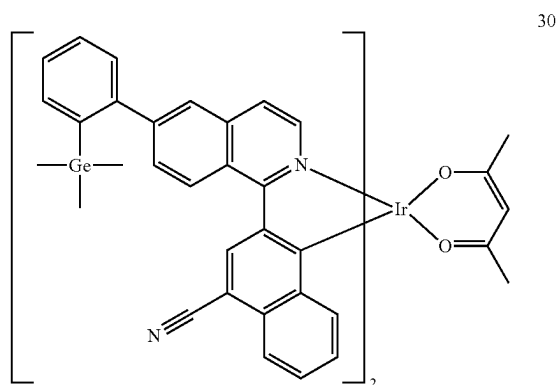
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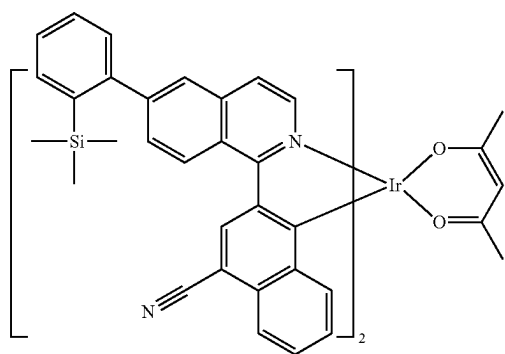
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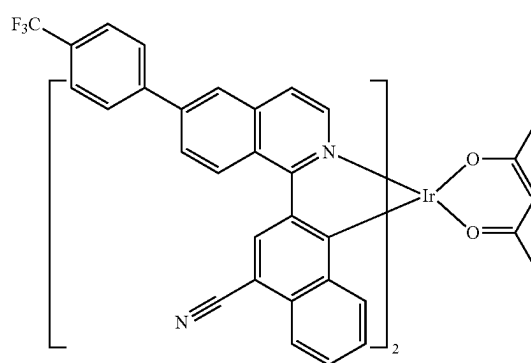
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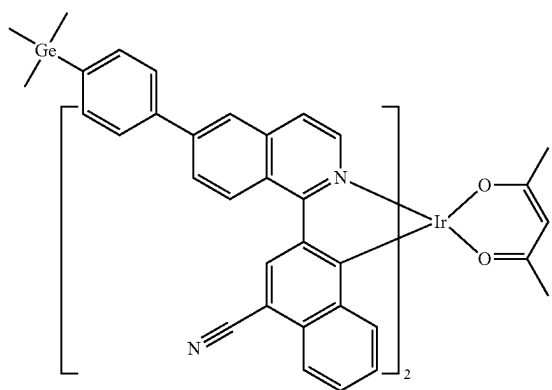
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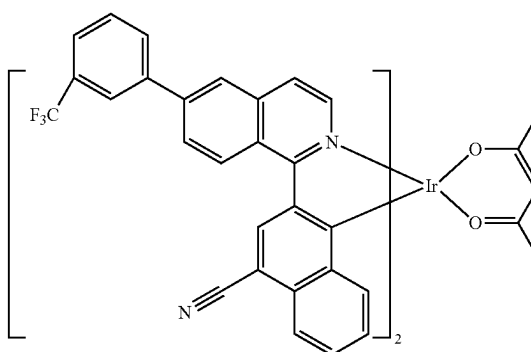
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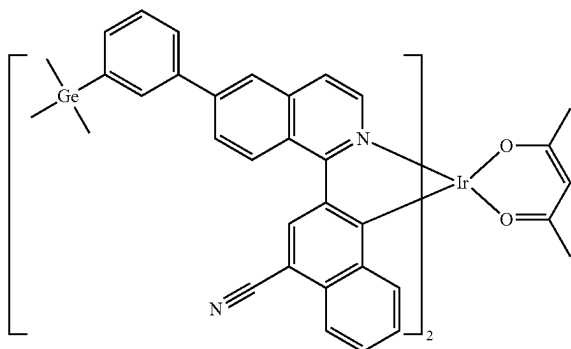
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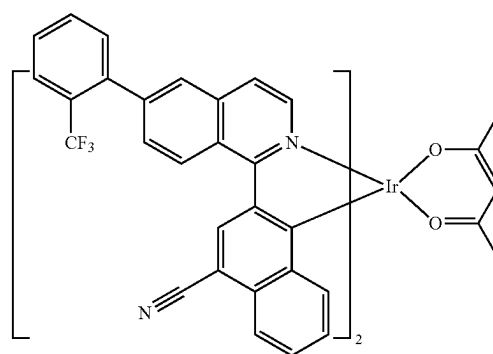
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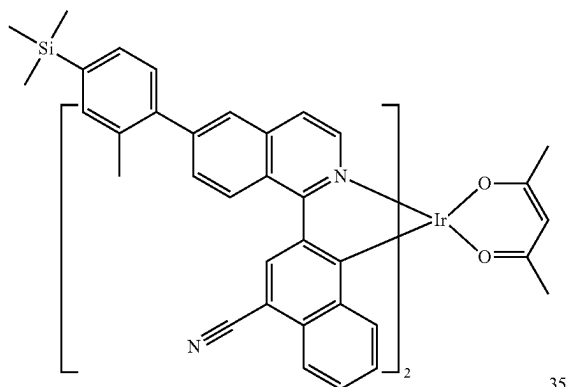


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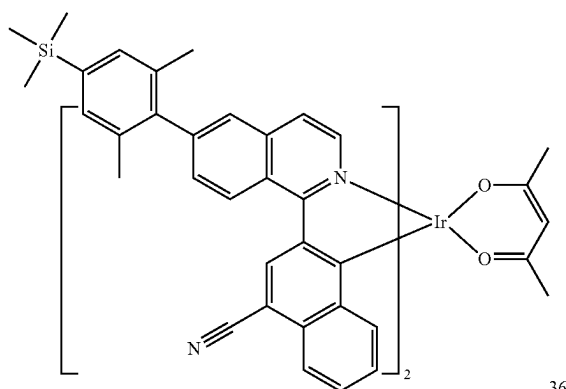


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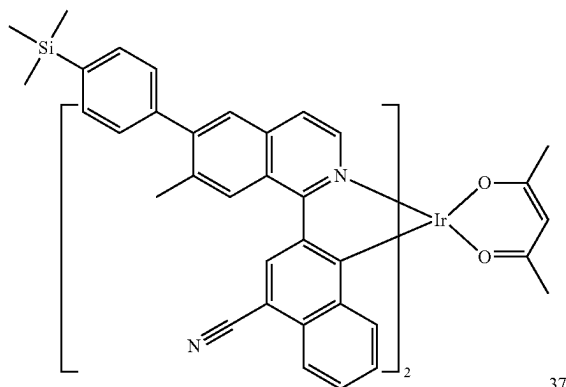
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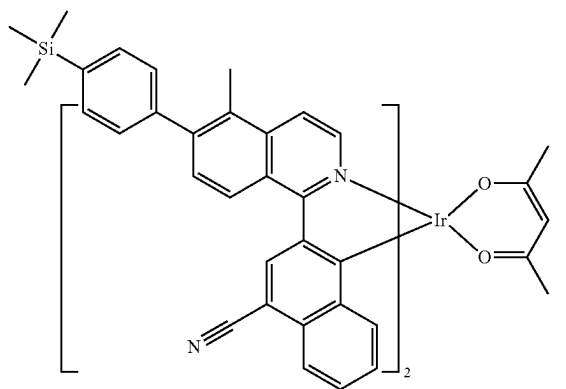
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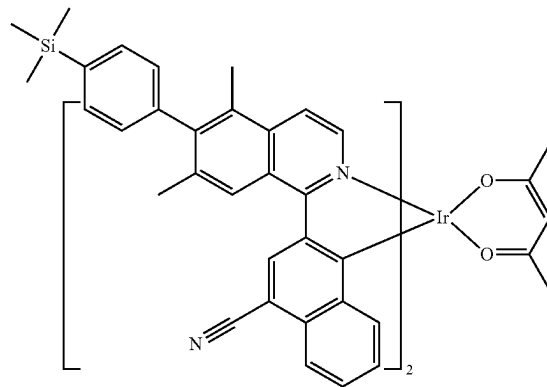


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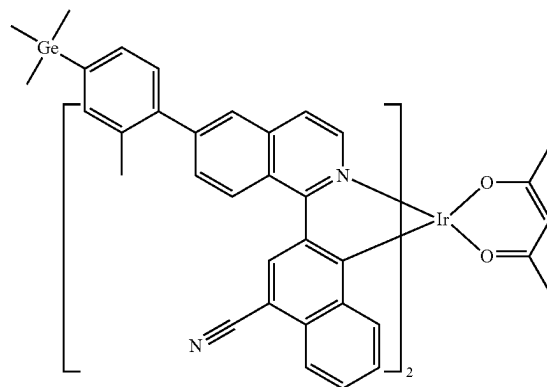


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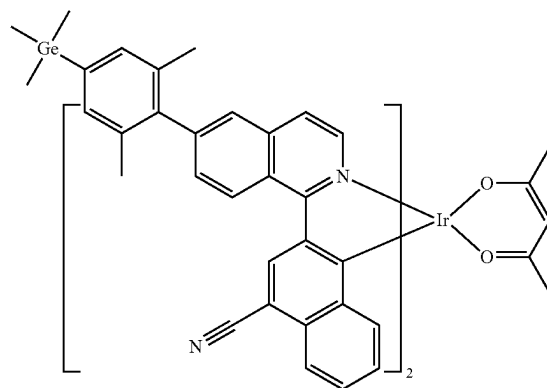
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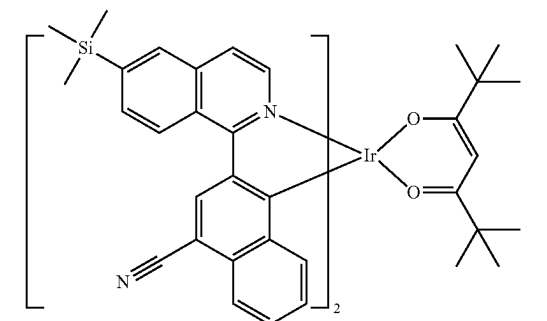
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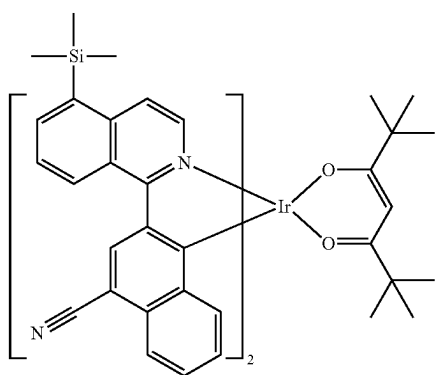
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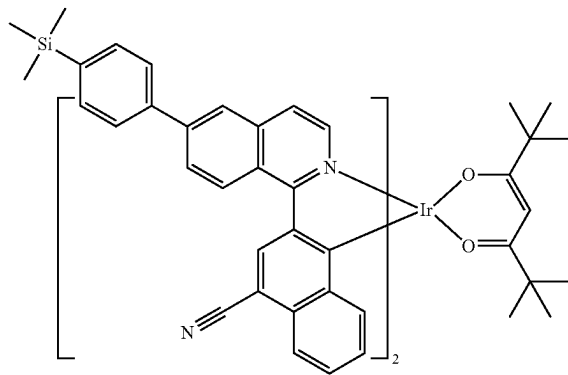


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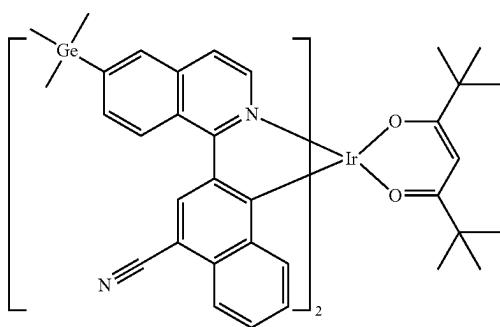


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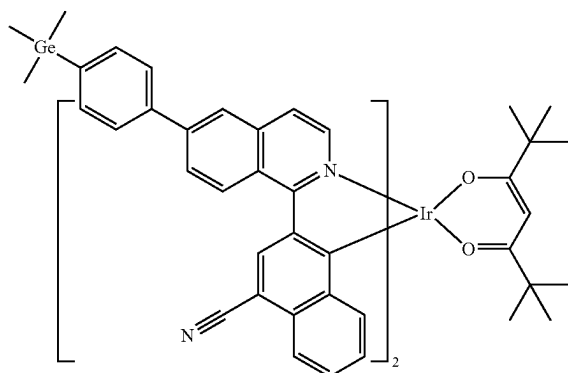
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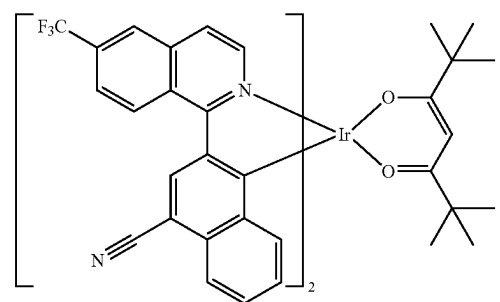
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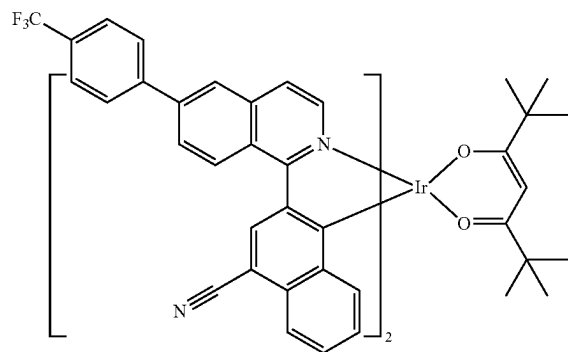
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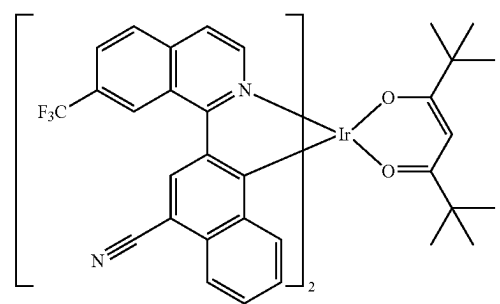
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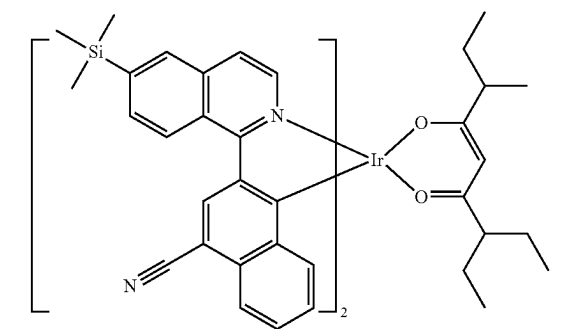
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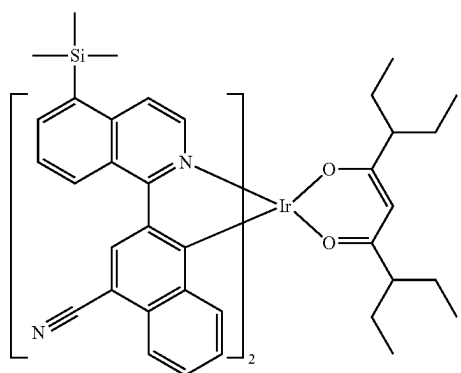


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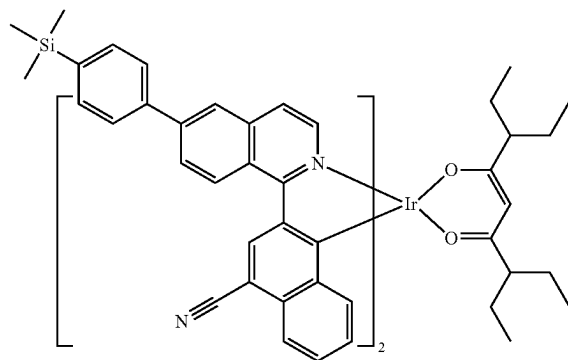
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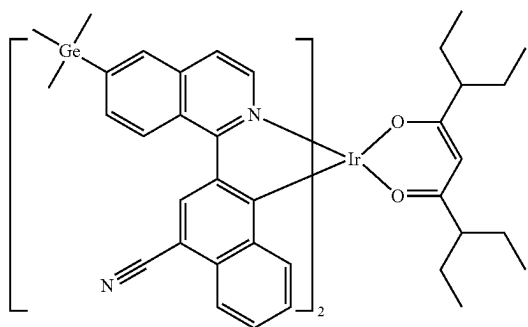


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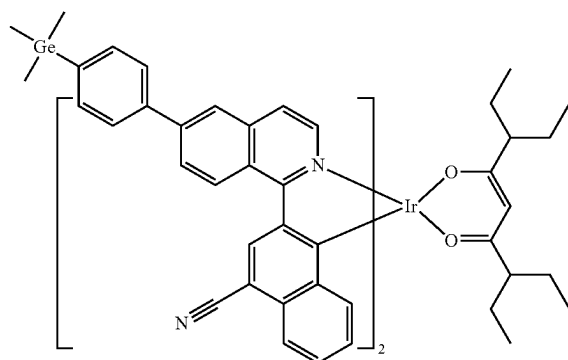
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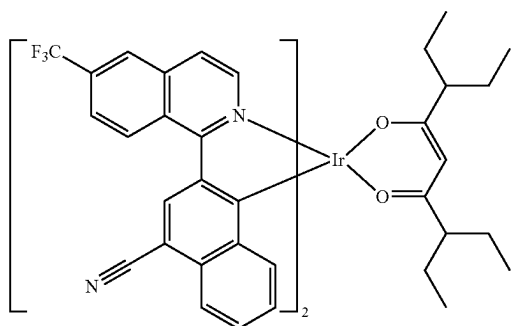
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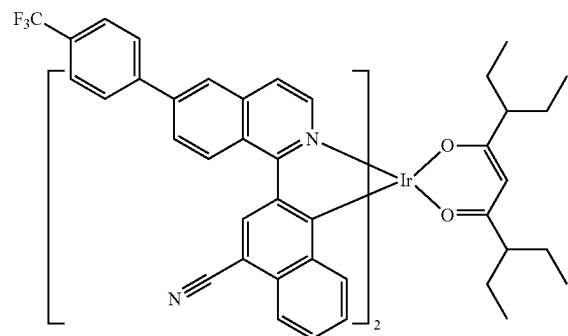
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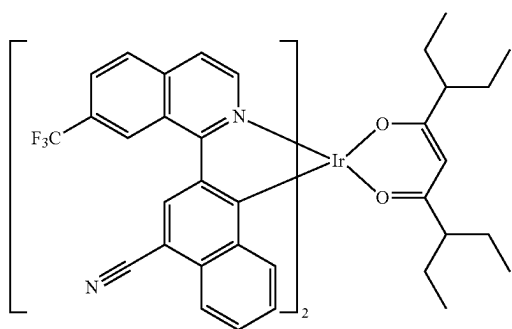
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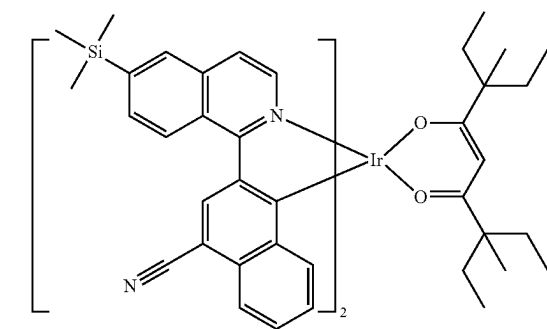
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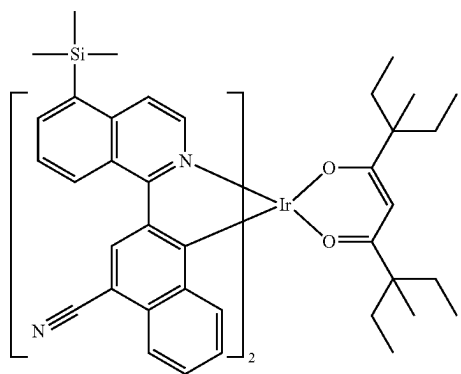


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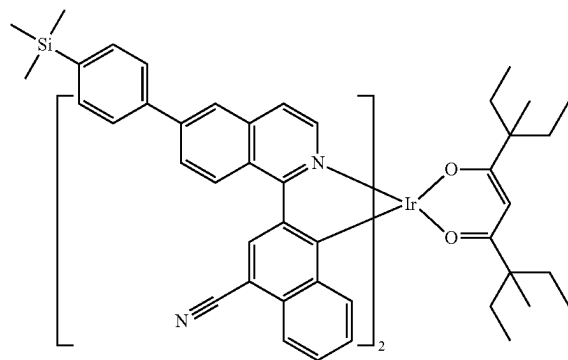
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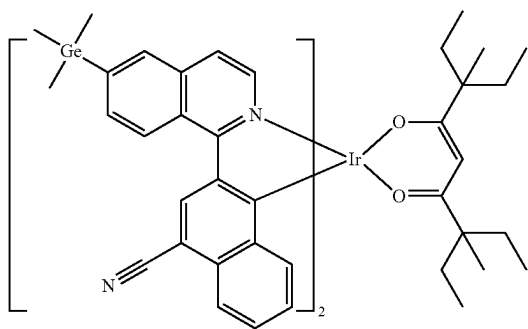


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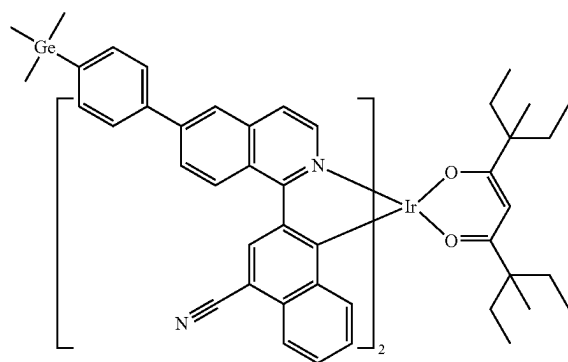
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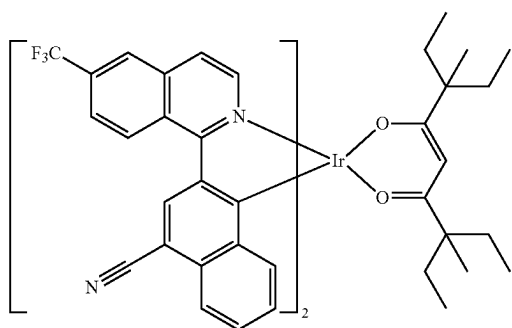
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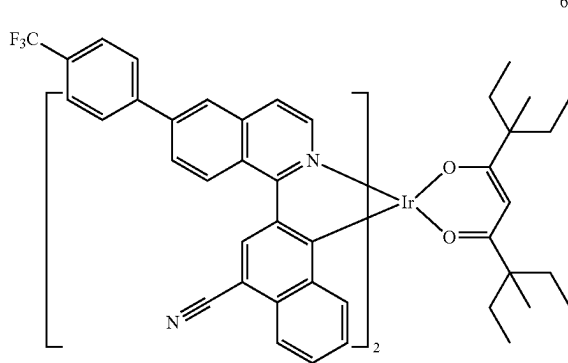
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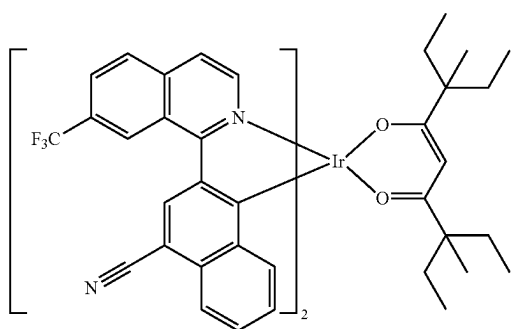
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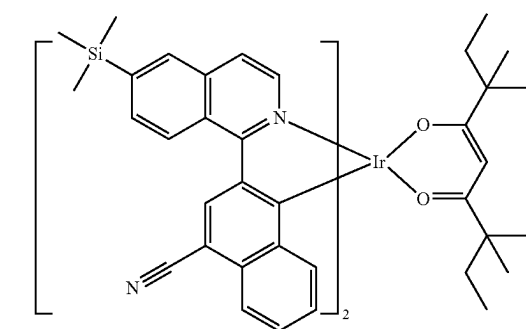
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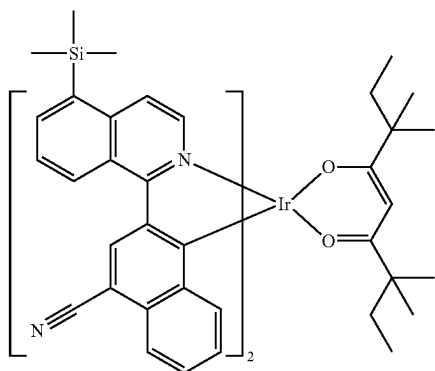


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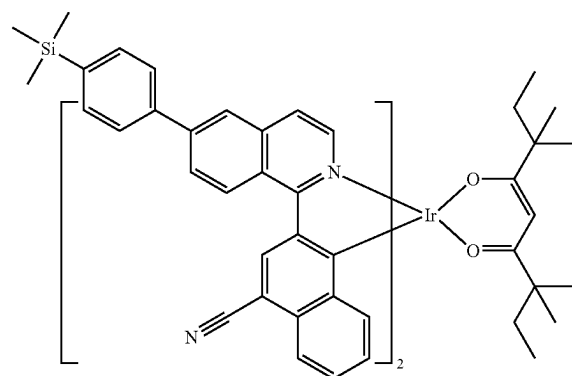
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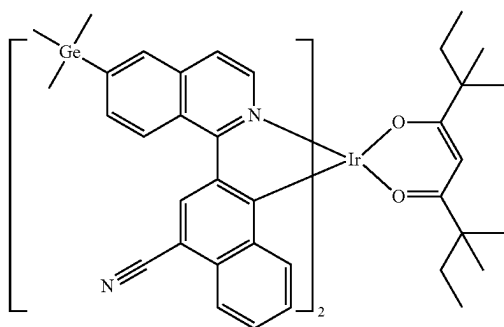


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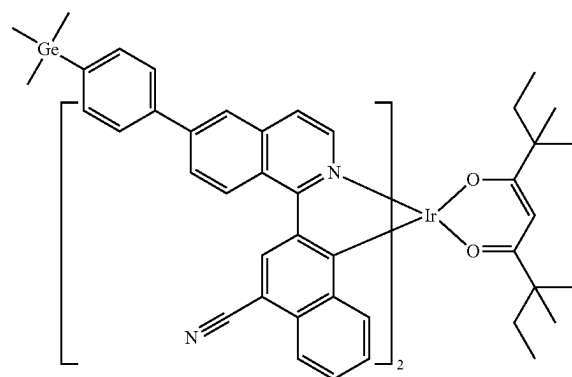
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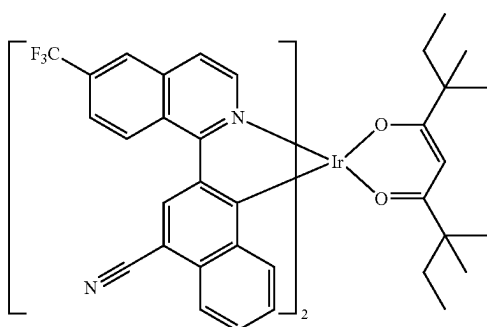
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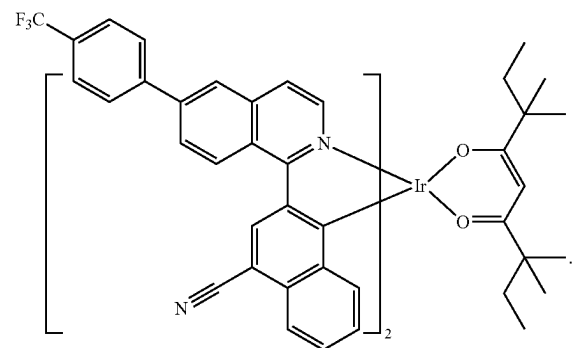
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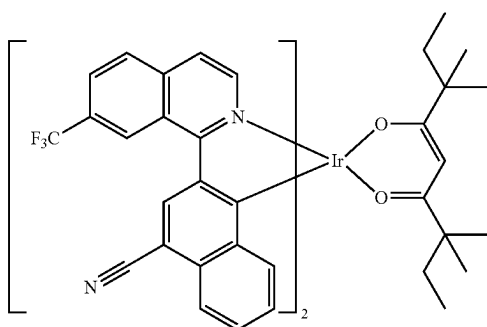
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15. An organic light-emitting device, comprising:

a first electrode;

a second electrode; and

an organic layer arranged between the first electrode and the second electrode,

wherein the organic layer comprises an emission layer, and

wherein the organic layer further comprises at least one of the organometallic compound of claim 1.

16. The organic light-emitting device of claim 15, wherein the emission layer comprises the at least one organometallic compound.

17. The organic light-emitting device of claim 15, wherein the emission layer further comprises a host, and an amount

of the host in the emission layer is greater than an amount of the at least one organometallic compound in the emission layer, on a weight basis.

18. The organic light-emitting device of claim **16**, wherein the emission layer emits a red light having a maximum emission wavelength of about 590 nanometers to about 650 nanometers.

19. The organic light-emitting device of claim **15**, wherein the first electrode is an anode,

the second electrode is a cathode,

the organic layer further comprises a hole transport region

arranged between the first electrode and the emission

layer, and an electron transport region arranged

between the emission layer and the second electrode,

the hole transport region comprises a hole injection layer,

a hole transport layer, an electron blocking layer, a

buffer layer, or a combination thereof, and

the electron transport region comprises a hole blocking

layer, an electron transport layer, an electron injection

layer, or a combination thereof.

20. An electronic apparatus, comprising the organic light-emitting device of claim **15**.

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